Zero Ties: The Complete Series

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Why it was written. What was done. What was asked. What the evidence shows.

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Why it was written. What was done. What was asked. What the evidence shows. — the full account, in the order that tells it: the purpose, the attack, the letter, and the seven investigations beneath them.

The Complete Edition

Foreword: Why This Was Researched and Written As It Was: The Method and Purpose of the Zero-Ties Series

An account of the reasoning behind seven documents — four device audits, an app audit, a war reconstruction, and the framework that binds them — written by the process that produced them.

1. The Question That Started Everything

Every investigation in this series began with a single sentence, given as a criterion rather than a curiosity: **a device with zero ties to Israel — in any way, foreign or domestic.**

The wording matters, and this document preserves it exactly, because every method choice downstream flows from it. "Zero ties in any way" is a far stricter standard than "not made in Israel" — it reaches past factories and assembly lines into design authorship, corporate ownership, venture portfolios, government contracts, the software inside the software, and the biographies of the people who wrote the code. The user did not ask which products felt clean. They asked which products *are* clean, under a definition that admits no hiding places.

The private motives behind the criterion are the user's own, and this document does not invent them. What can be documented is the record the series itself built around the question: by the fourth document, the Lebanon file had established — with names, dates, chemistry, and court-adjacent corporate records — that a supply chain can be a weapon, that

the authorship layer is the layer that matters, and that in September 2024 it was a kill chain. From that point forward, the criterion stopped being only a consumer standard and became something the evidence itself had already vindicated: **when the holder of a device cannot verify who authored it, the device is a standing risk.** The series was built to answer the question; the world answered it first.

2. Why the Method: Research Before Writing, Always

The working protocol, maintained across all seven documents, was simple to state and punishing to keep: **research exhaustively before writing anything; write only when told; then survive an adversarial audit before delivery counts.**

The reasoning is epistemological. A question like "is there a clean laptop?" invites two failure modes: the advocate's failure (finding what confirms the premise) and the marketer's failure (declaring cleanliness where only absence of evidence exists). The protocol attacks both. Research-only passes force the map to be drawn before the verdict is chosen. Evidence grading (audit-grade, filing-grade, on-record, reported, flagged, no-evidence-found) forces every claim to carry its own weight — so that "MediaTek has no Israeli office" and "no evidence was found of one" are never allowed to blur, because they are different statements with different certainty. And "we gather clues and evidence; we do not adopt the theories of others" — the user's own standing instruction — means Israeli government claims, Hezbollah claims, boycott-movement claims, and

corporate marketing are all treated as *sources*, never as findings.

The multi-pass structure exists because no single pass sees the whole board. The laptop took four passes; the Lebanon file took five. Each pass had a job: breadth, gap-closure, primary-source deep reading, adversarial verification. The user learned to weaponize this deliberately — "dive deeper," "no stone left unturned," "improve your searches as you go" — and the documents improved in direct proportion to how hard the passes were pushed.

3. Why the Voice: A Friend Who Refuses to Flatter You

The style instruction, restated across every document, was precise: like a really good journalist or documentary narrator — clear, authoritative, warm, defining every term as it appears, explaining *why* before *what*, no fluff, no invented stories, nothing omitted regardless of length.

The reasoning is respect. A reader who knows nothing about semiconductors deserves to understand PETN, shell companies, Article 7(2), and SBOMs without being made to feel stupid — and a reader who knows everything deserves to never be pandered to. Plain language is not simplification; it is the refusal to hide behind jargon, which is itself a kind of evidence discipline. If a claim cannot be said plainly, that is information about the claim.

The "no made-up stories" rule deserves its own line, because it is the series' most radical stylistic commitment. There are no composite characters, no illustrative anecdotes, no dramatized

scenes in any of the seven documents. The nine-year-old in the Lebanon file, Fatima, appears because the New York Times reported her; the 830-volt pacemaker shock appears because Barnaby Jack performed it on a test bench. When the record is silent, the documents say so — the assembly site of the rigged pager batteries is unconfirmed in every file, and stays unconfirmed. The drama in these documents is entirely a property of the facts, which proved sufficient.

4. Why the Structure: The Same Map, Drawn at Different Altitudes

Every document organizes its evidence the same way, because the evidence kept organizing itself that way. Five layers of entanglement: **authorship** (who designed the silicon or the code), **tooling** (what designed it — the EDA floor), **ownership** (who bought whom), **contracts** (who sells to whom, including militaries), and **people** (the talent pipeline no purchase can exclude). Two separable axes for products: corporate-level ties versus silicon-level ties — which produced the four-quadrant ledger in the phone document and held in every audit after. And one recurring finding, stated in every conclusion without exception: **the clean answer is never a better company; it is a shorter wire** — local-first over cloud, physical over eSIM, offline over connected, procedural protections over imaginary shields.

The Lebanon document sits inside the series for the same structural reason: it is the same map at a lethal altitude. The brand-licensing laundering vector the laptop audit found as a

corporate phenomenon (Gold Apollo's name over BAC's contract over Norta's money) is the identical machinery the attack used as a *weapons-delivery system*. The audit framework and the attack are one map; that realization is the series' intellectual spine, and it is why the beeper file ends by reaching back to the three device audits rather than standing alone.

5. Why the Audit Loop: Delivery Is Not the End

No document counted as delivered until the user had audited it and the audit had been survived. This was not politeness; it was quality engineering. The record of those rounds is now part of the series' own documentation: the user's repeated "Are you sure you included everything?" produced **seven genuine omissions and one self-inflicted error** in the Lebanon file, **three more omissions** in the router/car/medical trio, and **a false positive that my own patch briefly duplicated and the next audit round caught.** Every correction is logged in the conversation; none is hidden.

The deeper reason for the loop is epistemic humility as procedure. A single-pass document encodes the author's blind spots permanently. An adversarially audited document encodes them only provisionally — every error has a mechanism for discovery, and the mechanism proved itself repeatedly. The most valuable product of this series was not any single finding; it was the demonstration that an evidence claim subjected to hostile re-examination gets *more* accurate, not merely more defensive.

6. What the Series Is — and What It Refuses to Be

Taken together, the seven documents are a supply-chain audit conducted against a stated criterion, on the public record, with grades attached. They are **not** an advocacy brief: they record exculpatory evidence with the same care as inculpatory — AMD's clean silicon, MediaTek's audited Tier-D status, the diabetes-device stack's absence of ties, Bitdefender and QNX and Biotronik, the Kape counterweight (no misuse proven, audits exist), Anthropic's absence of an Israeli office, the United States' own "adaptation in advance" understanding that complicates the Lebanon legal case. A finding that cuts against the framework's dramatic arc — the clean column — appears in every document, because the column is real and the record requires it.

And they are not prophecy. Each document closes with an honestly bounded open-items list: what could not be verified, what no source addressed, which claims rest on single B-grade sourcing. An unresolved item in this series is not a failure of nerve; it is the map correctly marking the edge of the surveyed territory.

7. The Series in One Inventory

Document	Subject	Passes	The finding that defines it
The Beeper War	The Lebanon device attacks	5	The authorship layer was already a kill chain
The Zero-Ties Laptop	PCs and their stack	4	No clean laptop exists; ASUS/Acer + AMD is the floor
The Zero-Ties Phone	Handsets, carriers, prepaid	6	Honor is the only Tier E company; the modem has no clean Western maker
The Zero-Ties App	The app ecosystem	3	Three invisible infrastructure pillars, all Israeli-founded
The Zero-Ties Router	Home networking	5	The one category where near-total local control survives
The Zero-Ties Car	The connected automobile	5	Structural entanglement; minimization is the only answer
The Zero-Ties Implant	Medical devices	5	The hardest asymmetry: the device you cannot put down

Written 2026-09-20 as the series' account of itself. The methods described here are the methods that produced this document: nothing asserted above is not in the record of the seven documents and their audit rounds.

Part One: The Beeper War: The Complete Reconstruction of the September 2024 Lebanon Device Attacks

A fully-sourced, five-pass investigation into the pager and walkie-talkie explosions of September 17–18, 2024 — how the devices were rigged, who built the supply chain that delivered them, the legal and political war that followed, and what the event means for everyone who carries electronics.

1. What This Document Is

On the afternoon of September 17, 2024, thousands of pagers exploded almost simultaneously across Lebanon. The next day, walkie-talkies and home solar-energy systems joined them. It was the first time in history that a supply chain — the global system that designs, brands, ships, and sells our electronics — was weaponized in bulk, at scale, against human beings holding the finished products.

This document reconstructs that event end to end: the two days and their death toll, the chemistry and electronics inside the devices, the shell companies that carried them to their buyers, the decade of planning behind them, the legal argument that is still unresolved, and the afterlife of the attack in Israeli doctrine through 2026. Every claim carries a source and a grade. Where sources disagree — and several genuinely do — the disagreement is preserved rather than smoothed over. Nothing here is invented, and nothing that could not be verified is presented as fact.

How to read this document

- **Parts 2–3** give the event in brief and the framework for reading it.
- **Parts 4–7** cover the two days: the timeline, the casualty record (with its unresolved discrepancies), and the human stories.
- **Parts 8–10** are the engineering: exactly how the devices were rigged, the complete corporate supply chain, and the planning timeline.
- **Part 11** answers the question the law turns on: who was actually holding these devices.
- **Part 12** is the deep history — the same method's twenty-eight-year lineage.
- **Parts 13–15** are the legal war, the political record, and the attribution trail.
- **Part 16** is the cascade the attack set off, through 2026.
- **Part 17** is the expert verdict on what changed permanently.
- **Part 18** connects the event to the zero-ties device investigations that preceded this document.
- **Parts 19–23** cover method, open questions, key figures, glossary, and conclusion.

A thirty-second orientation for readers new to the subject

Three terms do most of the work in this document. **PETN** is a high explosive — chemically related to nitroglycerin, more stable, and the material of choice for small concealed charges since a terrorist tried to blow up an airliner with it hidden in his shoe in 2001. A **detonator** is the small initiator that sets off the main charge; making it non-metallic is what let these devices pass airport-style scanners. **The supply chain** is everything that happens to a product before you hold it — design, brand licensing, shipping, middlemen — and this attack's central discovery was that whoever controls the supply chain controls the product completely, invisibly, and, as it turned out, lethally.

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2. The Event in Brief

September 17, 2024, between 3:30 and 4:30 pm local time: thousands of pagers carried by members and affiliates of Hezbollah detonated within roughly an hour across Lebanon and parts of Syria. Day one: at least **12 dead — including two children — and roughly 2,700–2,800 injured**, with Iran's ambassador to Lebanon among the wounded.

September 18: hundreds of walkie-talkies exploded — some at a funeral for the previous day's dead — and home **solar energy systems** began exploding too. Day two killed at least **14–25 more** and injured 450–600 more, depending on the source.

Combined tolls across credible sources range from roughly 30 to 42 dead and 3,000–3,500 injured. United Nations experts

cited at least **32 killed and 3,250 maimed or injured, including 200 in critical condition and around 500 with severe eye injuries**. Lebanon's health minister reported that of about 1,800 hospital admissions, **460 required operations on eyes, faces, or limbs; at least 300 victims were blinded**. One ophthalmologist reported that of his roughly 40–50 pager patients, **every single one lost at least one eye**.

The dead included two children — a nine-year-old girl and, per Lebanese state media, an eleven-year-old boy — a hospital orderly, doctors, nurses, paramedics, teachers, charity workers, and office administrators: the civilian wing of Hezbollah's organization. Israel did not formally claim responsibility for two months; a US official confirmed Washington was briefed afterward; in November 2024 Prime Minister Netanyahu confirmed it publicly and later presented a golden pager to President Trump.

3. The Framework: This Was a Supply-Chain Attack

Every previous large-scale attack on electronic infrastructure — Stuxnet's destruction of Iranian centrifuges, the NSA's documented interception of a Cisco router bound for Syria, a car hacked on a highway for a magazine demonstration — attacked the device *in use*. This attack was different in kind: the weapon was installed *before the product existed as a product*, and the product itself was the delivery mechanism. The pager was not hacked. The pager was the bomb, from the factory (or the mid-route modification shop) onward.

That is why the reconstruction below spends as much space on Hungarian shell companies and fictional battery model numbers as on PETN: **the corporate layer was the weapons system.** The chemistry just rode along.

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4. Day One — September 17, 2024

The detonation window ran from approximately **3:30 pm to 4:30 pm**. Pagers vibrated, emitted a tone, and displayed a message — survivors report texts including "Error," "Press OK," and, in the Iranian ambassador's account, "**You have an important message**." The encrypted message required two buttons pressed simultaneously — both hands on the device — and the blast followed on the press, or on pickup, or on any button press depending on the device. Lebanese MP Tracy Chamoun reported the devices emitted a sound specifically designed to encourage users to pick them up and lift them toward their faces.

The geography was total: Beirut, the southern suburbs (Dahiyeh), the south, the Bekaa, and across the border in Syria. Emergency rooms were overwhelmed within the hour — the American University of Beirut Medical Center was treating patients on mattresses in its car park — its ophthalmology department alone performed more than 80 vision-saving surgeries in the first four days; Mount Lebanon's emergency room filled with screaming, bloodied arrivals; Nabatieh's public hospital alone treated about forty. More than 200 were in critical condition on day one. **Seventy-plus fires** burned in vehicles,

homes, and shops. Iran's ambassador Mojtaba Amani lost an eye.

A 9-year-old girl, **Fatima Abdullah**, was in the kitchen when a pager on the table began to beep; she picked it up to bring it to her father and was holding it when it exploded. Her aunt told the New York Times: "Fatima was trying to take courses in English. **She loved English**." A hospital orderly at Al Rassoul Al Azam Hospital was among the dead. The son of Hezbollah MP Ali Ammar was killed; the sons of MP Hassan Fadlallah and of Hezbollah security chief Wafic Safa were wounded.

Why then: the timing was not a calendar. CNN reported from an Israeli national-security source that the attack was launched **after Israel believed Hezbollah had discovered the capability** — an imminent-risk-of-burn trigger. The same day, hours before detonation, Israel's security cabinet formally expanded the war's aims to include the safe return of northern residents; minutes before the attack, Israel announced it had foiled a Hezbollah assassination plot against a former security chief. The Israeli source framed the operation as a "surgical" strike and posed its sharpest rhetorical question: "You have to ask why the Iranian ambassador to Lebanon was holding a Hezbollah beeper." Defense Minister Gallant told his US counterpart the diplomatic window was closing on Monday.

5. Day Two — September 18, and the Weeks After

The second wave began around 5 pm the next day, striking during funerals for day-one's dead. The devices were walkie-talkies bearing the branding of Japan's **ICOM** (the IC-V82 model), and — in a detail that revealed how far the compromise ran — **home solar energy systems** exploded in homes in Beirut and the south, wounding at least one girl.

ICOM's response became a case study in supply-chain ambiguity. The company confirmed it **halted IC-V82 production in October 2014**, discontinued its batteries, manufactures exclusively in Wakayama, Japan, never outsources, and had warned for years that "**almost all IC-V82s on the market are counterfeit**." The exploded units lacked ICOM's hologram security labels, and blast damage concentrated at the battery compartment. So the second wave likely rode either counterfeit supply chains or long-out-of-production stock modified in transit — the same interception logic as the pagers, one decade deeper.

For weeks afterward, further devices detonated — phones, laptops, fingerprint devices — and Lebanon's population operated under a single rule: *no electronic equipment is considered safe*. Hezbollah was reported to be destroying "thousands and thousands" of its remaining pagers.

6. The Casualty Record — Discrepancies Preserved

No official final accounting exists. The credible figures, side by side:

Source	Day 1 (Sept 17)	Day 2 (Sept 18)
Lebanon Health Ministry via AP (initial)	12 dead, ~2,800 injured	14 dead, 450+ injured
Lebanese state media/NBC	12 dead incl. 8-year-old girl, 11-year-old boy; ~2,700 injured, ~10% critical	20 dead, 450+ injured
UN experts, Sept 19	32 dead, 3,250 injured/maimed combined, 200 critical	—
GlobalSecurity compilation	37 dead incl. the two children; >3,000 injured; 70+ fires	—
Simple Wikipedia synthesis, Sept 22	42 dead total, incl. at least 12 civilians	30 killed, 750+ injured (health-ministry figure)

AOAV's briefing records **37 killed including two children and four healthcare workers**. The responsible statement is the range: roughly **30–42 dead, 3,000–3,500 injured**, with the lower figures earliest and the UN figure the most-cited middle estimate. A single B-grade source records 19 IRGC members killed and 150 injured in Syria's Deir ez-Zor on day one; it is flagged and not adopted into the totals.

7. The Human Record

The medical statistics had faces. **Sarah Jaffal, 21**, in the Associated Press's one-year retrospective: **45 surgeries and counting**, four fingers gone, two fused, a prosthetic eye pending — "God only burdens us with what we can bear." **Mustafa Choeib**, a preacher: blinded; his two young daughters used to play with his pager, and he would sometimes find it in their toy box. Two 12-year-old children, each having lost one eye with the other damaged. Every interviewed survivor had lost fingers; the men were blinded; shrapnel remained under their skin; their ears still rang. All expressed continued support for Hezbollah; all blamed Israel; **none had been consulted by any investigation** — theirs is the only testimony on record.

Iran's Red Crescent evacuated **95 of the wounded to Iran** and flew in two teams of doctors including eye specialists. The response detonated a parallel debate inside Iran, where citizens circulated videos noting that the Islamic Republic had **blinded hundreds of its own protesters in 2022 and denied them treatment**, while Hezbollah's wounded now received Iranian hospital beds.

An unnamed Hezbollah official told **Reuters** that **1,500 Hezbollah fighters were taken out of action** — Hezbollah's own acknowledgment of the operation's permanent cost.

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8. How the Devices Were Rigged — the Complete Engineering Picture

The most rigorous open-source reconstruction is the paper *A Closer Look at the Pager and Walkie-talkie Attacks* (arXiv, January 2025), corroborated by Reuters' forensic graphics desk. The mechanism, in plain terms:

The battery bomb. The standard rechargeable battery pack was rebuilt as a three-layer sandwich: two genuine, functional lithium-ion cells with — between them — **a sheet of PETN explosive plus a strip of highly flammable non-metallic material serving as the detonator**. The non-metallic choice was the critical tradecraft: a conventional metal detonator shows on X-ray; PETN behind two working battery cells with a plastic-strip initiator does not. The whole assembly sat in a black plastic sleeve inside a matchbox-sized metal casing, indistinguishable from a battery in use — **except it held less charge**. That reduced capacity was the operation's tell: after delivery, Hezbollah members complained the new pagers were bulky and that their batteries died quickly and heated up.

The scanner test they passed. When the devices arrived around February 2024, Hezbollah ran them through airport-grade explosive scanners. They passed — precisely because of the non-metallic detonator. Months later, the devices were fully distributed.

How much explosive — a genuine three-way discrepancy, preserved: up to ~3 grams per a Lebanese security source via Reuters ("up to three grams of explosives were hidden in the new pagers… undetected for months"); **30–60 grams (one to**

two ounces) per two US officials briefed on the operation, via the New York Times; **10–20 grams** in early BBC-cited estimates. A related variant from the same Lebanese source describes "a board inside the device that has explosive material that receives a code" — a circuit-board implant rather than purely a battery sandwich. Both architectures may be true of different batches. The honest statement: the quantity and layout are not settled across credible sources.

The trigger. The detonation signal traveled through the devices' own communication channels. For the pagers: a **specifically coded message transmitted via the commercial paging network** — the pager decoded it, sparked in the battery, fired the detonator. Unmodified pagers receiving the same signal were unaffected, proving the modification was device-specific. Experts, including Bryson Bort, assess the controller was registered within Lebanon's own paging infrastructure rather than transmitted from Israeli towers, which would have required enormous power and been traceable. The user-facing sequence — vibration, a "special beep," an urgent-looking message, a two-button confirmation requiring both hands — was engineered so the detonation would find the device at the user's face and fingers.

For the walkie-talkies: **CTCSS/DTCS tone-coded squelch** (the radio opens its receiver only on a specific tone) plus **DTMF codes** (touch-tone sequences) enabling precise, per-device remote activation.

The cover story planted in advance: multiple outlets initially reported detonation "by increasing the batteries'

temperature" — a thermal-runaway narrative that made the event look like a battery defect. The security community largely rejected it (a lithium cell cannot be made to produce PETN-like detonation force remotely), but it bought hours of confusion. A former Israeli intelligence officer explained the deeper doctrine behind the product legend: "**You want to make sure that if they look, they find something. Not finding anything is not good**" — the tradecraft of agent legends applied to consumer electronics.

9. The Supply Chain — Company by Company

The attack's true innovation was corporate. Four entities carried the devices from concept to the victims' pockets:

Gold Apollo (New Taipei City, Taiwan; ~40 employees; founded 1995). The legitimate manufacturer whose brand covered the operation. The AR-924 "rugged" pager — 85-day battery life, exactly what an organization feared phone-tracking and lived with failing electricity wanted — was the bait. Reuters' reconstruction (S-grade) established that **the AR-924 and its battery model "LI-BT783" did not exist on any market**: the product, including a fictional battery with a searchable backstory, was manufactured as legend. Chairman **Hsu Ching-kuang** was approached about three years before the attack by **Teresa Wu — a former Gold Apollo employee** — acting with her "big boss, called Tom." Hsu licensed the brand to Wu's operation, took payment from a Middle Eastern bank account that his own Taiwanese bank repeatedly blocked as "strange," and noted the exploded devices used a chip Gold

Apollo doesn't use. Whether Wu or "Tom" knew they were working with Israeli intelligence remains unresolved. **Taiwan's investigation concluded no Taiwanese company was involved**; a lawsuit by 13 Lebanese victims was dismissed by a Taiwan court in early 2026 for non-compliant filings.

BAC Consulting Kft (Budapest, Hungary; registered May 2022; €7,840 in capital; 2022 revenue $725,768; 2023 revenue $593,972). The paper contractor. CEO **Cristiana Bársony-Arcidiacono** — a UCL physics PhD who, per a former collaborator, "did no scientific work afterward" — described herself as "just an intermediary." BAC listed **118 business activities** including sugar production, retail jewelry, and natural gas extraction, operated from a duplex whose window displayed multiple company names on printer paper, and spent roughly 55 percent of its 2023 revenue on "equipment." Hungary's government confirmed the pagers were never on Hungarian soil; its security services cooperated internationally. The EU Commission said it was "looking into" her claimed affiliations with it, UNESCO, and the IAEA; the outcome never surfaced.

Norta Global Ltd (Sofia, Bulgaria; registered April 2022 — the same three-week window as BAC). Per the Hungarian investigative outlet Telex, **Norta was the real spine**: it imported the pagers from Taiwan and sold them to Hezbollah while BAC signed the paper contract with Gold Apollo. Founder-CEO **Rinson Jose** — Indian-born, **Norwegian citizen**, 37, an Oslo resident and NHST media-group employee whose Founders Nation profile described an entrepreneur seeking partners on a site connecting founders with **Israeli start-ups** —

left Oslo on a "pre-planned trip" on the day the pagers detonated and has not been heard from since. Oslo police opened a missing-person investigation; journalists found his blinds drawn and grass overgrown. Bulgaria's security agency (DANS) found **no approval for the relevant sales** and documented only a **€1.6 million transfer to Hungary**, stating it "did not detect any shipments" on Bulgarian territory — Norta was a financial pass-through. Its registered address hosts **196 companies**. A Chinese registry summary links both shells to the Mamram association of former IDF cyber commanders — B-grade sourcing, flagged, not adopted.

The interception that almost stopped it: days after the attack, Turkey's MIT intercepted **1,300 pagers and 700 chargers** at Istanbul Airport bound for Lebanon; experts found explosive and flammable liquids inside.

Mossad's own account. In a December 2024 60 Minutes interview, former Mossad officers ("Gabriel" and "Michael") described: the operation **began a decade earlier with walkie-talkies — 16,000-plus explosive radios sold over the years**, explosives in the batteries, dormant until activation; Mossad **hired the Gold Apollo saleswoman Hezbollah was already used to**; offered the first pager batch "free of charge" as an upgrade; produced fake marketing videos; and **quoted inflated prices to every non-Hezbollah customer**, ensuring only Hezbollah bought. "We make like 'Truman Show' — everything is controlled by us behind the scene. Everything was 100% kosher — businessman, marketing, engineers, showroom, everything." (An Israeli outlet dubbed the operation "Khaybar"

and noted the pager's own inventor, Al Gross, had conceived the device after WWII as an act of revenge for a missed date — social-media-grade color, flagged, not part of the factual record.)

Order numbers, reconciled: Sky News and Mossad's account say **5,000 pagers**; a Hezbollah official told AP the order was **15,000, of which only 8,000 arrived**, nearly half distributed. Combined with 16,000 radios, the network covered the organization.

10. The Planning Timeline

- **~2014–2018:** per intelligence correspondent Ronen Bergman, the concept originated with a Mossad officer — a woman whose name remains confidential — proposing under Mossad chief Yossi Cohen that Israel follow Hezbollah into the low-tech communications realm it was moving toward. Syntheses describe a ten-year horizon.
- **February 2024:** after losing ~170 fighters to Israeli intelligence penetration, Hezbollah draws up a war plan to address communications security; on February 13 Nasrallah orders members to "break, bury or lock in an iron box" their cellphones and go low-tech. Israel, anticipating exactly this, has BAC and Norta registered within weeks of each other since spring 2022.
- **~February 2024:** rigged devices arrive; Hezbollah scanner-tests them; they pass; complaints about bulk and battery heat are logged and ignored; distribution proceeds — to fighters, medics, and administrators across the organization's branches.

- **September 16, 2024:** a test signal activates **3,500 devices**. Bergman describes Lebanon's first confused reports as "the Israeli Air Force is bombing Lebanon in thousands of places."
- **September 17, 3:30 pm:** detonation — pulled early, per CNN, because Israel believed the capability had been discovered.

11. Who Actually Held the Devices — the Evidence the Law Turns On

The legal question — were these military targets? — has a documented answer from the target's own side:

- **Nasrallah (Sept 19):** the exploded pagers went to **lower-ranking members**; leaders did not use that model.
- **Lebanon's Health Minister:** most ER patients were **in civilian clothing**; affiliation unclear; casualties included the elderly and young children.
- **Foreign Minister Bou Habib (Sept 26):** most carriers were **not fighters — civilians such as administrators**.
- **Hezbollah expert Qassim Qassir:** the attacks mostly struck **civilian workers**; the military wing was largely unaffected.
- The concrete dead: a hospital orderly, teachers, medics, charity workers — and a 9-year-old girl holding a device that was not hers.
- Hezbollah's own internal investigation concluded the penetration resulted from **negligence** and cleared its officials of collaboration.

Reading professor Marko Milanovic's legal synthesis: given that some pagers were issued mere **hours** before detonation, it

was **almost impossible for Israel to have known** whether they were in military or civilian hands — making the attacks, in his assessment, most likely **indiscriminate** under the laws of war.

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12. The Deep History — Twenty-Eight Years of the Same Idea

The WWII floor. The US Defense Department's own Law of War Manual illustrates the prohibition this attack would later be tested against with a Second World War case: Italian **booby-trapped communications headsets** — equipment a soldier puts on expecting function, engineered to kill. The parallel to a pager that begs to be picked up is nearly verbatim.

Operation Mongoose, 1960–65 — the CIA's disguised-object laboratory. After the Church Committee, eight confirmed assassination plots against Fidel Castro stand on the record (Cuban intelligence chief Fabián Escalante later claimed 634–638; the verified number is eight). Declassified files describe a special task force — with a stage-magician consultant — that developed: a **C-4 exploding cigar**, tested as reliable, "enough to remove the target's head," never delivered; an **exploding conch shell** ($11,000 developed and tested, abandoned over diving-location uncertainty); a **wetsuit contaminated with Madura-foot fungus**; a **hypodermic poison pen** dispatched to Havana, never used; **botulinum-toxin cigars** so potent that one in the mouth would kill (dispatched August 16, 1960 through an intermediary, never

chewed); and the **1963 milkshake plot** — the closest call, foiled when the poison capsule froze and ruptured in the Havana Libre's freezer. The Mafia (Giancana, Roselli, Trafficante) served as CIA proxies for several plots. After the 1975 Church exposure, President Ford's executive order banned assassinating foreign leaders. The methods were prank-shop novelties — exploding gag cigars were standard jokes; Ulysses S. Grant allegedly gifted one, Hemingway pranked a general's bodyguards — engineered into weapons. September 2024 sat at the end of that sixty-year arc.

Yahya Ayyash, January 5, 1996 — Israel's founding case. Hamas's chief bombmaker, "the Engineer," was killed by the Shin Bet with a rigged Motorola phone, in an operation whose details matter because Hezbollah's turn to pagers was itself a reaction to it:

- The technology unit packed explosive into the phone's battery compartment (**sources disagree on the charge: 11 grams per Israeli security accounts, 15 grams per standard references, up to 50 grams per biographer Samuel Katz — preserved as a discrepancy**). The **original plan was a rigged fax machine**; the phone was chosen for size.
- Delivery ran through **Kamil Hamad**, uncle of Ayyash's host, who unwittingly passed the phone to his nephew; Kamil was reportedly paid $1 million, a fake passport, and a US visa, then vanished.
- **The first attempt failed** — a wire had disconnected. The device was recalled through the collaborator, diagnosed, repaired, and returned. On Friday, January 5, 1996, Ayyash

used it to call his father; an Israeli aircraft overhead confirmed his voice; the charge fired with such precision that the **left side of his face and the hand not holding the phone were untouched** — the surviving right hand became a Palestinian symbol.

- Retaliation came within weeks: **four suicide bombings killed 78 Israelis in February–March 1996**. Shin Bet director Carmi Gillon confirmed responsibility in the 2012 documentary *The Gatekeepers*. Hamas named its Ayyash-250 rocket after him.

The 2000 coda: per Bruce Schneier's New York Times essay, Israel killed **a Fatah activist in 2000 by the same method — a remotely detonated booby-trapped cellphone**. This case is thinner documented than Ayyash and is flagged as such.

September 2024: the same idea, industrialized — one device in 1996, thousands in 2024, a twenty-eight-year production learning curve from artisanal to mass-market.

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13. The Legal War — Both Sides, Fully Stated

The legal center of gravity is **Article 7(2) of Amended Protocol II of the Convention on Certain Conventional Weapons**, binding on both Israel and Lebanon (Israel submitted no reservation): "*It is prohibited to use booby-traps or other devices in the form of apparently harmless portable objects which are specifically designed and constructed to contain explosive material.*"

The case for violation: William Boothby (Lieber Institute) provisionally concluded the devices fall within Article 7(2) and are prohibited. Brian Finucane (Just Security, ex-State Department legal adviser): there is "**a strong basis to conclude Israel violated Article 7(2)**" — noting the batteries were mass-produced for thousands of devices over years, "the opposite of field-improvised." Human Rights Watch called the means indiscriminate. Article 36's Richard Moyes: "a direct breach." Notre Dame's Mary Ellen O'Connell: weaponizing a civilian-object is "strictly prohibited." UCLA's Jessica Peake: "quite blatant" violations of proportionality and indiscrimination. At the UN Security Council — meeting at Algeria's request — South Korea cited Article 7(2); rights chief Volker Türk demanded an independent investigation. UN human rights experts (September 19) called the attacks "**terrifying violations of international law**" that "violate the human right to life, absent any indication that the victims posed an imminent lethal threat," requiring investigation for "**the crime of murder**." Amnesty International and AOAV (Action on Armed Violence) demanded international investigations; AOAV's warning: without accountability, "the precedent set by these attacks could lead to more widespread use of such tactics, endangering civilian lives worldwide."

The case for legality, at full strength: John Spencer (West Point's Modern War Institute): the operation "**passes all fundamental laws of war — necessity, proportionality, and distinction**." Boothby himself conceded planners could reasonably assume military-issued pagers would be with military

users. Professors Cohen and Shany doubted Article 7(2) covers remotely detonated devices planted on military objects; scholar Michael Sari argues the treaty's drafting deliberately excludes "adapted" devices; a Swedish military-law thesis concluded the pagers are best classed as "other devices," a likely **lawful means of warfare** whose legality turns on targeting facts that remain unknown. A complicating fact for the prosecution case: the **United States' own formal understanding** of Article 7(2) permits "adaptation in advance" of objects as booby-traps — an interpretation Israel never needed, having filed no reservation.

The blinding dimension: with at least 300 blinded and one ophthalmologist reporting every one of his 40–50 pager patients losing at least one eye, analysts additionally raised the prohibition on weapons causing superfluous injury — the same principle behind the ban on blinding lasers (Protocol IV) — noting the injuries were not incidental to the blast but its design intent.

Status at compilation: no party claimed responsibility on the record at the time; Israel confirmed responsibility in November 2024; no transparent investigation has occurred; no one has been charged.

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14. The Political Record — Washington and the World

The White House — distance without condemnation: Press secretary John Kirby, repeatedly: "**We were not involved in yesterday's incidents or today's in any way**" — while refusing to confirm Israeli responsibility ("I'm not going to get into intelligence assessments") and declining to say whether detonating wireless devices is an acceptable form of warfare. Blinken stated categorically the US had no prior knowledge. The record is genuinely two-sided and preserved as such: CNN reported (three sources) that Israel notified Washington an operation would occur Tuesday — without details — including in the Austin–Gallant call; Israel's KAN broadcaster reported **two Gallant–Austin calls, one "minutes before the first wave,"** and asserted coordination; Washington denied both foreknowledge and involvement. Kirby's later Sunday framing: support for Israel continues "**but how they do it matters**… as precise and as discriminant as possible." White House press secretary Karine Jean-Pierre said she was "afraid and concerned" about regional escalation.

Congress: Representative Ocasio-Cortez — "this attack **clearly and unequivocally violates international humanitarian law**" — demanding a full accounting "including an answer from the State Department as to whether any US assistance went into the development or deployment of this technology"; her office was subsequently vandalized with blood-splattered signs. Senator Sanders filed joint resolutions of disapproval against US arms sales to Israel; Senator Fetterman

publicly gloated support.

The world: UN Secretary-General Guterres was "deeply alarmed"; Lebanon's Prime Minister Mikati called it "a collective crime that defies humanity"; Foreign Minister Bou Habib warned of "a dangerous precedent… signal[ing] a wider war." Iran's ambassador declared Tehran "reserves the right to respond," and the **IRGC ordered its personnel to stop carrying pagers and other communications devices** — while, by late 2025, analysts noted no direct Iranian retaliation for the pagers had materialized, leaving the operation, in the deterrence ledger, operationally unanswered. Russia and China condemned the attack at the Security Council.

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15. Attribution — the Trail of Confirmation

Israel maintained formal silence for two months. A US official confirmed Israel **briefed Washington after the fact**; Defense Minister Gallant declared "we are at the start of a new phase in the war" the day of the second wave; IDF Chief of Staff Herzi Halevi stated that plans had been drawn up for additional action; Axios reported the final go-decision was taken by **Netanyahu, his top ministers, and the heads of the IDF and intelligence agencies** after indications Hezbollah had grown suspicious. In **November 2024 Netanyahu confirmed Israeli responsibility** publicly. He later presented President Trump with a **golden pager** — the attack elevated from covert operation to state trophy. Mossad veterans' on-record accounts (60 Minutes, December 2024) completed the confirmation from the inside.

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16. The Cascade — What the Attack Set Off

The pagers were the opening strike of the campaign that broke Hezbollah. **September 19:** Nasrallah's "declaration of war" speech, claiming Israel had rigged 4,000 pagers "with the clear intent of killing 4,000 people in a single minute" — the death toll limited, he said, because many devices were undistributed or out of service — and announcing "investigative committees" tracing the chain "**from the factory… to how they ended up in the hands of the users**": the victim conducting the same audit this document conducts. **September 22 onward:** Hezbollah rocket escalation. **September 27:** Nasrallah assassinated. **October 1:** Israeli ground invasion. **November 27, 2024:** ceasefire. **June 2025:** Israel struck Iran's nuclear program and command. The pager attack thus opened a sequence that ended, within nine months, with the decapitation of Hezbollah and a direct Israel–Iran war.

The 2026 benchmark. The operation has become Israel's internal unit of measurement. In April 2026, after an IDF strike killed 250 Hezbollah operatives across three areas in sixty seconds, spokesman Lt. Col. Nadav Shoshani assessed it against the original: "**The beeper had more… effective injuries. That was the purpose of it**" — adding that both operations shared the deeper goal of "creating disarray, breaking their chain of command… tilting the organization out of balance." A retired Israeli general's verdict on the April strike: "far more devastating than the pager attack." The beeper is now the benchmark.

The wounded as permanent infrastructure. The ~1,500 fighters Hezbollah acknowledged taking out of action — plus the

civilian-wing dead — are a lifelong care burden: blinded men, fingerless hands, shrapnel under skin, prosthetic eyes awaiting fitting. The attack's costs compound monthly, in clinics, not headlines.

17. The Expert Verdict — Schneier's Anatomy

The definitive technical assessment is Bruce Schneier's New York Times essay (September 22, 2024). His argument is an evidence argument, matching this document's method: **none of the components was new; what was new was the combination, scale, and publicity.**

Component	Prior precedent
PETN implanted in a carried object	Shoe-bomber Richard Reid, 2001 — "that's what all those airport scanners are designed to detect"
Assassination by personal device	Ayyash, 1996, and a Fatah activist, 2000 — both booby-trapped cellphones
Supply-chain interdiction	NSA's documented interception and modification of a Cisco router bound for Syria (Snowden archive); Stuxnet, 2010; Bloomberg chip allegations, 2018; Huawei bans
Front companies selling rigged devices	The FBI's 2019 fake secure-phone company (ANOM) — eavesdropping, not killing
Remote hardware destruction	Idaho National Lab's 2007 generator explosion; CIA Vault 7 car-hacking files; the 2015 live hijacking of a Wired reporter's car

His conclusions: "Our international supply chains for computerized equipment leave us vulnerable. **And we have no good means to defend ourselves.**" A Hungary-based company sourcing from Taiwan raised no flags "because that sort of thing is perfectly normal"; changing that "would require bringing global economies back to the 1980s." And the forward warning:

"now that the line has been crossed, other countries will almost certainly start to consider this sort of tactic as within bounds… developed countries like the United States will be especially vulnerable, simply because of the sheer number of vulnerable devices we have."

The counterpoint inside the same community: several experts, including Brookings' Michael O'Hanlon, questioned not the legality but the wisdom — "most of these people are just sort of run-of-the-mill Hezbollah operatives… I am not sure it's wise" — and noted Hezbollah's likely response: never trusting networked equipment again, which may have been a primary goal in itself.

18. What This Means for the Zero-Ties Investigations

This document was produced within a series of supply-chain audits — *The Zero-Ties Laptop*, *The Zero-Ties Phone*, *The Zero-Ties App* — that asked whether any consumer device can be purchased with zero ties to Israel. Those investigations found that end-to-end clean Western devices **do not exist**: Israeli engineering sits inside every flagship chip architecture, every Western modem, the chip-design tools themselves, the billing software of nearly every carrier, and the advertising SDKs inside half of all apps.

The Lebanon operation demonstrated what that finding means at its absolute extreme. Every mechanism documented in those reports — brand licensing as an authorship-laundering vector (Gold Apollo's name over BAC's contract over Norta's money),

shell-company supply chains, the invisibility of true authorship, scanner-evasion engineering — appeared there as *corporate entanglement*. In September 2024 the same machinery functioned as a **weapons-delivery system**. The audit framework and the attack are the same map read at different altitudes: the authorship layer is not merely a boycott ledger. In Lebanon it was a kill chain — one that passed through a Budapest duplex, a Norwegian entrepreneur's shell, a Taiwanese brand's good name, and a nine-year-old girl's kitchen.

AOAV's closing warning belongs to the whole series: without accountability and stronger legal protections, "the precedent set by these attacks could lead to more widespread use of such tactics, endangering civilian lives worldwide."

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19. Methodology and Evidence Grading

How this document was built. Five sequential research passes: the event and casualties; the engineering forensics; the corporate supply chain; the legal and political record; the expert consensus and deep history. Primary sources opened in full include the arXiv forensic reconstruction (January 2025), Reuters' graphics investigation, Mossad veterans' 60 Minutes account, Ronen Bergman's reporting, Schneier's NYT essay, UN expert statements, the Just Security and Lieber Institute legal analyses, and AP's one-year survivor reporting.

Evidence grades: S-grade (forensic/academic and wire-service investigations — the arXiv paper, Reuters' reconstruction); **A-grade** (major-outlet reporting with named

sources — AP, NYT, BBC, CNN); **on-record** (named or pseudonymous first-person accounts — Mossad veterans); **official** (governments, UN, courts); **B-grade** (single-source or registry-derived claims — the Mamram link, the Deir ez-Zor figures, the "Khaybar" operation name); **flagged** (recorded but not adopted into the factual spine).

The house rules, applied throughout: no theory adopted from any party — Israeli, Hezbollah, American, or academic — without independent structure; every discrepancy preserved (PETN quantity, casualty ranges, the 2000 Fatah case); every gap named (see Part 20). First-person quotations are reproduced only from published interviews and official statements.

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20. What Remains Unresolved — the Bedrock List

1. **The physical assembly site** of the rigged battery packs/boards (Taiwan components → shells → Lebanon): unconfirmed.
2. **"Tom" and Teresa Wu's knowledge** of the operation: unresolved by Reuters.
3. **Rinson Jose's whereabouts:** missing since September 17, 2024; the Oslo police case remains the only open official investigation into the corporate layer.
4. **Explosive quantity:** 3g (Reuters/Lebanese source) vs 30–60g (NYT/US officials) vs 10–20g (early BBC): preserved, unresolved.
5. **The Mamram-association link** to the shell companies: B-grade, unverified.

6. **The "Operation Khaybar"** name: social-media grade, unverified.
7. **The EU Commission check** on Bársony-Arcidiacono's claimed affiliations: outcome never surfaced.
8. **Israel's internal legal review** of the operation: no document has surfaced.
9. **Hezbollah's investigative-committee report:** announced, never published.
10. **The 2000 Fatah activist case** (Schneier-attested): thinner documentation than Ayyash, flagged.
11. **The final casualty figure:** no official final accounting exists; the ranges in Part 6 stand.

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21. Key Figures at a Glance

Figure	Value	Context
Day-one toll	12 dead, ~2,700–2,800 injured	Incl. 2 children
Combined toll (ranges)	30–42 dead; 3,000–3,500 injured	No final official accounting
Blinded	≥300	One ophthalmologist: every patient lost ≥1 eye
Surgical operations	460	Eyes, face, limbs (of ~1,800 admissions)
Hezbollah fighters out of action	~1,500	Hezbollah official via Reuters
PETN per device	3g / 30–60g / 10–20g	Three credible-source estimates, preserved
Walkie-talkies sold	16,000+	Explosives in batteries, per Mossad veterans
Pagers ordered	5,000 (Sky/Mossad) / 15,000→8,000 arrived (AP)	Turkey later intercepted 1,300 + 700 chargers
Norta→Hungary transfer	€1.6M	Only documented transaction per Bulgaria's DANS
BAC capital	€7,840	118 listed business activities
Gold Apollo brand license	3 years before the attack	Payment from blocked Middle East account
Ayyash precedent charge	11g / 15g / 50g RDX	Sources disagree, preserved
Church Committee plots	8 confirmed (Escalante: 634–638)	Operation Mongoose, 1960–65
UN experts' verdict	"Terrifying violations of international law"	Investigation for "the crime of murder" demanded
Israeli confirmation	November 2024	Netanyahu; golden pager gifted to Trump
April 2026 benchmark	250 killed in 60 seconds	IDF: "The beeper had more effective injuries"

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22. Glossary

- **PETN:** pentaerythritol tetranitrate — a stable high explosive; the shoe-bomber's material; hidden in the pager battery sandwiches.
- **Detonator / initiator:** the small component that sets off the main charge; making it non-metallic defeated standard scanners.
- **CTCSS/DTCS:** continuous-tone-coded squelch — the radio only "opens" on a specific tone; combined with DTMF (touch-tone) codes, it enabled per-device remote triggering of the walkie-talkies.
- **POCSAG:** the global paging protocol the detonation signal exploited.
- **Booby-trap / "other devices":** treaty terms: hidden weapons triggered by an unsuspecting person; "other devices" are manually/remote-triggered variants.
- **CCW / Amended Protocol II:** the 1980 Certain Conventional Weapons convention; its landmine protocol's Article 7(2) bans booby-traps disguised as harmless portable objects — the legal center of the dispute.
- **Distinction / proportionality / necessity:** the laws-of-war triad — attack only military targets; civilian harm not excessive to military gain; force only where needed.
- **Superfluous injury:** the rule against weapons designed to cause unnecessary suffering (the basis of the blinding-weapons ban, invoked for the eye injuries).
- **Shell company / front:** a corporate vehicle with no real operations, used to move money or goods while hiding their

true controller — BAC, Norta.

- **Legend:** an intelligence term for the fabricated identity/backstory given to an asset or, here, a product ("if they look, they find something").
- **Entity List:** the US Commerce blacklist — invoked here only by analogy; no company in the pager chain was listed.
- **AOAV:** Action on Armed Violence, the UK charity whose casualty briefing is the NGO record of the attack.

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23. The Conclusion, in One Paragraph

On September 17, 2024, the supply chain — the invisible global machinery that decides what is inside the objects we hold — killed thirty to forty people in an hour, blinded three hundred, and maimed thousands, by delivering them pagers and radios that had been bombs since before they were products. This document has reconstructed that event completely as open sources allow: the battery sandwiches and their fictional model numbers, the Hungarian duplex and the missing Norwegian and the Taiwanese chairman who never made the device, the decade of planning and the twenty-eight-year lineage from a rigged Motorola in 1996 to thousands of rigged pagers in 2024, the legal argument still unresolved over whether Article 7(2) covers a weaponized consumer product (with the US Defense Department's own manual listing booby-trapped headsets among the prohibited), the White House's distance without condemnation, the UN's demand for a murder investigation no one has begun, and the 2026 doctrine that measures new strikes against "the beeper." Two judgments survive every pass. Schneier's: this capability cannot be defended against, only deterred, and the line is now crossed. And the narrower, harder one that belongs to this series of documents: when the holder of a device cannot know who authored it, the device is not a product but a standing risk — and in a world where the authorship layer has already been a kill chain once, asking who made what is no longer paranoia. It is the minimum.

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Compiled 2026-09-20. Five research passes; S/A/B-grade sources separated throughout; all casualty ranges and quantity discrepancies preserved; all open questions named. First-person quotations from published interviews and official statements only.

Part Two: An Open Letter to Israel

September 20, 2026

To the People of Israel

Re: The attacks of September 17–18, 2024; accountability for the weaponization of civilian supply chains; and a question put to a nation

To the people of Israel,

This letter comes from an independent researcher, following a fully documented investigation into the events of September 17–18, 2024 — the detonation of thousands of pagers, walkie-talkies, and home solar-energy systems across Lebanon and parts of Syria — and into the record of which those events form a part. The findings are public, compiled from forensic analyses, corporate registries, court-adjacent records, first-person accounts, and the statements of Israeli officials themselves. This is not addressed to a ministry, because the questions that remain belong to more than a ministry. They belong to a nation — its conscience, its courts, its press, its classrooms, and its standing in the community of peoples. So this letter is addressed to all of you.

What the record shows. On September 17, 2024, between approximately 3:30 and 4:30 in the afternoon, thousands of pagers carried by members and affiliates of Hezbollah detonated within a single hour across Lebanon and parts of Syria. At least twelve people were killed on that day — including two children — and roughly three thousand were injured, around five hundred of them with severe eye injuries; at least three hundred were left

blinded. On September 18, walkie-talkies exploded, some during funerals for the previous day's dead, and home solar-energy systems began exploding in homes. Credible tallies place the combined dead between thirty and forty-two. Among the dead were a nine-year-old girl holding a device that was not hers, a hospital orderly, doctors, nurses, teachers, and charity workers. Responsibility was confirmed by your own Prime Minister in November 2024. A golden pager was later presented to the President of the United States as a trophy.

The supply chain your services built. The operation's true innovation was corporate. A Taiwanese brand's good name — its chairman approached three years earlier by a former employee bearing an unseen "big boss" — licensed to a Hungarian intermediary with 118 listed business activities and under eight thousand euros in capital, financed through a Bulgarian pass-through whose Norwegian-Indian founder left Oslo on the day of the detonations and has not been seen since. Devices were scanner-tested by the buyers and passed; their reduced battery capacity was the operation's only tell, and it was ignored. On the recorded account of your own former intelligence officers, the first batch was offered free as an "upgrade," marketing videos were fabricated, and non-Hezbollah buyers were quoted prices high enough to refuse. "We make like 'Truman Show,'" one of them said — everything controlled behind the scene.

Why this belongs to the nation and not only to its government. Three questions remain open, and each of them exceeds any single administration.

The first is legal. Article 7(2) of Amended Protocol II to the Convention on Certain Conventional Weapons — a treaty Israel ratified without reservation — prohibits booby-traps and other devices in the form of apparently harmless portable objects specifically designed to contain explosive material. The devices at issue were consumer objects modified in the supply chain before reaching their users. Serious analysts conclude the provision was violated; serious analysts conclude it was not; the United States' own formal interpretation permits "adaptation in advance." What no one can honestly claim is that the question was ever transparently examined. United Nations human rights experts called the attacks "terrifying violations of international law" and urged investigation into what they termed "the crime of murder." No investigation — Israeli or international — has occurred. Nations that pride themselves on the rule of law do not leave such questions in the hands of their adversaries' tribunals.

The second is factual. No public record establishes where the devices' battery assemblies were physically built, who within your institutions reviewed the operation's legality, or what process assessed the foreseeable presence of children, patients, and bystanders in the blast radius of a weapon delivered by a beeping pager. A nation's intelligence services act in its name; their secrets are also its inheritance.

The third is normative, and it is the largest. The method demonstrated on September 17, 2024 is now part of humanity's permanent public record: the supply chain itself — the trust between a buyer and a product — can be converted into a delivery system for explosives, at scale, against people who

cannot see the weapon they are holding. The leading security analysts of our time state that there is no technical defense against this tactic, only law and deterrence. And your own military conceded the principle in November 2025, when it excluded Chinese connected vehicles from its tenders on precisely these supply-chain grounds. A precedent that endangers every nation's civilians — yours included, perhaps yours most of all, given how much of the world's technology passes through your own engineers' hands — cannot responsibly be left without an authoritative legal and moral accounting. The first author of a tactic inherits its reputation forever.

What the world felt — and what it is doing about it.

September 17 was watched live around the world. People saw, in the hours that followed, what it means when the ordinary objects of daily life — the pager on a belt, the radio in a hand, the solar panel on a roof — detonate without warning. The footage and the testimony did what no argument could: they transferred suspicion onto every device, everywhere. A parent now looks at a phone on a kitchen table differently. A hospital administrator looks at a networked monitor differently. A driver looks at a dashboard that phones home. This is not speculation; it is the documented behavioral consequence of the precedent your services established, and it will not fade, because the underlying fact does not fade: nobody can verify, by looking, what their device is.

The commercial consequence follows the sentiment. Purchasers are asking a question that did not exist at scale before September 2024: not "where was this made" but "who

authored it, who owns the company that built it, and what else have they built." They are choosing local-first over cloud-dependent, offline over connected, the shorter wire over the shinier one. This response attaches to deeds, not to origin or faith — and any attempt to dismiss it as prejudice misunderstands it completely. It attaches to the documented, admitted, and unaccounted commission of an act that turned the world's supply chains into a delivery system for explosives: an act committed without regard for the nine-year-old in the kitchen, the orderly in the corridor, or the mourner at the funeral. Where identity enters at all, it enters only as the maker's mark on the deed.

And it attaches, further, to the wider campaign of destruction of which that act formed a part. Until those responsible are completely revealed and brought to justice, devices carrying any affiliation to the apparatus that built this weapon will be assessed as standing threats — not from prejudice, but from precedent: the identical precedent your own military invoked when it excluded Chinese connected vehicles from its tenders on supply-chain grounds. People around the world who buy, carry, or promote the products of Israeli companies increasingly feel made complicit in a genocide unfolding — and they want no part in aiding aggression, past or present. They will exercise the one power that is fully theirs — what they purchase, what they carry, and what they recommend — to ensure they are not part of it. A nation that has asked the world for decades to separate its people from the deeds of those acting in their name cannot now refuse the world the same separation, in reverse, when the deeds

are its own.

What is asked of Israel. Not of a minister — of the nation, in the tradition of self-examination that produced the Knesset's hardest hearings, your Supreme Court's unsparing reviews, your press's own investigations, and a documentary in which your own former security chiefs confessed their service's sins on camera:

That Israel publish, or transmit to an appropriate international body, the legal review — if any — that preceded the operation.

That Israel consent to, or commission, an independent investigation into the attacks, addressing Article 7(2) and the questions of distinction and proportionality that the death of children in kitchens necessarily raises.

That Israel's institutions — judicial, legislative, journalistic — examine the doctrine the operation established: whether explosive devices concealed in civilian supply chains are to be Israeli policy, and if so, under what limits the nation is prepared to defend them.

That the nation remember the names it can still learn: Fatima, nine, who loved English; the orderly of Al Rassoul Al Azam; the blinded and the fingerless who now live on both sides of the line this operation drew. Their care, like the precedent, is permanent.

This is not written expecting agreement. It is written because the record — your officers' own words, your government's own confirmation, the public wreckage of the supply chain your services built — deserves an answer worthy of the country that produced a High Court willing to restrain its own executive, and

a press willing to print what the censor disliked. That country can examine this. The question is whether it will.

Respectfully,

— Independent Researcher

Enclosures: *The Beeper War: The Complete Reconstruction of the September 2024 Lebanon Device Attacks*; *The Zero-Ties Laptop*; *The Zero-Ties Phone*; *The Zero-Ties App*; *The Zero-Ties Router*; *The Zero-Ties Car*; *The Zero-Ties Implant*; *Why This Was Researched and Written As It Was*.

Part Three: The Zero-Ties Laptop: A Complete Audit of Israeli Entanglement in the Personal Computer Supply Chain

A fully-sourced investigation into whether a laptop with zero ties to Israel — in any way, foreign or domestic — can exist, and what the closest achievable alternatives are.

1. The Question

The question sounds simple: **is there a laptop on the market with zero ties to Israel — no Israeli-designed components, no Israeli corporate involvement, no Israeli software, nothing?**

It turns out to be one of the hardest "simple" questions in consumer technology, because the answer is not found in any single part, brand, or factory. The answer is buried across five different structural layers of the global electronics industry, most of which are invisible to the buyer — and invisible to most of the industry itself.

This document audits every layer, names the specific companies, dollar figures, acquisitions, contracts, executives, and engineering centers involved, and then builds — piece by piece — the closest thing to a clean laptop that can actually be purchased, along with the only literal zero-tie paths that exist.

How to read this document

- **Part 2** gives the short answer.
- **Part 3** explains the five-layer framework — the mental model for everything that follows.

- **Parts 4–12** are the layer-by-layer audit (the evidence): CPU, GPU, memory and storage, power, connectivity and sensors, software, the PC brands, the chip-design tooling, and manufacturing.
- **Part 13** is the practical result: the minimum-ties build.
- **Part 14** covers the literal zero-tie alternatives.
- **Part 15** examines the people layer — the one thing no purchase decision can reach.
- **Part 16** explains how we know what we know (methodology and evidence grading).
- **Part 17** lists what remains unresolved.
- **Part 18** is a key-figures table for quick reference.
- **Part 19** is a glossary of every technical term used.
- **Part 20** is the conclusion, in one paragraph.

A note on framing before we begin: this document does not argue for or against any political position. The user's requirement is a supply-chain criterion — zero ties to Israel — and this document measures the market against that criterion. Where Israeli involvement is found, it is documented with sources. Where it is *not* found, that is documented too, with the same care. Where evidence is ambiguous, the ambiguity is preserved rather than resolved by opinion.

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A thirty-second orientation for readers new to computer hardware

A laptop is a stack of roughly a dozen independently made technologies that happen to snap together. There is one main chip — the **CPU**, the computer's brain — plus memory (short-term workspace), storage (long-term files), a Wi-Fi radio, a webcam, power-management chips, firmware (software etched into the hardware itself), an operating system, and finally the brand-name shell. Each of those pieces is typically **designed in one country, manufactured in another, and assembled in a third**, often by companies the buyer has never heard of. Two facts from that sentence drive this entire document: first, "designed in" matters as much as "made in," because design authorship is invisible in the finished product yet embedded in everything it does; and second, the brand on the lid is often the *least* informative thing about the machine. Keep those two ideas in mind and every section below will read clearly.

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2. The Short Answer

No commercially available laptop has zero ties to Israel. Not one. The reason is structural, not incidental:

1. **Every viable laptop CPU architecture has Israeli engineering at its core.** Intel's modern processor lineage — the chips inside the majority of the world's PCs — was designed in Israel, from the 8088 in 1979 to the Core Series 4 shown at CES 2026. Apple's M-series chips (M1 Pro, M1 Max, and successors) were physically designed and validated in

Israel. The ARM architecture underlying nearly all remaining chips has Israeli-developed components, including its flagship GPU. The only mainstream exception is AMD's Zen architecture — and AMD the *corporation* still maintains documented Israeli ties, even though its silicon does not appear to.

2. **The tools used to design chips are also entangled.** The two companies whose software designs virtually every commercial chip on Earth — Synopsys and Cadence — both operate Israeli development centers. Even a hypothetical "clean" chip is authored with Israeli-engineered tooling.
3. **The major PC brands are entangled at the corporate level.** HP has supplied the Israeli military with PCs since 2009. Dell won a two-year Israeli Ministry of Defense server tender worth over $150 million in January 2023. Lenovo completed a $500M+ acquisition of an Israeli storage company in April 2026, embeds Israeli-founded security software in new PCs, and now owns one of the major BIOS firmware vendors (Phoenix Technologies, acquired April 2026).
4. **The dominant operating system has the deepest verified entanglement of all.** Microsoft's relationship with Israeli military intelligence is documented at audit level: Unit 8200 — Israel's signals-intelligence agency — operated a customized Azure cloud environment storing and analyzing intercepted Palestinian phone calls, a relationship that followed a 2021 meeting between Microsoft's CEO and Unit 8200's commander. Microsoft's own follow-up review in

September 2025 found evidence supporting elements of that reporting, and it disabled specified Israeli Ministry of Defense services.

5. **The people layer cannot be excluded by any purchase decision.** An estimated ~1,400 veterans of Israeli intelligence units work across US technology companies (per MintPress/Drop Site News research), concentrated in exactly the companies that design laptop silicon and software. Talent is not a component, and it cannot be specified out of a bill of materials.

The closest achievable mainstream machine — detailed in Part 10 — is an ASUS or Acer laptop with an AMD Ryzen CPU, Realtek Wi-Fi, SK Hynix memory, a Kioxia-based SSD, an OmniVision webcam on the USB path, and Linux running Firefox. Every one of those choices is individually defensible as clean or nearly clean. The residual ties (AMD's corporate-level Israeli presence, the EDA tooling authorship, partial Linux sponsorship taint) are irreducible.

The only literal zero-tie paths — detailed in Part 11 — are Chinese domestic Loongson/LoongArch machines and immature RISC-V machines, each with major practical tradeoffs.

Even the global boycott movement's own materials implicitly concede the point: no BDS-aligned guide claims a certified "Israel-free" laptop exists. Ethical Consumer's 2024 laptop guide recommends Framework and TCO-certified ASUS — recommendations based on *corporate conduct*, while the silicon problem underneath remains acknowledged but unresolved.

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The Plain-Language Version (Start Here If Chip Terminology Is New to You)

Everything in this document can be understood with one analogy: **a laptop chip is like a building.**

The factory that pours the concrete is only the last step. Before that, an architect draws the plans — the floor-by-floor layout, the wiring, the load-bearing walls. In chip terms, that drawing stage is called **design**, and the drawing itself is called a **tape-out** (because the finished design used to be sent to the factory on reels of magnetic tape). A chip can be designed in one country, manufactured in a second, assembled into a laptop in a third, and sold by a brand in a fourth. Each stage can have a different nationality.

This matters because of how the industry has evolved. Most people assume a product "comes from" wherever it is physically made. But semiconductors inverted that logic decades ago: the *design* is where the value and the authorship live. A processor designed in Haifa, Israel, and manufactured in Taiwan is Israeli-designed and Taiwanese-made — and the Israeli contribution travels inside every one of the millions of copies made, because it is baked into the plan itself. You cannot unscrew it, the way you could unscrew a foreign-made part.

Now add one more twist: even the architect doesn't draw by hand. Modern chips contain tens of billions of transistors — switches far too numerous for any human to place individually. Architects use specialized software to draw them, called **EDA** (Electronic Design Automation). There are effectively only two companies that make this software: Synopsys and Cadence. And

both of those companies employ engineers in Israel. So even a chip with no Israeli designers on the team was drawn with tools containing Israeli engineering — the drafting software has its own nationality problem.

Then there is the hidden computer inside your computer. Every laptop ships with firmware — software frozen into the hardware itself. Intel builds one called the **Management Engine** (ME), AMD one called the **Platform Security Processor** (PSP). These are tiny computers that run *below* your operating system, with access to everything — screen, keyboard, network — and that the owner cannot inspect or disable. This document doesn't claim they contain Israeli code; it claims something more basic: every Intel Management Engine runs on Intel silicon, and Intel's silicon is Israeli-designed. The hidden layer and the design layer meet inside the same chip.

Finally, the cloud. When this document says Israeli military intelligence used Microsoft's Azure, here is what that means in everyday terms: Microsoft rents out computing power the way a landlord rents offices, over the internet, in enormous warehouses of computers called data centers. An Israeli intelligence unit (Unit 8200 — think of it as Israel's equivalent of the NSA's most elite department, staffed largely by soldiers recruited straight out of high school) reportedly rented a private, walled-off section of that computing power and stored intercepted phone calls in it. Journalists reported this; Microsoft first said its leadership didn't know, then ran its own review and announced it had found evidence supporting parts of the story, and switched off some of the services. Why does a laptop buyer

care? Because that landlord also makes Windows — the operating system that would run on most laptops — and this record describes the *company*, not one product line.

One last decoding tool: the audit **scores**. You will see numbers like "BRS 712, Tier B (Severe)." These come from a research project called OpenIntel, which builds dossiers on major technology companies and scores how directly each is involved with Israel across four areas: military, digital, economic, and political. A higher number means more documented, more direct involvement. "Tier B (Severe)" is the second-highest of its bands. These scores are one framework's judgments, and this document quotes them the way it quotes everything else — as a documented summary of named facts, not as an endorsement of the framework's politics. Every underlying fact is also stated on its own, so you never have to take the score on faith.

With those five ideas — design vs. manufacture, EDA tooling, firmware, the cloud, and the audit scores — the rest of this document reads as a single connected story: who drew the plans for your laptop, who owns the companies on the lid, who their customers are, and how close to "zero ties" a real purchase can honestly get.

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3. The Five Layers of Entanglement: A Framework

Think of a laptop not as a product but as a stack of layers, each of which can be independently audited. When people ask "is there Israeli involvement in this laptop?", they are usually imagining one layer — usually the assembly plant, or perhaps the brand name on the lid. The entanglement, however, lives in five distinct places:

- **Layer 1 — Authorship (who designed the silicon).** A chip can be fabricated in Taiwan, assembled in China, and shipped by an American brand — and still be *designed* in Haifa or Herzliya. Design authorship is the least visible and most consequential layer, because it is embedded in the chip's architecture itself and follows the product everywhere it goes.
- **Layer 2 — Tooling (what designed the silicon).** No human draws a modern chip by hand. Chips are designed using Electronic Design Automation (EDA) software — effectively, the CAD tools of the semiconductor world. If the EDA software itself contains Israeli engineering, every chip built with it carries that connection regardless of where the chip company is headquartered.
- **Layer 3 — Ownership (who owns whom).** Acquisitions, subsidiaries, venture investments, and voting control. A Taiwanese laptop brand can be clean in every factory and still be entangled if its supplier is majority-owned by a company with deep Israeli operations.
- **Layer 4 — Contracts (who sells to whom).** Military tenders, government cloud contracts, security-agency relationships. This is the layer that forensic audits (like the

OpenIntel BDS-1000 assessments cited throughout this document) examine most closely, because it is the layer where corporate relationships become state relationships.

- **Layer 5 — People (who built it).** Engineers move between companies and countries. Unit 8200 — the Israeli military's elite signals-intelligence unit, roughly analogous to the NSA's Tailored Access Operations combined with a national talent academy — feeds thousands of alumni into global tech. This layer cannot be audited from a spec sheet at all.

The audit that follows walks the *physical* laptop layer by layer (silicon, memory, power, connectivity, sensors, firmware, software, OEM, manufacturing), and overlays the five entanglement layers onto each. Every claim carries a source. Where the evidence is strong, it is stated strongly. Where it is thin, it is labeled thin.

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4. The CPU Layer: The Fatal Layer

The central processing unit is the laptop's brain, and it is where the entanglement is deepest. There are effectively three viable paths to a laptop CPU in 2026: Intel, Apple, and the ARM ecosystem (Qualcomm, Samsung, MediaTek, and everyone else). AMD is the fourth path — and the only one with no Israeli silicon authorship.

4.1 Intel — the deepest entanglement in the industry

Intel's relationship with Israel is not a footnote in company history; it is, by several measures, the single most important bilateral relationship in the semiconductor industry. The verified facts, in sequence:

1974 — the founding. Dov Frohman — a Holocaust survivor's son, Technion graduate, and the inventor of the EPROM (Erasable Programmable Read-Only Memory, an early chip that could be rewritten) — convinced Intel co-founder Andy Grove to open Intel's first design center outside the United States, in Haifa, initially with four workers. Gordon Moore, Intel's other co-founder, called Frohman's EPROM "as important in the development of the microcomputer industry as the microprocessor itself." *Why this matters: the EPROM made it possible to revise chip logic after manufacturing — a foundational enabler of the entire iterative design culture that later produced Intel Israel's design dominance.*

1978–1979 — the 8088. The Haifa team designed the 8088, a cost-reduced variant of Intel's 8086 processor with an 8-bit external bus, which IBM selected for its first personal computer in 1981. Every x86 PC architecture in existence descends from this decision. The Haifa center also designed Intel's first math coprocessor (1979) and later the MMX multimedia instruction set (mid-1990s), the Pentium MMX, the Pentium 4, and the Centrino platform — which integrated Wi-Fi into the laptop for the first time and, in Intel's own description, made Intel Israel "the headquarters for the corporation's global R&D for wireless technology." The Centrino line is what ignited the Wi-Fi laptop

boom of the 2000s — meaning the Israeli design center midwifed the very concept of the modern laptop.

1984–1985 — Jerusalem fab. Intel built its first fabrication plant outside the US — in Jerusalem (Fab 8). A "fab" is the factory where chips are physically manufactured: a clean-room facility costing hundreds of millions to billions of dollars, printing circuit patterns onto silicon wafers. The Jerusalem fab cost $700 million — a project that included the first-ever Israeli government grant to Intel ($188 million). By the early 1990s, the Jerusalem fab was producing roughly **75% of global output of the 386 processor**, the chip that carried the PC through its first decade of mass adoption. The Jerusalem fab closed in 2008, briefly reopened as a die-preparation facility (the step where finished silicon wafers are cut apart into individual chips) in 2009, and closed permanently in 2016.

1995–2008 — Kiryat Gat. Intel built Fab 18 in Kiryat Gat in 1995 — the site Intel itself describes as its most advanced manufacturing facility — then Fab 28 in 2008 — a $3.5 billion project described at the time as the largest construction project in the history of the state of Israel. In 2007, Intel Israel's exports reached $1.4 billion, equal to 8.5% of Israel's total electronics exports.

2003–2015 — the modern Core lineage. The Banias project (released as the Pentium M in 2003) saved Intel's laptop roadmap and was developed in Israel; its successor Merom became the Core 2 Duo, the chip that made "Core" the dominant brand in computing. Sandy Bridge and Ivy Bridge — at their peak accounting for roughly 40% of Intel's global sales — were

developed in Israel. Haswell and Skylake followed the same path; Skylake was, per Israeli industry sources, "produced for the most part in Israel at Intel's Haifa research center." Intel Israel also developed Thunderbolt connectivity.

2023–2024 — Fab 38, the grant, the freeze. In December 2023, Intel finalized a $25 billion expansion of its Kiryat Gat complex (Fab 38), paired with a **$3.2–3.25 billion Israeli government grant**, preferential tax treatment, and a commitment to purchase approximately NIS 60 billion (roughly $16–17 billion) in goods and services from Israeli suppliers over the following decade. In June 2024, Intel halted construction, citing "changing schedules, business conditions, market dynamics and capital management." Israeli business outlet Globes reported the expansion frozen "for the foreseeable future." The underlying operations — fabs, development centers, ~9,000–11,700 employees — continued, but the freeze rippled immediately through the local supplier ecosystem: publicly traded Israeli contractor Lesico reported that delays to contracts worth over NIS 100 million would materially affect its performance, and Calcalist's reporting noted that equipment giants Applied Materials and KLA maintain Israeli operations precisely because of Intel's physical presence there.

2024–2026 — the design lineage continues. Even Intel's newest architecture is Israeli-developed: Israeli industry reporting confirms that the Core Ultra Series 3 and Core Series 4 processors unveiled at CES 2026 were "largely developed in Israel." Apple's hardware chief Johny Srouji, in a rare on-record interview, put it directly: "By no coincidence, the [Apple M1]

Max and the Pro were launched at the same time as our competitor, Intel's new Alder Lake Core 12 processor, which was also developed in Israel. So, the crème de la crème of the world's computer processors are made in Israel."

Scale, per Intel's own corporate reporting and Israeli business press. Intel's Israel page reports 9,335 employees; Calcalist's September 2024 reporting put the figure at ~11,700 — approximately 10% of Intel's entire global workforce. Roughly 3,900 of them work in manufacturing at Kiryat Gat. Intel Israel reported $27 billion in cumulative capital investment through 2021 and $86 billion in cumulative exports through 2023. In 2022 alone, Intel's Israeli operations were responsible for almost 2% of Israel's entire GDP; its $9 billion in exports represented roughly 6% of all Israeli high-tech exports; it indirectly employed an estimated 42,000 people in secondary and tertiary roles and purchased $3.5 billion in goods and services from Israeli businesses. Calcalist's description: "Unlike many international companies that only have development centers or limited activities in Israel, Intel Israel operates like a small Intel USA." Intel is Israel's largest private-sector employer and its single biggest exporter. The Israeli corporate footprint includes registered entities such as Intel Israel (74) Ltd., Intel Electronics Ltd., Hampton Acquisition Ltd., Intel Semi Conductors Ltd., Intel Mobile Communications Israel Ltd., and Granulate Cloud Solutions Ltd. — a structure documented in Intel's own filing-grade disclosures. The development staff (~7,000 of the total) work across communications networks, data centers, PC processors, and AI — which is to say, across every division whose

output ends up in a laptop.

Corporate acquisitions (all Israeli). Mobileye ($15.3 billion, 2017 — the autonomous-driving company Intel retains control of to this day: per Mobileye's 2025 Form 10-K, Intel held approximately 77% of common stock and 96.9% of voting power as of February 2026 — voting power being what actually commands a company: it elects the board and approves major decisions, so Intel's roughly 77% economic stake understates its roughly 97% control); Habana Labs (~$2 billion, December 2019, AI chips); Moovit (transit app); Granulate (~$650 million reported, cloud optimization). The proposed $5.4 billion acquisition of Tower Semiconductor was terminated in August 2023 after Chinese regulators withheld approval — had it closed, Israel would have become Intel's second-largest site worldwide after the US. Intel Capital had invested $120 million across 14 Israeli companies by 2018, and Intel joined Team8's cybersecurity syndicate in 2017 (Team8 is a venture group founded by a former Unit 8200 veteran).

Military-adjacent integration (downstream, verified at audit level). Elbit Systems' ETC MK7 and MK7+ tactical computers — ruggedized command-and-control systems used for tactical data dissemination, with series-level use by the Israel Defense Forces — incorporate commercial Intel Core i7 processors (the MK7+ brochure identifies an Intel Core i7-1370). This is commercial off-the-shelf silicon embedded in a military product; the audit (OpenIntel, BDS-1000 methodology) found no direct Intel–IDF or Intel–Israeli Ministry of Defense procurement contract, and explicitly records "no public evidence identified" of

such a contract.

Governance and politics. The BDS Movement launched a named global #BoycottIntel / "Apartheid Chips" campaign in March 2024. Intel's board opposed — and shareholders rejected — an Ethical Impact Assessment proposal concerning Israeli operations in the 2025 proxy cycle (a proxy vote is the annual mechanism by which shareholders formally instruct corporate management; rejection means the policy was not adopted); the board also opposed a human-rights due-diligence proposal in the 2026 cycle. Intel's then-CEO Pat Gelsinger (a devout Christian who frequently quoted Torah verses and described himself as having a mission of loyalty to Israel) wrote to employees on October 8, 2023 expressing concern for Intel's Israel-based staff; Intel Israel and the Intel Foundation reported approximately □4 million in urgent aid plus □4.4 million in employee-directed donations and matching grants for 2023, including a one-time $5,000 wartime grant per Israel-based employee. Context worth preserving alongside: in July 2024 the US government announced a $3.5 billion grant for Intel to produce chips for the Department of Defense — a marker of how strategically important Intel's production had become to two states at once — and Israeli industry sources speculated that Tower Semiconductor (the company Intel failed to acquire) could one day buy the aging Kiryat Gat fab, which would keep the site's Israeli character intact under new ownership.

Audit verdict. OpenIntel's human-reviewed V6.5 assessment assigns Intel a BDS-1000 score of **712 — Tier B ("Severe Complicity")** — driven principally by the economic domain

(Mobileye control, manufacturing scale, the state-grant relationship). One caveat the audit preserves: Kiryat Gat lies within Israel's pre-1967 territory, not in West Bank/East Jerusalem/Golan settlements, though NGO sources describe the site as associated with Palestinian villages depopulated in 1948 — a land-history finding the audit treats as distinct from post-1967 settlement activity.

Bottom line for a laptop buyer: every Intel laptop CPU from the 8088 to the Core Series 4 carries Israeli design authorship at the architecture level. There is no Intel CPU generation exempt from this.

4.2 Apple Silicon — the M-series, in Apple's own words

Apple's Israeli operation is younger than Intel's but, in chip-design terms, arguably more concentrated.

The center. Apple opened its Herzliya R&D center (12 Hamaskit Street) in March 2012 — its first outside the US — with a single employee: Aharon Aharon, Srouji's former lecturer at the Technion, who served as CEO of the center until 2017 (when he left to run the Israel Innovation Authority, the government's tech-funding body). By 2022 the center employed 1,800 people; Rony Fried has run it since 2017. Apple added a Haifa site and, later, a Jerusalem development center specifically earmarked for future Mac processors. In 2026, Apple elevated Johny Srouji — Haifa-born, Technion-educated, a former Intel Israel engineer (1992–1999) — to the new role of Chief Hardware Officer under incoming CEO John Ternus, and Sribalan Santhanam took over leadership of Apple Silicon

development in Israel. The operation is growing.

What the center builds — from Srouji's own interview (Ynet, on the record):

- "Every storage controller in every Apple product comes from Israel." The storage-controller line began with Apple's first Israeli acquisition, Anobit, whose team produced the first Apple-designed storage controller — in a Mac — nine months after the acquisition closed, later adapted for iPhone and iPad.
- "The M1 Max has 57 billion transistors. Both [M1 Pro and M1 Max] were developed in Israel. Building the chip, assimilating it, its physical design, validation — was all conducted in Israel."
- The Apple Watch's entire SoC — "the watch's electronic brain" — plus its Wi-Fi and Bluetooth capability, were developed in Herzliya.
- PrimeSense (acquired 2013, the Israeli depth-sensing company) built the core technology behind Face ID; the LiDAR core technology integrated into iPhones and iPads is Israeli; Cinematic Mode (iPhone 13) is based on the same technology; Israeli teams built hand- and eye-tracking and depth-sensing components for Vision Pro.
- The Mac facilities program — the decision, effectively, to move the Mac off Intel — was architected around the Israeli design operation.

A separate MintPress/Drop Site News investigation identified dozens of Unit 8200 veterans inside Apple's chip teams, as part of a larger mapping of approximately 1,400 Israeli intelligence-agency veterans across US technology companies.

A structural note on disclosure. Apple's public communications have never named Israel as the location of its second-largest R&D operation outside the US — Israeli site expansions are announced only as happening "all over the world." The design authorship is real; the branding obscures it.

The wider chip family. Beyond the M1 Pro/Max, Israeli teams are publicly tied to the **M2** generation, the **R1** chip (the sensor-fusion processor inside the Vision Pro headset), the **C1** modem (Apple's first in-house cellular modem, which shifts modem design — long a Qualcomm domain — into the Israeli operation), and the **N1** wireless chip. A 2026 statement by Apple Israel's site lead, Rony Fried, distilled the pattern: "It's hard to find an Apple product that doesn't have some kind of touch from Apple employees in Israel."

Bottom line for a laptop buyer: every Apple laptop sold today carries silicon physically designed, integrated, and validated in Israel, plus Israeli storage controllers in every unit. MacBooks are among the *most* Israeli-authored laptops on the market.

4.3 AMD — the cleanest mainstream silicon, with corporate-level caveats

AMD is the only mainstream CPU vendor whose actual silicon appears free of Israeli authorship. The full record:

The silicon is US-designed. AMD's Zen microarchitecture — the design family inside every Ryzen laptop processor — was rebuilt under CTO Mark Papermaster (ex-IBM, ex-Apple, ex-Cisco), who joined in October 2011 and, per the public record, "restructured the CPU design team to build Zen." AMD's known

CPU design sites are Austin, Boston, Fort Collins, and Bellevue — none in Israel. No evidence was found in any pass of this investigation of Israeli design involvement in Zen silicon, in the AGESA firmware, or in the PSP (Platform Security Processor, AMD's analog to Intel's Management Engine).

The corporate ties are real but commercial-grade. Per the OpenIntel forensic audit: the Graphic Remedy acquisition (~$5 million, October 2010) seeded a Ramat Gan/Tel Aviv R&D center (opened June 2011, focused on parallel computing and AMD's Fusion APU software); **Xilinx Israel Ltd. (Holon) is a 100%-owned AMD subsidiary**, per AMD's own FY2023 subsidiary schedule; AMD Ventures participated as a co-investor in Jerusalem-based Teramount's **$50 million Series A (July 2025)** — explicitly a participant, not the lead investor. AMD has a reported AI-inferencing collaboration with Israeli startup NeuReality, and its technology surfaces in Israeli product contexts (e.g., Kameleon Security's ProSPU). A 2020/2023 export-control incident involving AMD and Israeli firm CNG Labs was investigated and did not result in AMD enforcement action. An Israeli corporate registry entry for an "AMD Advanced Micro Devices Israel Ltd." entity shows as voluntarily eliminated — a low-confidence detail, preserved here for completeness.

What the audit explicitly did *not* find: any direct AMD or Xilinx Israeli military contract, any Project Nimbus role, any verified surveillance deployment, any settlement operation. The audit characterizes AMD's Israeli nexus as "economic rather than military or digital."

Bottom line for a laptop buyer: an AMD Ryzen CPU carries no Israeli silicon authorship — the design lineage is American. AMD the *corporation* maintains a small, documented, commercial-only Israeli presence (an R&D office, a wholly-owned subsidiary inherited from Xilinx, a venture stake). For a buyer applying a "zero ties in any way" standard strictly, this corporate residue is a tie; for a buyer applying a "nothing Israeli inside the machine" standard, AMD is the clean choice.

4.4 The ARM ecosystem — the architecture beneath everything else

Everything that isn't x86 (Intel/AMD) is, with rare exceptions, ARM. And ARM itself is entangled:

- **ARM (the company):** its Ra'anana development center's hardware teams oversee development of the flagship **Immortalis GPU** and chip-security components, inherited via the 2015 acquisition of Israeli security-IP firm Sansa Security. ARM licenses its processor designs and architecture to Qualcomm, MediaTek, Samsung, Apple, and effectively everyone else — so ARM's Israeli-engineered IP propagates into every non-x86 laptop chip.
- **Qualcomm:** two Israeli R&D centers (Haifa, Hod Hasharon) operating since 1993; five Israeli acquisitions between 2010 and 2016. Its Snapdragon laptop chips therefore carry both direct Israeli R&D and ARM-derived IP.
- **Samsung:** its Tel Aviv R&D center (SIRC) is responsible for development of next-generation flash/SSD/UFS/eMMC architecture and image sensors — Samsung-made storage, not just Samsung chips, has Israeli design involvement at the

platform architecture level. (Samsung DRAM appears Korea-designed.)

- **Nvidia:** acquired Israeli chipmaker Mellanox for $6.9 billion in 2019; Nvidia Israel now accounts for over a tenth of Nvidia's global workforce and has been aggressively hiring ex-Intel Israel engineers.
- **Marvell:** three Israeli offices leading Ethernet/NIC development — relevant to docking stations and wired connectivity.
- **Historical footnote:** National Semiconductor operated a chip design center in Herzliya in the 1980s that co-developed hardware-description-language technology with Tel Aviv University — evidence that Israeli chip-design DNA in the US industry is generational, not recent.

Bottom line: the only CPU paths without Israeli architectural authorship are AMD's x86 (corporate ties only) and non-ARM, non-x86 architectures — Chinese Loongson or open RISC-V (see Part 11).

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5. The GPU Layer

- **Intel Arc and Intel integrated graphics:** same Intel Israel design pipeline as the CPUs. Maximal entanglement.
- **Nvidia (GeForce/RTX in gaming laptops):** consumer GPU silicon is US-designed; the Mellanox acquisition adds Israeli networking IP (more relevant to datacenter than consumer laptops) and a large Israeli workforce.

- **AMD Radeon:** clean. The ATI lineage is Canadian (acquired 2006); leadership and engineering are US-based; no Israeli design centers were found in any pass.

Practical consequence: the GPU question resolves the same way as the CPU question — AMD is the clean path.

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6. Memory & Storage

- **SK Hynix: clean.** No Israeli R&D center, acquisition, or design presence was found in any pass. Korean-designed, Korean-manufactured. Its external partnerships (e.g., MRAM co-development with Kioxia) do not involve Israel.
- **Kioxia: clean.** Japanese; its diversification partnership is with Taiwan's Nanya Technology.
- **Micron: not clean.** Micron's official R&D location list explicitly includes Israel, with an active Israel ASIC design team currently hiring (an E3-level ASIC Design Engineer posting for the Israel ASIC team). Historically, Micron operated a fab in Kiryat Gat — a plant Intel built and leased to Micron for NOR flash — until Intel repossessed it in 2010, absorbing roughly 800 Micron workers when Micron shut the line. Separately, SanDisk (now part of Western Digital) was co-founded by Israeli-born Eli Harari.
- **Samsung NAND/flash storage: compromised at the architecture level** via SIRC (above). Samsung DRAM appears Korea-designed and cleaner.
- **SSD controllers: clean.** Phison and Silicon Motion (Taiwan) dominate third-party controller supply; Realtek has entered

the category. None has documented Israeli ties.

- **Dell/Lenovo storage software (ecosystem note):** Dell's PowerFlex storage platform is built on technology from Israeli company ScaleIO; Lenovo now owns Israeli storage vendor Infinidat (Part 9). These are enterprise products, not laptop components, but they document how deeply Israeli storage engineering runs through the OEM ecosystem.

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7. Power & Battery Management

A laptop's power circuitry — the charger IC, the PMIC (power management integrated circuit), the battery fuel gauge — is a genuine blind spot in most supply-chain thinking. The audit found it mostly clean:

- **Intersil (now Renesas):** the dominant laptop battery-charger IC family (ISL6255, ISL9237, ISL88731, ISL95522, used across Dell, Lenovo, and HP boards) is US-origin (Milpitas, California), now owned by Japan's Renesas. No Israeli ties found.
- **Texas Instruments:** TI's Israeli chapter is real but irrelevant to its power products. TI's Israel operations were built on two 1990s acquisitions (Libit, $365 million; Betterplay, $50 million) focused on short-range communications (Wi-Fi/Bluetooth/NFC "Butterfly" technology); TI shut its Israel development center in 2013, laying off 250 staff — and **Apple hired 100–150 of those engineers**, who became core contributors to Apple's Ra'anana design center. So TI's old Israeli chip DNA literally flowed into Apple's silicon operation,

while TI's laptop power ICs themselves show no Israeli authorship.
- **Monolithic Power Systems (MPS) and Richtek:** clean (US and Taiwan respectively).
- **Battery cells** (ATL, Panasonic, LG Energy Solution): clean.

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8. Connectivity, Sensors, Firmware, and the Component Long Tail

8.1 Wi-Fi / Bluetooth / Ethernet

- **Avoid on a minimum-ties build:** Intel (Israeli-designed silicon), Qualcomm (Haifa/Hod Hasharon), Marvell (three Israeli offices), Broadcom (Israeli R&D).
- **Realtek: the practical clean choice.** Taiwan's leading fabless (chip-design-only; manufacturing outsourced) designer, in-house engineering, extremely widely used in AMD-based laptops. Caveat: Realtek SoCs license ARM processor cores and are built on Synopsys EDA platforms — an *indirect* taint through shared tooling and licensed IP, not Israeli-authored silicon.
- **USB controllers:** ASMedia and Genesys Logic (Taiwan) — clean.
- **Emerging vector to watch:** Astera Labs — a US connectivity-chip firm whose PCIe/USB4 retimer chips are entering new laptop designs — opened Ramat Gan and Haifa R&D offices in February 2026 by hiring roughly 60 hardware engineers from failed Israeli startup Pliops.

8.2 Webcam, fingerprint, touchpad, audio

- **Webcam sensors: OmniVision — clean of Israeli ties.** US-founded, majority-owned by China's Will Semiconductor since 2018–2019; ubiquitous in laptops (OV9734, OV02C10). **Critical caveat:** on Intel platforms, the sensor typically feeds an **Intel IPU6** imaging processor — an Israeli-designed chip. On AMD platforms, the same OmniVision sensor runs over a plain USB path with no Israeli imaging processor. This is a concrete, purchasable distinction.
- **Fingerprint sensors: Goodix — clean** (Chinese; used by Lenovo, Framework, and others). Synaptics (US): no Israeli footprint found.
- **Touchpads:** Synaptics (US) or Elan (Taiwan) — clean.
- **Audio codecs:** Realtek or Cirrus Logic — clean.

8.3 Firmware

- **Intel ME (Management Engine) / AMD PSP (Platform Security Processor):** the closed-source subsystems that run below the operating system (sometimes called "Ring -3" — a privilege level even the OS kernel cannot see into). No public source attributes ME development to Israel specifically — but every Intel ME instance runs on Israeli-designed silicon, and the early independent security research on ME came from Russian (Positive Technologies) and Polish (Invisible Things Labs) researchers, not from the vendor. The deeper point: the ME/PSP layer is un-auditable by the owner *regardless of nationality*, which is itself a finding worth stating plainly.

- **BIOS/UEFI vendors:** AMI, Insyde (US/Taiwan) — clean. **Phoenix Technologies' BIOS unit was acquired by Lenovo, deal closed 26 April 2026** — meaning one of the major firmware vendors is now inside a Chinese OEM with deepening Israeli ties (Part 9). This does not taint Phoenix's firmware *per se*; it is a corporate-structure note.
- **Embedded controllers:** ITE, ENE, Nuvoton (Taiwan) — clean.

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9. Software: The Operating System Is the Heaviest Layer

9.1 Microsoft Windows — the deepest verified entanglement in this document

Microsoft's Israeli operation predates most: the company began operating in Israel in 1989, and its Israel R&D center (established 1991) was **Microsoft's first development center outside the United States**. Today Microsoft runs five Israeli sites (a 46,000-square-meter Herzliya campus housing both commercial and R&D functions, plus Haifa, Tel Aviv, Be'er Sheva, and Nazareth), thousands of employees, and a dedicated Israel Azure cloud region launched November 2023. Israeli acquisitions include Aorato (2014), Equivio (2015), Adallom (2015), Secure Islands (2015), Hexadite (2017), and Peer5.

The verified military-digital record, from the OpenIntel audit and the underlying Guardian/+972 Magazine/Local Call investigation:

- In late 2021, Unit 8200 commander Yossi Sariel met Microsoft CEO Satya Nadella at Microsoft headquarters regarding the migration of sensitive intelligence workloads to Azure.
- From 2022, **Unit 8200 operated a customized, segregated Azure environment storing, translating, transcribing, searching, and analyzing intercepted Palestinian phone calls from Gaza and the West Bank** — reportedly supporting intelligence used in planning operations, including target research and West Bank arrests. Microsoft engineers collaborated with Unit 8200 on the security measures needed to host the workloads.
- Microsoft initially said its leadership had not known intercepted civilian calls were being stored; in **September 2025, its own follow-up review "found evidence supporting elements of the reporting,"** and Microsoft ceased and disabled specified IMOD cloud-storage and AI subscriptions. The restriction was partial — other Microsoft–Israeli government and cybersecurity relationships continued, with audited contracts scheduled through 2026–2028.
- The commercial record alongside it: an approximately **$133 million, three-year Israeli Ministry of Defense agreement beginning 2021**; approximately **19,000 hours of Microsoft engineering support purchased by the IMOD between October 2023 and June 2024** (reported ~$10 million); a circa-2002 ~$35 million IMOD software/cybersecurity licensing agreement (partly financed

through US Foreign Military Sales); Elbit Systems' OneSim military simulation infrastructure rebuilt on Azure with Microsoft engineering collaboration; Azure reportedly powering the Civil Administration's "Al Munaseq" permit-application system and Microsoft applications associated with the "Rolling Stone" population-and-movement system — both used in the administration of Palestinian movement; documented Microsoft services to Ariel University (occupied West Bank) into 2025, and a 2019 education partnership in Ma'ale Adumim settlement schools. An IAI tank prototype used retail Xbox controllers — off-the-shelf repurposing, not a defense product, but recorded in the audit for completeness.

- **AnyVision:** Microsoft's venture arm M12 invested $74 million in a 2019 round for Israeli facial-recognition company AnyVision; Microsoft divested in 2020 after an independent review said the evidence did not substantiate the alleged mass-surveillance program, and announced it would stop minority investments in facial-recognition companies.
- **Governance:** in December 2025, a shareholder proposal seeking a public report on human-rights due diligence received more than 26% support. Employee protests over the Israeli military contracts were followed by terminations Microsoft attributed to conduct and safety-rule violations; no labor-board finding against Microsoft resulted.
- **Audit verdict:** OpenIntel assigns Microsoft a BDS-1000 score of **711 — Tier B (Severe)** — with the maximum domain score in Digital (7.80), driven by the Azure/Unit 8200 record.

BDS-movement literature accordingly describes Microsoft as "perhaps the most complicit tech company" in its category. The audit's exculpatory record is preserved here with the same weight as its inculpatory record: no evidence Microsoft manufactured weapons, built Israel's named targeting models ("Gospel," "Lavender," "Where's Daddy"), or collected the underlying intercepted data; the interception and analytical tooling are attributed to Israeli agencies operating on general-purpose Microsoft infrastructure.

9.2 Linux — mostly clean, with a sponsorship asterisk

The Linux kernel is community-authored, but development is employer-sponsored. The Linux Foundation's own reporting lists the top-10 sponsoring organizations as **Intel, Red Hat, Linaro, IBM, Samsung, SUSE, Google, AMD, Renesas, and Mellanox** — Mellanox being the Israeli company Nvidia acquired. So the kernel that would run on a "clean" laptop is partly funded by the exact companies this document flags. The code is open and inspectable; the sponsorship is real; both facts belong in the record. **Linux remains the cleanest practical OS layer by a wide margin.**

9.3 Browsers and applications

- **Firefox/Mozilla:** open source; no documented Israeli R&D or acquisition ties — the cleanest mainstream browser.
- **Chrome/Chromium:** Google operates large Israeli R&D (Google acquired Israeli companies including Waze; Project Nimbus — the Google/Amazon cloud contract with the Israeli government — is extensively documented separately).

- **macOS/Windows:** both covered above.

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10. The OEM Layer: The Brand on the Lid

This is the layer most buyers actually see, and it is where corporate entanglement concentrates.

10.1 Dell — BRS 691, Tier B (Severe)

Structure: two Israeli legal entities (Dell Technology & Solutions Israel Ltd.; EMC Israel Advanced Information Technologies Ltd.), sites in Herzliya, Haifa, and Beersheba; cumulative Israeli M&A exceeding **$1 billion since 2006**.

The acquisition ledger:

- Kashya (2006) — data replication
- Cyota (2005, via RSA) — anti-fraud
- Exanet (2010) — storage
- **XtremIO ($430 million, May 2012)** — Herzliya flash-storage company founded by Israeli storage veterans including M-Systems co-founder Aryeh Mergi; EMC's sixth Israeli acquisition; EMC already ran a 700–800-person Israel R&D center including VMware staff
- **ScaleIO (~$200 million, June 2013)** — founded by Boaz Palgi, Erez Webman, Lior Bahat, Eran Borovik, and Erez Ungar; became Dell's PowerFlex storage platform ("Dell's PowerFlex software is built on technology developed by the Israeli company ScaleIO")
- VMware's Israeli stack during Dell ownership: B-hive, Wanova, Digital Fuel

- **Cloudify (January 2023, est. ~$100 million)** — orchestration
- Dell Technologies Capital's Israel portfolio includes VAST Data, Redis, Pecan, JFrog, GuardiCore, Elastifile, Descope, Entro, Iguazio, LightBits, LayerX, and Quantum Source

Military procurement (verified reporting): January 2023 — a two-year IMOD tender **reported above $150 million**, funded through US foreign aid, covering servers and maintenance for the Israeli military and security bodies; 2006 — tender for ~50,000 army computers (~$35 million); 2010 — EMC storage contract (~$300 million); VMware's NIS 100 million IMOD agreement (2016–2019, during Dell ownership); NIS 22.56 million in Israeli R&D incentives (2019–2024) tied to National Cyber Park activity in Beersheba, adjacent to an IDF technology campus; Dell mentored a 2018 military cyber hackathon attended by the IDF Computer and IT Directorate. Workforce cited at ~1,200 staff (circa 2019 figure, now dated).

Political record: Michael Dell's January 2024 social-media post alongside Israeli President Isaac Herzog ("an honor to stand with Isaac Herzog and Israel"); a **$1.8 million Friends of the IDF scholarship donation** by Michael and Susan Dell (2014). Michael Dell has separately been quoted in Israeli business press describing himself as "deeply committed to Israel."

Audit verdict: BDS-1000 score **691, Tier B (Severe)** — driven by the acquisition stack and the IMOD contract record.

10.2 HP — deep, current, and historical

Structure: HP Indigo — manufacturing in Kiryat Gat (an 11,000 m² ink plant) and R&D headquarters in Ness Ziona — employs roughly 2,400–3,000 workers and is cited at approximately **$1 billion in annual exports**. Indigo itself is Israeli-born: founded in Rehovot in 1977 by Benny Landa, acquired by HP for approximately $829 million in 2001; HP Scitex (also Israeli-origin) merged into Indigo in 2022/23.

Military and security contracts: HP has been the **exclusive PC provider to the Israeli military since 2009**, with the most recent such contract signed in 2014. Historical entanglements (largely discontinued by 2016–2017) include the "Basel" biometric checkpoint system and exclusive IDF PC/server contracts. **Critically for 2026: HPE holds renewed Israeli security-body contracts — Israel Police (through December 2026), Israel Prison Service (through February 2026), and the Population & Immigration Authority's "Aviv" population-registry system (through May 2028)** — with renewals occurring after the July 2024 ICJ advisory opinion and November 2024 ICC warrants. NGO sources (AFSC's Investigate database) list HP in the UN OHCHR settlement database; the audit notes the entry details are unverifiable due to database access restrictions.

Acquisition history: Mercury Interactive ($4.5 billion, 2006); NUR Macroprinters ($457 million, 2008).

10.3 Lenovo — rapidly deepening (2025–2026)

- **Infinidat:** acquisition announced January 16, 2025, **completed April 9, 2026**. Herzliya-based enterprise storage company founded in 2011 by storage pioneer Moshe Yanai; ~300 of its 500 staff are Israel-based; reported value above $500 million (market estimates ran to ~$1 billion). Infinidat operates as a business unit inside Lenovo's Infrastructure Solutions Group, with its Herzliya development center expanding — Lenovo's Israel GM stated the acquisition is expected to **double Infinidat's local sales and headcount**.
- **Security ecosystem:** a Cybersecurity Innovation Center with Ben-Gurion University (2023), evolving into a Tel Aviv Digital Trust Lab (February 2026) — located in the Gav-Yam Negev/CyberSpark ecosystem that co-locates IDF intelligence and technology units including Unit 8200. **SentinelOne — an Israeli-founded cybersecurity company — has its Singularity platform embedded in new Lenovo PC shipments via ThinkShield**, under an expanded multi-year collaboration.
- **Firmware:** Lenovo's acquisition of **Phoenix Technologies' BIOS unit closed 26 April 2026** (per TechCrunch/MassDevice deal reporting).
- **Corporate residue:** a registered "Lenovo (Israel) Ltd." entity; venture exposure via Lenovo Capital into Canaan Partners Israel and (2012) Vertex's Israel fund.
- **Not found:** any Lenovo military contract with Israel; Lenovo does not appear on BDS target lists. The trajectory, however, is unmistakably toward deeper entanglement — two

Israel-linked acquisitions and an Israeli security-software embedding in consumer PCs, all within ~15 months.

10.4 ASUS and Acer — the cleanest major OEMs

Across all audit-grade sources, **no Israeli R&D centers, acquisitions, or subsidiaries were found for either ASUS or Acer.** Both operate under a general Taiwan–Israel R&D cooperation framework (signed 2015), but neither has planted Israeli operations. For a buyer applying any tie-based standard, this makes ASUS and Acer the default chassis choice — and notably, even BDS-adjacent consumer guides converge here: Ethical Consumer's 2024 laptop guide recommends Framework and TCO-certified ASUS (recommendations made on corporate-conduct grounds, which implicitly acknowledges the silicon layer this document documents).

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11. The Meta-Layer: EDA Tooling, and Why Even "Clean" Chips Aren't

Before a chip exists as hardware, it exists as a design inside software. **Electronic Design Automation (EDA)** is the category of software used to design, simulate, verify, and lay out every modern integrated circuit — the CAD of the semiconductor world, but vastly more complex: a single chip can contain tens of billions of transistors, and no human team could place them by hand.

The EDA market is effectively a duopoly: **Synopsys and Cadence**, whose tools are used to design "virtually any"

commercial chip. And both have Israeli development operations — Cadence lists Israel explicitly in its stated geographic footprint (its development center is in Netanya); Synopsys is Realtek's "primary EDA partner," with Realtek licensing Synopsys' Galaxy Implementation Platform and DesignWare IP. Synopsys' own Israeli presence (via acquisitions including Virage Logic) is documented in the public record, though this investigation cites it conservatively.

Why this matters: even the cleanest component choice in this entire document — a Taiwanese Realtek Wi-Fi chip with no Israeli employees, no Israeli subsidiary, no Israeli contract — is *designed using tooling that contains Israeli engineering*. This is the floor of the problem. A "zero ties in any way" standard, applied with total literalism, bottoms out here: there is no modern chip, from any country, designed without these tools. This finding does not negate the component-level distinctions elsewhere in the document — choosing AMD over Intel removes *Israeli-authored silicon*, which is categorically different from *shared tooling* — but it must be stated, because it defines the theoretical limit of what procurement can achieve.

***

12. Manufacturing: The Clean Layer

For completeness: the *physical production* of laptops is essentially Israel-free.

- **Assembly (ODMs):** Quanta, Compal, Wistron, Foxconn, Pegatron — Taiwanese/Chinese contract manufacturers. No Israeli ties.

- **Displays:** BOE (China), AUO and Innolux (Taiwan), LG (Korea). No documented Israeli ties.
- **Fabrication equipment:** ASML (Netherlands), Applied Materials, Lam Research, KLA (US), Tokyo Electron, Nikon, Canon, Advantest (Japan). No laptop's manufacture depends on Israeli equipment. (Applied and KLA do maintain Israeli *operations* — they "came because of Intel," per Calcalist's reporting on the Kiryat Gat ecosystem — but this is downstream service activity, not an equipment dependency.)
- **Fabs:** TSMC (Taiwan), Samsung (Korea), Intel (US/Israel — Kiryat Gat does fabricate Intel CPUs, which is Layer 1 authorship, distinct from equipment).
- **Tower Semiconductor** (Israel) is a specialty/analog foundry — not in the laptop supply chain.

The choke point of this entire investigation is **design authorship**, not tooling, not assembly, not materials. Chips are conceived in Israel; they are not (outside Intel's Kiryat Gat) generally *made* there.

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13. The Minimum-Ties Build: The Practical Result

Putting every layer together, this is the closest a buyer can get to the stated criterion with a modern, usable, warranty-backed machine:

Layer	Choice	Why	Residual taint
OEM / chassis	**ASUS or Acer**	No Israeli R&D, acquisitions, or subsidiaries found (audit-grade)	None found

Layer	Choice	Why	Residual taint
CPU	**AMD Ryzen**	Zen architecture US-designed; no Israeli silicon authorship found	Corporate only: Ramat Gan R&D office, Xilinx Israel Ltd. (100% subsidiary), $50M Teramount co-investment
GPU	**AMD Radeon (integrated)**	ATI Canadian lineage; no Israeli design centers found	None found
Wi-Fi / Bluetooth	**Realtek**	In-house Taiwanese design	Indirect: ARM IP licensing; Synopsys EDA tooling
SSD	**Kioxia NAND + Phison controller, or SK Hynix**	Japanese/Korean/Taiwanese chain	None found
RAM	**SK Hynix**	Korean design and manufacture	None found
Charger / PMIC	**Intersil-Renesas, TI, or MPS**	US/Japan power ICs	None found
Webcam	**OmniVision sensor on the AMD USB path**	Clean sensor; on Intel it would feed an Israeli-designed IPU6 imaging processor	None found on this path
Fingerprint reader	**Goodix**	Chinese vendor, widely used	None found
Touchpad / audio	**Synaptics or Elan; Realtek codec**	Clean vendors	None found
BIOS / EC	**Insyde/AMI (or Phoenix, noting Lenovo ownership); ITE embedded controller**	Clean vendors; structural note that Phoenix is now Lenovo-owned	None found in firmware itself
OS	**Linux**	Community-authored; inspectable	Partial: kernel sponsorship top-10 includes Intel, Mellanox (Israeli→Nvidia), AMD, Samsung, Google

Layer	Choice	Why	Residual taint
Browser	**Firefox**	Open source; no documented Israeli ties	None found

The irreducible floor — stated plainly, because honesty requires it:

1. **AMD corporate ties.** Even the cleanest CPU vendor has an Israeli office, a wholly-owned Israeli subsidiary, and an Israeli venture stake. If "zero ties in any way" means zero corporate relationships, no x86 or ARM machine qualifies.
2. **EDA authorship.** Every chip in the machine was designed with Synopsys/Cadence tools containing Israeli engineering (Part 11).
3. **Linux sponsorship.** Partial funding taint at the kernel-sponsor level.
4. **The people layer.** ~1,400 Israeli intelligence-unit veterans work across US tech (MintPress/Drop Site mapping), concentrated in exactly the firms that design chips and OSes. No purchase decision touches this layer.

That is the honest floor of a mainstream machine. It is dramatically cleaner than an Intel/HP/Windows laptop — it removes Israeli *silicon authorship* entirely — but it is not, and cannot honestly be called, zero.

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14. The Literal Zero-Tie Paths

Three paths satisfy the criterion as literally as the physical market allows. Each carries real tradeoffs that must be stated with the same honesty as the entanglement findings.

14.1 Loongson / LoongArch (China)

What it is: the Chinese domestic CPU program (its current mainstream laptop/desktop processor is the 3A6000). Loongson processors use the LoongArch instruction set architecture — a fully Chinese-designed ISA (the fundamental language a CPU speaks), not x86 and not ARM. Complete laptops — current desktop-class models include the 3A6000 — ship with Chinese domestic operating systems (Loongnix, UOS, or Kylin).

Why it qualifies: at the CPU level, this is the only literal zero-Israeli-R&D path found in the entire investigation. No Israeli design involvement, no Israeli subsidiary, no Israeli venture relationship. Fabrication runs through TSMC or SMIC; SMIC's fabs run on ASML tooling (Netherlands) — and per Part 11, even this bottoms out at the EDA layer unless the Chinese toolchain (which exists in domestic form) was used end-to-end.

The tradeoffs, stated without exaggeration: performance and compatibility well behind x86/ARM (x86 software requires translation layers); the Chinese-state trust question inverts the original criterion for many buyers (a machine with zero Israeli ties that is, by design, transparent to a different state); and ecosystem immaturity (driver support, peripheral compatibility).

14.2 RISC-V open-hardware laptops

What it is: RISC-V is an open-source instruction set — free to implement, no license fees, no ARM. As of 2026, fully functional RISC-V laptops exist: the Framework 13 with a RISC-V mainboard, and the DC-ROMA II (DeepComputing), built around the 8-core ESWIN EIC7700 (P550) SoC — a Chinese-designed RISC-V chip.

Why it qualifies: no Israeli design involvement in the ISA or the ESWIN silicon; no ARM IP; no Israeli corporate presence in the vendor chain.

The tradeoffs: performance far below x86/ARM contemporaries; an immature software ecosystem; small vendors with limited warranty infrastructure. This is a real, purchasable category — not a concept — but it is early-adopter territory.

14.3 Vintage (pre-1979)

Machines predating the Israeli-designed 8088 (i.e., pre-IBM-PC) avoid the Intel Israel lineage — but a 1978-era computer is a museum piece, not a tool. It is listed for completeness, not as a recommendation.

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15. The People Layer: What No Spec Sheet Can Exclude

One finding deserves its own section because it reframes the entire question: the *talent pipeline*. Unit 8200 is the IDF's signals-intelligence unit — Israel's equivalent of the NSA's most elite operations, combined with a national talent academy that selects teenagers out of high school, trains them intensively, and releases them into industry. The MintPress/Drop Site News mapping identified approximately **1,400 veterans of Israeli intelligence agencies across US technology companies** — with concentrations at exactly the firms that author laptop silicon and software: Apple, Intel, Microsoft, Google, Amazon, Nvidia, Meta.

Johny Srouji himself is a product of this pipeline-adjacent ecosystem (Technion → IBM Haifa → Intel Israel → Intel Austin → Apple), and the pattern repeats at every company in this document: the Israeli design centers are staffed substantially by veterans of Israeli military technology units. This is not an accusation of anything illegal — it is a structural description of how the industry actually works, confirmed by the companies' own executives ("If you're looking for engineering talent, Israel is the place," Srouji says, on the record).

Implication: a "zero ties" laptop criterion can be applied to components, corporations, contracts, and tooling. It cannot be applied to the biography of every engineer who touched the supply chain — and any framework that pretends otherwise is not evidence-based. This document therefore treats the people layer as context that sharpens the other findings, not as a

purchasing criterion.

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16. Methodology and Evidence Grading

How this investigation was conducted. Four sequential research passes: (1) breadth mapping of all supply-chain layers; (2) targeted gap-closure on memory, power, OEMs, and firmware; (3) full consolidation; (4) primary-source deep reading — full articles and audit dossiers rather than search snippets. Sources include named-executive interviews (Srouji/Ynet), Israeli business press (Calcalist, Globes, CTech), SEC filings (Mobileye 10-K, AMD subsidiary schedules), company press releases (Lenovo/Infinidat), forensic corporate audits (OpenIntel's BDS-1000 assessments of Intel, Microsoft, and Dell, cited with their methodology and their exculpatory findings), NGO databases (AFSC Investigate, Who Profits), and hardware-teardown reporting.

Evidence grades used implicitly throughout:

- **Audit-grade:** documented in a forensic dossier with named sources and explicit "not found" statements (OpenIntel entries).
- **Filing-grade:** corporate disclosure (SEC, subsidiary schedules).
- **On-record:** named executive or company statement.
- **Reported:** quality press, single or convergent sourcing.
- **Unresolved:** flagged where evidence is thin, dated, or contested.

Caveats the record requires:

- "No public evidence identified" (the audits' formulation) records the result of specified public-source checks; it does not prove non-public activity impossible. This applies symmetrically — to Intel's lack of a direct IMOD contract and to AMD's lack of military ties alike.
- Some figures are dated cumulative measures (e.g., Intel's $86B cumulative exports through 2023; Dell's ~1,200 Israel staff circa 2019) and are labeled as such.
- BDS-1000 scores (Intel 712, Microsoft 711, Dell 691 — all Tier B "Severe") are one framework's human-vetted output, reproduced here as evidence of *documented activity*, not as endorsements of the framework's politics. The underlying documented facts stand independently of the scoring.
- This document adopts no theory of anyone's intent. It gathers clues and evidence: who designs what, who owns whom, who contracts with whom, and what remains unproven.

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17. What Remains Unresolved

Stated for completeness, in descending order of significance:

1. **Intel ME/AMT firmware authorship at team level.** Whether Haifa teams specifically developed ME or AMT firmware is internal and likely unresolvable from open sources. Moot in practice — every ME runs on Israeli-designed silicon anyway.
2. **Micron's Israel ASIC team scope.** Active hiring is confirmed; whether its output ships in consumer products or enterprise-only is unresolved.

3. **Synopsys Israel specifics.** Cadence's Israel footprint is confirmed; Synopsys' Israeli R&D is documented in the public record but this investigation cites it conservatively (no site-level detail).
4. **MediaTek's direct footprint.** None found; its indirect taint (ARM IP, Synopsys tooling) is quantified and that is the honest statement.
5. **BDS-framework blind spots.** Scoring frameworks optimize for their own criteria; a company can score low while having meaningful ties, or high while lacking direct military contracts. The scores are used here as documentary summaries, not oracles.

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18. Key Figures at a Glance

Figure	Value	Context
Intel Israel workforce	~9,335 (Intel) / ~11,700 (Calcalist 2024)	~10% of Intel's global staff
Intel cumulative exports from Israel	$86 billion (through 2023)	~6% of Israeli high-tech exports in 2022
Intel Israel share of Israeli GDP (2022)	~2%	Largest private employer in Israel
Fab 38 expansion	$25B announced; $3.2–3.25B state grant; NIS 60B local procurement pledge	Construction halted June 2024
Intel acquisitions in Israel	Mobileye $15.3B (2017, 96.9% voting control), Habana ~$2B (2019), Moovit, Granulate ~$650M	Tower $5.4B deal blocked by China (2023)
Intel BDS-1000 score	712 — Tier B (Severe)	OpenIntel V6.5 audit
Microsoft BDS-1000 score	711 — Tier B (Severe)	Driven by Digital domain (7.80)
Microsoft–IMOD agreement	~$133M, three years, from 2021	Plus ~19,000 engineering hours Oct 2023–Jun 2024
Dell BDS-1000 score	691 — Tier B (Severe)	Israeli M&A >$1B since 2006
Dell–IMOD tender	>$150M, two years, January 2023	US foreign-aid funded
HP Israel workforce (Indigo)	~2,400–3,000; ~$1B annual exports	Exclusive IDF PC supplier since 2009
Lenovo–Infinidat	>$500M (est. to ~$1B), completed April 2026	~300 of 500 staff Israel-based
Apple Herzliya center	Opened 2012; 1,800 staff by 2022	Every Apple storage controller Israeli-built
AMD Israeli ties	R&D office, 100%-owned Xilinx Israel Ltd., $50M Teramount co-investment	No Israeli silicon authorship found
Unit 8200 alumni in US tech	~1,400 (MintPress/Drop Site mapping)	People layer; not purchasable
Linux kernel top-10 sponsors	Includes Intel, Mellanox (Israeli→Nvidia), AMD, Samsung, Google	Sponsorship taint, not code authorship

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19. Glossary

- **ISA (Instruction Set Architecture):** the fundamental language a CPU speaks — the set of operations it can execute. x86 (Intel/AMD), ARM, LoongArch, and RISC-V are ISAs. Software must be translated or rewritten to move between them.
- **SoC (System on Chip):** a single chip integrating CPU, GPU, memory controllers, and I/O. Apple's M-series chips are SoCs.
- **Fabless:** a chip company that designs but does not manufacture (Realtek, Qualcomm, Nvidia). Manufacturing is outsourced to fabs.
- **ODM (Original Design Manufacturer):** the contract manufacturers that actually build laptops (Quanta, Compal) — usually invisible to buyers behind the brand.
- **EDA (Electronic Design Automation):** the software used to design chips. Synopsys and Cadence dominate.
- **IP (Intellectual Property) cores / IPU:** licensed chip-design building blocks (ARM sells CPU cores); IPU here also means Intel's Image Processing Unit (IPU6), a camera-interface chip.
- **BIOS/UEFI & EC (Embedded Controller):** firmware that runs before the OS (BIOS/UEFI) and the microcontroller managing keyboards, fans, and power (EC).
- **ME / PSP:** Intel Management Engine / AMD Platform Security Processor — closed subsystems running below the OS with full hardware access.
- **PMIC:** Power Management Integrated Circuit — the chip(s) governing charging and power distribution.

- **Retimer:** a signal-conditioning chip for high-speed interfaces (PCIe, USB4); Astera Labs' product.
- **Unit 8200:** the IDF's signals-intelligence unit; primary feeder of Israeli tech talent to global industry.
- **IMOD:** Israel Ministry of Defense.
- **BDS / BDS-1000:** the Boycott, Divestment, Sanctions movement, and OpenIntel's audited 1,000-point complicity-scoring methodology (Tier B = "Severe").
- **Tier B (Severe Complicity):** OpenIntel's second-highest complicity band. Intel 712; Microsoft 711; Dell 691.
- **Tape-out:** the moment a chip design is finalized and sent for manufacturing — the "plans handed to the construction crew" stage.
- **Data center:** a warehouse of computers that rents computing power over the internet; Microsoft's Azure and its Israeli region are examples.
- **Transistor:** the microscopic on/off switch that is the basic building block of all chips; modern processors contain tens of billions of them.

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20. The Conclusion, in One Paragraph

A laptop with zero ties to Israel cannot be bought — not because of one component, but because Israeli engineering sits inside the design of the world's dominant CPU architectures, the tools used to design every chip, the acquisition stacks of three of the four biggest PC brands, the cloud contracts of the dominant operating-system vendor, and the talent pipeline of the entire industry. The closest a buyer can honestly get is an ASUS or Acer machine with an AMD processor, Realtek connectivity, Korean or Japanese memory and storage, an OmniVision camera on the USB path, and Linux with Firefox — a configuration that eliminates Israeli *silicon authorship* entirely while retaining small, documented, corporate-level ties that no consumer choice can remove. The only literal zero-tie machines are Chinese Loongson systems and early RISC-V systems, each carrying tradeoffs that must be weighed against the criterion itself. That is the complete, evidence-backed answer — every layer audited, every stone listed, nothing adopted from anyone's theory, and every unresolved item marked as unresolved.

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Compiled 2026-09-20. Four research passes; primary sources opened in full. All monetary figures as reported by the cited sources; dated figures labeled. "No public evidence identified" statements are preserved from the underlying audits and mean exactly what they say.

Part Four: The Zero-Ties Phone: A Complete Audit of Israeli Entanglement in the Cell Phone and Prepaid Market

A fully-sourced investigation into whether a cell phone — including the prepaid market, the carriers underneath it, and every brand on sale as of September 2026 — can have zero ties to Israel, foreign or domestic.

1. The Question

The question from the laptop investigation returns, harder: **is there a cell phone on the market — at any price, in any category, including the prepaid burner market — with zero ties to Israel?**

A phone is a more difficult object to audit than a laptop, for three structural reasons explained in Part 3. But the question also reaches further than the device itself: a phone without service is a camera-shaped brick, so this investigation audits the entire stack a working phone depends on — the silicon, the modem, the operating system, the brand, the prepaid carrier, the billing system behind the carrier, the SIM card, the point of sale, and even the apps and advertising code the user touches afterward.

How to read this document

- **Part 2** gives the short answer.
- **Part 3** explains why phones are harder than laptops — a short orientation for any reader.

- **Parts 4–10** are the hardware audit: silicon, modems, memory, cameras, security chips, and the last component layers.
- **Parts 11–14** are the software and corporate audit: operating systems, the spyware ecosystem, every brand on the market, and the clean-phone teardowns.
- **Parts 15–18** are the service stack: prepaid, SIM cards, carriers, and the back office.
- **Part 19** is the app and advertising-code layer — the deepest software finding of the investigation.
- **Part 20** covers Israel's domestic market, for the "domestic" half of the criterion.
- **Part 21** is the practical result: the minimum-ties phone.
- **Part 22** is the complete market ledger.
- **Part 22** explains how we know what we know (methodology and evidence grading).
- **Part 23** lists what remains unresolved.
- **Part 24** is a key-figures table for quick reference.
- **Part 25** is a glossary of every technical term used.
- **Part 26** is the conclusion, in one paragraph.

A note on framing, identical to the laptop report: this document argues no political position. The user's criterion is a supply-chain standard — zero ties to Israel — and this document measures the market against it. Where Israeli involvement is found it is documented with sources; where it is not found, that too is documented; where evidence is thin or disputed, the ambiguity is preserved rather than resolved by opinion.

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A thirty-second orientation for readers new to phones

A modern smartphone is effectively **two computers wearing one case**. The first is the one you see — the touchscreen, the apps, the operating system. The second is the **cellular modem**, a completely separate computer with its own closed software whose only job is talking to the cell towers, and which sits in a privileged position the main computer cannot fully supervise. On top of that, a flagship phone is built around a single chip — the **SoC** (system-on-chip) — that fuses the processor, graphics, camera processing, security hardware, and modem interface into one piece of silicon. A laptop lets you swap a tainted part for a clean one; a phone, fused this way, largely does not. Keep that image — one fused chip, plus a secret second computer — in mind and every section below will read clearly.

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2. The Short Answer

No mainstream smartphone has zero ties to Israel — and a phone is entangled in ways a laptop never can be. Five structural findings hold at audit grade:

1. **Every flagship silicon path is tainted.** Qualcomm (two Israeli R&D centers operating since 1993). Apple (its Herzliya team designs the performance CPU cores — Firestorm, Avalanche, Everest — plus every storage controller, the Face ID core, and now the C1X modem and N1 wireless chip). Samsung (its Israeli centers contribute to Exynos chips and design the UFS storage architecture and ISOCELL camera line used in hundreds of millions of phones). Google Tensor

(built on Samsung silicon and ARM designs, while Google assembles new chip teams inside Israel). The only audited clean silicon path is **MediaTek — BDS-1000 score 248, the lowest audited score in this entire investigation — and it is "clean with an asterisk": a venture-capital trail into Israeli chip startups, including one defense-adjacent AI firm, and distributor channels into Israel, but no office, no subsidiary, no contract.**

2. **The modem layer has no clean Western option.** Every iPhone modem ever shipped is either Qualcomm (Israeli R&D) or Apple C1/C1X (Israeli-designed). Samsung's modems come from the same company whose Exynos and UFS architecture carry Israeli authorship. The only Israeli-clean modems in any phone are Chinese — made by BroadMobi, Quectel, or integrated into Unisoc and MediaTek designs.
3. **The software forks do not escape the hardware.** GrapheneOS — independently audited as clean software, with no Israeli vendors anywhere in its infrastructure — runs *only* on Google Pixel phones whose Tensor silicon is triply tainted. The best private operating system in the world sits on the most entangled mainstream phone hardware.
4. **The service stack is entangled at its core.** Every US carrier — the networks every prepaid SIM must ride — carries documented Israeli ties: AT&T runs a 500-engineer Israeli R&D operation that develops 5G network software and the FirstNet public-safety system; Verizon maintained an Israeli venture office for six years; T-Mobile's parent Deutsche Telekom has invested in Israeli funds since 2002. And the

billing software underneath essentially all of them is **Amdocs, an Israeli-founded company holding contracts with roughly 90 percent of the world's top carriers, with AT&T alone accounting for about a quarter of its revenue.**

5. **Israeli code reaches a typical phone through three doors the buyer cannot fully control:** the apps themselves (Meta, Spotify, Waze, Israeli mobile games), the advertising-and-attribution **SDKs embedded inside other people's apps** (AppsFlyer — Israeli-founded, inside roughly half of all Android apps), and the **offensive-cyber industry** (NSO's Pegasus, Intellexa's Predator, QuaDream's Reign) that exists specifically to attack these devices under Israeli Ministry of Defense export licensing.

The closest achievable mainstream machine (Part 21): an **Honor** phone with a MediaTek Dimensity chip — Honor is the only Tier E ("Minimal/No Complicity") company audited in this investigation, scoring **192**, and its upcoming Magic 8 Mini is tipped to carry a flagship MediaTek Dimensity 9500 — running stock Android with Meta apps uninstalled, on a T-Mobile-based prepaid plan, activated online, with a physical SIM. **The maximal-cleanliness machine**: the PINE64 PinePhone, which contains no Israeli-authored component found anywhere in its build — at hobbyist performance, riding a carrier certification system that barely tolerates it.

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3. Why Phones Are Harder Than Laptops

Three structural differences, stated plainly:

Integration. A laptop is a federation — you can choose the CPU, the Wi-Fi card, the memory, the SSD independently. A flagship phone is one SoC that fuses everything. When that one chip is entangled, the whole device is entangled; there is no AMD-versus-Intel choice inside a Snapdragon phone.

The secret second computer. The cellular modem runs its own closed operating system — a real-time OS the user never sees, cannot audit, and often cannot update independently. Security researchers treat the modem as the single most privileged attack surface in mobile hardware. And the modem question has a bitter answer: every Western modem is documented as Israeli-touched, and the clean ones are exclusively Chinese.

Lockdown. Mainstream phones ship with locked bootloaders, meaning the owner cannot fully replace the vendor's software. The vendor's choices — preinstalled apps, cloud backups, telemetry — follow the device for its usable life unless the buyer deliberately works around them.

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4. The SoC Layer — The Fatal Layer, at Audit Grade

4.1 Qualcomm — the Intel of phones

Two Israeli R&D centers (Haifa and Hod Hasharon) operating since 1993; five Israeli acquisitions between 2010 and 2016; and in May 2023 a definitive agreement to acquire **Autotalks**, an Israeli vehicle-connectivity chip startup. Snapdragon sits inside most premium Android phones and, historically, inside every iPhone's modem. Its silicon carries the same category of taint as Intel's CPUs — Israeli-authored engineering at the core of the product. Notably, **Qualcomm Ventures is also an investor in Nothing** (a phone brand examined below) — a corporate link recorded for completeness even though the money itself is American.

4.2 Apple — the deepest silicon authorship of any phone maker

The laptop investigation established Apple's Herzliya dossier; the phone investigation extends it:

- **The performance CPU cores themselves are Israeli-designed.** Semiconductor-analysis reporting on Apple's chip organization is unambiguous: "Israel is where Apple's CPU performance leadership is designed. The Herzliya team, many recruited from Intel Israel — which created the Pentium M, Core, and Sandy Bridge architectures — designs the Firestorm, Avalanche, and Everest cores that dominate benchmarks." Apple runs 8,000-plus chip engineers across more than fifteen design centers; the performance cores — the ones that win benchmarks — are the Israeli file.

- **Every storage controller in every Apple product is Israeli-designed**, per Johny Srouji, Apple's hardware chief, on the record: it began with the Anobit acquisition ("our first Israeli company"), whose team produced Apple's first in-house storage controller nine months later, later adapted for iPhone.
- **Face ID's core** is PrimeSense (Israeli, acquired 2013 for about $345 million); Apple also acquired Israeli facial-recognition firm RealFace (2017) and AR-camera firm Camerai; the January 2026 acquisition of **Q.ai — reported at roughly $1.5–2 billion** — adds facial micro-movement and "silent speech" AI. (Founders' Unit 81/8200 backgrounds trace to a single advocacy source and are flagged uncorroborated.)
- **The modem the company built to escape Qualcomm is Israeli-designed:** the C1 (iPhone 16e) and C1X (iPhone 17 series), along with the N1 wireless chip — per the OpenIntel Apple dossier, "Israel-designed silicon feeding directly into global Apple products."

OpenIntel's Apple audit: BRS 582, Tier C ("High") — Economic 8.00 (the acquisition stack plus Israeli-designed silicon in current products), Political 6.29 (explained below), Military 0.09 and Digital 0.20 (near-zero: no defense contracts, not in the UN OHCHR settlement database, not listed by PAX or Who Profits, and explicitly **not** a party to Project Nimbus). The political score comes from the **Benevity donation-matching controversy** — Apple matched employee donations to Friends of the IDF, the Jewish National Fund, One Israel Fund, HaYovel, and IsraelGives; 133–140 employees signed a June 2024 open

letter; Apple did not respond to press inquiries. Apple also disciplined employees over pro-Palestinian expression (a Chicago keffiyeh case; roughly 40 staff reprimanded over bracelets; both Muslim and Jewish employee-resource Slack channels suspended). And Apple's relationship with Israeli offensive cyber is adversarial, not cooperative: it sued NSO Group in November 2021, then moved to dismiss in September 2024 citing the risk of exposing vital security information.

4.3 Samsung — BRS 639, Tier B ("Severe")

- Two Israeli R&D centers since the **2007 acquisition of Transchip** — the first foreign semiconductor acquisition by a Korean conglomerate since 1997.
- The **Herzliya camera center** built the Galaxy S3's camera and, in the center's own description, all of the camera's processing technology — the wiping effects and special processing — done in Israel. Its technology is installed on hundreds of millions of smartphones.
- The **Ramat Gan semiconductor center** contributed directly to Exynos chips, including the Exynos 5 (32nm dual-core) and the transition to 14nm — "work on the chip was done both in Israel and South Korea."
- **SIRC Tel Aviv** (the Samsung Israel R&D Center) belongs to Samsung's flash-memory division and is responsible for the development of next-generation **SSD, UFS, and eMMC storage architecture** — meaning the flash interface standard inside essentially every Samsung phone (and sold to others) has Israeli design authorship at the platform level.

SIRC also develops image sensors.

- **Audit verdict: BRS 639, Tier B**, driven by the durable wholly-owned Israeli R&D footprint, the completed Transchip acquisition, and consumer-market scale. The audit explicitly preserves its caveat: unverified reporting about Cellcom device sales was *not* converted into proof of direct military contracting.

4.4 Google Tensor — quadruply tainted, and Google is building more in Israel

Tensor (Pixel chips) descends from Samsung semiconductor design, licenses ARM technology, and is developed inside a company whose Israeli operations keep expanding: in March 2021 Google hired ex-Intel corporate VP **Uri Frank** to build a server SoC design team in Israel (Frank had run a 2,000-person Intel design group spanning Israel, the US, and India); in September 2026 Google assembled a new edge-AI chip team in Israel led by former Hailo executive Guy Kaminitz, with ex-Pliops CEO Ido Bukspan leading data-networking chips. Separately, Google's security chip — the **Titan M2** — is designed and manufactured by Google in-house and runs on the open **RISC-V** instruction set (architecturally the one mainstream security chip not built on ARM IP); no public source attributes Titan authorship to Israel, but it sits inside Tensor silicon, so the enclave is clean while the host is not. Google is also, with Amazon, the contractor on **Project Nimbus** — the $1.2 billion Israeli government cloud program.

4.5 MediaTek — the cleanest audited semiconductor company in this investigation, full detail

OpenIntel's forensic dossier assigns **BRS 248, Tier D ("Moderate")** — the lowest score encountered across every company audited for either the laptop or the phone investigation. The complete ledger:

- **No Israeli office, R&D center, subsidiary, or state contract.** MediaTek's official office list spans the United States, China, Dubai, Finland, India, Japan, Korea, Singapore, Sweden, Taiwan, and the United Kingdom — no Israel entry. The audit *re-examined and rejected* a prior claim of a "MediaTek Semiconductor (Israel) Ltd." — it was an erroneous reading of an Ireland entry.
- **The venture portfolio (the asterisk):** at year-end 2024, via MTKC Global Holdings, MediaTek disclosed a **4 percent stake in NeuroBlade** (Tel Aviv AI chips; $5.539 million) and a **1 percent stake in proteanTecs** (Haifa; founded by ex-Mellanox executives; $3.846 million). Earlier disclosures: 1 percent each in **Valens** (Israeli HDBaseT — MediaTek also joined its 2017 round), **Innoviz** (Israeli lidar), and **VisIC** (Ness Ziona gallium-nitride power), plus exposure to **Viola Ventures VI** (an Israeli fund). In 2017 MediaTek co-led a round in **Corephotonics**, the Tel Aviv dual-camera startup whose optical-zoom technology defined the iPhone 7 Plus era and was licensed industry-wide.
- **The one defense-adjacent item:** in June 2020 MediaTek participated in the **$3.5 million seed round of OneView**, a Tel Aviv synthetic-data AI company, **alongside Rafael**

Development Corporation — the venture vehicle of the Rafael defense prime. OneView's founders hail from "elite intelligence units"; its site markets to defense and intelligence customers and still lists MediaTek as an investor. The audit's precision matters: this is **investment adjacency, not supply** — no verified MediaTek supply relationship with Rafael, the IDF, or the Israeli Ministry of Defense.

- **Distribution:** MediaTek products reach Israel only through third-party distributors (Rutronik's franchise covers Europe plus Israel; Mouser Israel; DigiKey Israel). No settlement-specific distribution was found.
- Post-2024 portfolio status is unresolved (2025 filings do not confirm continuation or exit) — preserved as an open item.

Bottom line: MediaTek silicon carries no Israeli design authorship and no Israeli corporate presence — only a documented venture-portfolio footprint, one item of which sits beside a defense-prime venture arm. For a buyer applying a "nothing Israeli inside the machine" standard, MediaTek is the answer; for a strict "zero relationships of any kind" standard, the venture portfolio is the residue.

4.6 Unisoc and HiSilicon Kirin

Unisoc (China): no Israeli office, investment, or partnership surfaced in any pass — its chips power Nokia's Unisoc feature phones, some Hisense models, and low-cost Androids worldwide. Caveat: unlike MediaTek, Unisoc has not been through an audit-grade dossier; the finding is "no evidence found," one tier weaker.

HiSilicon Kirin (Huawei): the Kirin silicon itself is designed in Shenzhen — no Israeli *silicon authorship* was found. But Huawei-the-corporation fails a zero-ties test at corporate level, through **Toga Networks** (full dossier in Part 13): Huawei acquired Toga Networks and HexaTier in 2016 (Toga reported at up to $150 million); Toga — founded 2009 in Hod Hasharon by Itzik Malobani — is Huawei's Israeli R&D arm with roughly 400–500 employees across Hod Hasharon and Haifa in 12–14 research groups (high-speed switching, cloud, storage — division closed 2023 — AI and data-center silicon, automotive, computer vision). The working model, per Globes: Israeli teams build architecture to proof-of-concept stage, "the accrued know-how is then transferred in its entirety to China"; Toga files 90–100 patents a year. Toga was conspicuously absent from the May 2019 Entity List and added in the August 2020 round — per former employees, it used the grace period to stockpile test equipment — and one export-license revocation pushed Huawei to move development to the UK. February 2026 brought layoffs of about 50 cloud staff. Huawei, like AMD in the laptop report, is "corporate-level ties, silicon clean" — with the added note that the corporate tie is a genuine R&D center, not merely an office.

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5. The Modem Layer — The Secret Second Computer, Fully Traced

The baseband modem is a second, independent computer inside every phone, running closed firmware in the most privileged position in the device. Its genealogy is now traceable end to end — and it contains one genuinely clean chapter:

- **2007–2010: Infineon (German).** The original iPhone's modems were German-designed — the only Israeli-clean chapter in iPhone modem history.
- **2010/11: Intel buys Infineon's wireless unit for $1.4 billion.** The Munich operation became Intel's modem hub.
- **2011–2016: Qualcomm's billion-dollar exclusivity.** Court testimony revealed Qualcomm *paid Apple* roughly $1 billion over three years to be the exclusive modem supplier, with clawbacks if Apple dual-sourced.
- **2016–2018: Intel's window.** iPhone 7 dual-sourced (Intel PMB9943); the iPhone XS/XR generation went all-Intel in 2018 because Qualcomm refused to support dual-sourcing (Apple COO Jeff Williams, sworn testimony).
- **April 2019: the settlement.** Apple paid approximately $4.5–4.7 billion and signed a six-year Qualcomm license; Intel exited 5G modems the same day (the unit was losing about $1 billion a year, with one customer).
- **July 2019: Apple buys Intel's modem unit — $1 billion, 2,200 employees, more than 17,000 wireless patents** — having already hired Qualcomm's decade-long modem chief and Intel's XMM 8160 lead.

- **2025–2026: the payoff.** The C1 (iPhone 16e) and C1X (iPhone 17 series) are both Israeli-designed per the OpenIntel Apple dossier. **The five-year, multi-billion-dollar effort to escape a tainted supplier ended in Herzliya.** Every iPhone modem ever shipped is now documented as Israeli-designed, one way or another.

Beyond Apple: Qualcomm's X-series modems carry the same Haifa/Hod Hasharon lineage; Samsung's Shannon modems inherit the Samsung/SIRC taint; and the only Israeli-clean modems found anywhere are Chinese — BroadMobi and Quectel (used in Linux phones) and the integrated modems in Unisoc and MediaTek designs. The strategic irony is total: the one component that talks to the cellular network — the component where a state-level actor would most want visibility — has no clean Western manufacturer.

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6. Memory, Storage, and Cameras

- **Samsung UFS/flash storage: tainted at the architecture level** via SIRC — the same Israeli center that designs SSD/UFS/eMMC platforms designs the storage inside Samsung phones.
- **Micron: not clean** (Israel R&D locations, an active Israel ASIC design team, and a former Kiryat Gat fab leased from Intel — full detail in the laptop report).
- **SK Hynix and Kioxia: clean** — Korean and Japanese design and manufacture, no Israeli footprint found.

- **Apple: every storage controller in every iPhone is Israeli-designed** (the Anobit line) — so even the NAND vendor matters less than the controller sitting in front of it.
- **Cameras:** Samsung's ISOCELL sensors and their processing are Herzliya-built (Transchip lineage, Galaxy S3 onward). **Corephotonics** — the Tel Aviv dual-camera startup in which Samsung Ventures, Foxconn, and MediaTek invested — supplied the optical-zoom architecture the whole industry licensed after the iPhone 7 Plus, meaning mainstream flagship zoom pipelines trace to Israeli IP regardless of brand. **Sony's correction:** standard Sony camera sensors remain Japan-designed and clean — but Sony Semiconductor Israel (formerly Altair Semiconductor, acquired for $212 million in 2016) developed the **AI digital signal processor inside Sony's IMX500/IMX501 — the world's first image sensor with on-sensor AI processing**; and on April 27, 2026, Sony spun Altair back out with $50 million led by Israeli VC Pitango at a reported ~$300 million valuation, **while retaining an undisclosed equity stake**. Sony therefore leaves the "clean corporate" column: retained stake, Israeli-authored AI layer in the intelligent-vision sensor line. **OmniVision** (US-founded, Chinese-owned) is clean of Israeli ties.

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7. The Security-Chip Layer

- **Google Titan M / Titan M2:** designed and manufactured by Google in-house (the company's own words), running on the open RISC-V instruction set; no public attribution of Titan authorship to Israel — but it lives on Tensor silicon, so the *enclave* is clean while the *host* is tainted.
- **Samsung Knox Vault:** a secure subsystem on Samsung's own SoCs — inherits the full Samsung taint.
- **Qualcomm SPU:** inside Snapdragon — tied by the same lineage.
- **MediaTek devices:** ARM TrustZone plus MediaTek in-house security IP — indirect ARM taint only.
- **Apple Secure Enclave:** provenance **unresolved** — no source was found in repeated attempts; recorded as an open item, not guessed.

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8. UWB, Satellite, and the Remaining Component Layers

- **Apple U1/U2 (ultra-wideband):** provenance unresolved (zero results) — noted, not guessed.
- **Android UWB:** Samsung's own or Qorvo (US, clean) / NXP-based — NXP carries its Yokne'am, Israel design-center tie.
- **Satellite:** iPhone Emergency SOS rides Globalstar (US, clean); competing Snapdragon Satellite is Qualcomm-tied.

- **Wi-Fi/Bluetooth:** in flagships, Qualcomm FastConnect (tied) or Apple-designed wireless (the N1 wireless chip is Israeli-designed per the OpenIntel Apple dossier); in MediaTek phones, MediaTek's own integrated Wi-Fi/BT (clean); Realtek (clean) in Linux phones.
- **NFC/eSIM secure elements:** NXP (Yokne'am tie) versus ST and Infineon (clean).
- **GNSS:** u-blox (Swiss, clean) versus Qualcomm/Broadcom (tied).
- **Audio, haptics, displays, batteries, assembly: clean** — the same finding as the laptop report.
- **Punkt MC02:** Dimensity 900 confirmed, but its RAM/flash vendors have never been identified in any teardown — open item (industry norms suggest Samsung or Micron; flagged as inference, not evidence).

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9. The Operating System Layer

- **iOS: maximal entanglement** — the Apple dossier applies verbatim (Israeli CPU cores, Israeli storage controllers, Israeli modem).
- **Android: heavy.** Google Israel's R&D (Waze lineage, Call Screen, search and AI features), Project Nimbus ($1.2 billion, Google and Amazon only — OpenIntel confirms Apple is not a party), and Google's new Israeli chip teams.
- **GrapheneOS — audited clean, with the hardware catch.** An OpenIntel digital-infrastructure audit found: no Israeli enterprise-security vendors anywhere in its stack; no Israeli

data-center locations; no Nimbus participation; no Israeli contracts, R&D, or investments; a Canadian non-profit. **But GrapheneOS runs exclusively on Google Pixel hardware** — so the cleanest audited phone *software* chain bottoms out in tainted silicon. This clean-software-on-tainted-hardware combination is the phone problem in miniature.

- **CalyxOS and /e/OS:** similar forks with similar device constraints; /e/OS supports a wider device list including some Samsung and Fairphone models — trading one taint for another.
- **KaiOS** (feature phones): a Hong Kong/San Diego company whose OS descends from Mozilla's Firefox OS (clean lineage), but whose app store preloads Google and Facebook apps; and its hardware mixes Qualcomm, MediaTek, and Unisoc by model — "KaiOS phone" is not an answer, the chipset is.

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10. The Offensive-Cyber Layer — Israeli Industry Built to Attack These Phones

Recorded here because it is the most direct form of "tie" between Israel and the devices in question, with framing stated precisely: this is not a supply-chain relationship — it is a state-licensed commercial ecosystem whose entire function is penetrating these devices. For a reader whose criterion is exposure to Israeli state-linked capability, it is the single most relevant finding; for a strict supply-chain reading, it is context. Both readings are presented; neither is adopted.

- **NSO Group / Pegasus:** founded 2010 (Niv Karmi, Omri Lavie, Shalev Hulio). Israel classifies Pegasus as a **weapon**; every export requires Israeli Ministry of Defense DECA approval. The New York Times documented Israel treating NSO as a "de facto arm of the state" — a diplomatic instrument: sales were cleared to advance the Abraham Accords, and **blocked to Estonia and Ukraine to protect relations with Russia** (Estonia's $30 million down payment was returned). Deployment requires only entering a phone number; setup takes about fifteen weeks; US court documents priced surveillance of 25 numbers in Ghana at **$8 million**. Israel required NSO to block US numbers from targeting; US intelligence officials reportedly presume Israeli state access to Pegasus-derived data (NSO denies any backdoor). The US placed NSO on the **Entity List in November 2021**; L3Harris acquisition talks in 2022 collapsed in part over Israel's insistence on retaining export-license control; Amnesty International's Israeli lawsuit to revoke the export license was rejected in July 2020.
- **Intellexa / Predator:** founder Tal Dilian served 24 years in Israeli military intelligence; an Intellexa founder was sentenced to **eight years in Greece** over Predator use. Notably, Predator's loader **self-terminates if the phone's locale is Israel or the number is +972** — it is not built to operate at home.
- **QuaDream / Reign:** founded by NSO alumni; its ENDOFDAYS exploit chain used the iCloud calendar for zero-click compromise; full chains exist for both iOS and

Android.

Why this layer cannot be dodged by hardware choice: Pegasus-class tools are delivered by phone *number*, not by account type or device brand — a prepaid burner defeats billing surveillance but not zero-click exploitation. Mitigation is behavioral and patch-discipline based, not purchasable.

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11. The Complete Brand Ledger — Every Phone Family on Sale, Examined

Approximately forty brand families across every category and price tier, grouped by the two governing axes: **corporate-level ties to Israel** and **silicon-level authorship**.

Quadrant I — No corporate tie found + clean silicon available (the achievable answer)

Brand	Ownership	Clean-silicon path	Grade
Honor	Shenzhen Zhixin (state-linked; 49.55% direct as of June 2025)	MediaTek X-series, 200 Lite; **Magic 8 Mini tipped with Dimensity 9500 — the first flagship-class clean-SoC path**	**Audit: BRS 192, Tier E — the cleanest entity in the investigation**
Transsion (Tecno/Infinix/itel)	Shenzhen; Shanghai STAR-listed	MediaTek throughout	No-evidence-found
Unihertz	Shanghai (2016)	MediaTek (Jelly 2E Helio A22; Jelly Max Dimensity 7300)	No-evidence-found
Mudita	Warsaw, Poland (founder: Michał Kiciński, CD Projekt co-founder)	MediaTek MT6761, de-Googled MuditaOS K, hardware kill switches, FCC/PTCRB certified	No-evidence-found
BLU	Miami (private; founded 2009)	MediaTek models	No-evidence-found + FTC/China flag (below)
HMD/Nokia	Finland	**Unisoc models**: 2660 Flip, 2760 Flip 4G (T107/T310); avoid 2760/2780 flips (Snapdragon 215)	No-evidence-found
TCL	China	MediaTek models	No-evidence-found
Punkt	Switzerland	Dimensity 900 (MC02)	No-evidence-found
PinePhone	PINE64 (Hong Kong)	Allwinner A64 — maximal cleanliness	No-evidence-found
Chinese rugged (Ulefone, Doogee, Oukitel, AGM)	Shenzhen	MediaTek Helio / Unisoc Tiger	No-evidence-found
Oppo/Vivo	BBK	Dimensity models in A/X lines	Consumer-guide grade

Quadrant II — No corporate tie found + tainted silicon

Oppo/Vivo/OnePlus/Realme/iQOO flagships (Snapdragon) · ZTE/Nubia/RedMagic (Nubia now 78.33% Shanghai Shenqi Jiyuan after ZTE's December 2025 transfer; gaming line Snapdragon throughout) · Meizu (79.09% owned by Geely founder Li Shufu's Xingji Shidai since July 2022) · ASUS Zenfone/ROG (Snapdragon; ASUS even co-built Qualcomm's own branded phone in 2021) · Nothing main line (Snapdragon; the Phone (4a) and CMF line run Dimensity) · Sharp (Foxconn-affiliated) · **Sony Xperia** (Snapdragon; corporate note: Sony's retained Altair stake and the IMX500 AI-DSP lineage) · Fairphone 5 (Qualcomm QCM6490) · Sonim and Kyocera rugged (every model Qualcomm) · Light Phone 3 (Qualcomm) · CAT (Bullitt Group, insolvent 2024).

Quadrant III — Corporate tie + clean silicon possible

Huawei — Toga Networks (~400–500 staff) plus HexaTier; Kirin silicon Shenzhen-designed. **Xiaomi** — **corrected**: the documented Israeli R&D center (Xiaoyi Multimedia Israel Ltd, Rehovot, 25 staff, face recognition, founded 2015) belongs to **Xiaoyi**, a Shanghai camera company and Xiaomi *ecosystem partner* — not to Xiaomi itself. Xiaomi's own direct footprint reduces to retail sales presence (Tel Aviv store since 2018); no Xiaomi-entity R&D center was found under its own name. The earlier "confirmed tie" was a listicle's shorthand collapsing two companies — exactly why evidence grades exist.

The Motorola distinction — the most important clarification in the brand ledger

Motorola Solutions is not Motorola Mobility, and the difference is the deepest corporate finding in this investigation. The two split in 2011. **Motorola Solutions** — the public-safety and radio company, not the consumer phone brand — is among the most deeply military-entangled Western technology firms operating in Israel:

- Operating in Israel **since 1964** — its first R&D subsidiary outside North America.
- **Sole supplier of the IDF's independent cellular network**: the Israeli military runs its own private 4G system, built and operated end-to-end by Motorola Solutions.
- **January 2014: a 15-year, $100 million Israeli Ministry of Defense contract** (half funded through US Foreign Military Sales) for encrypted Lex M20 military smartphones — Israeli-made secure handsets on MSI's network.
- The **Mountain Rose** encrypted network program (~$100 million, 2000–2004), and the **MotoEagle Wide Area Surveillance System** — NIS 400 million since 2005, installed in more than 25 settlements, feeding the IDF's monitoring of Palestinian movement.
- Israel Police's **Nitzan** tactical communications system; settlement contracts (Ariel, Jordan Valley, Mateh Binyamin); **listed in the UN OHCHR settlement database in both 2020 and 2023**; reportedly lobbying the IDF on a ~NIS 1.5 billion 5G upgrade, and in 2021 pressuring the IMOD to withdraw a 5G tender request for information. Its Israeli

Innovation Center (opened 2016) is headed by Hemi Pecker, a former prime minister's office cyber chief.

Motorola Mobility — the actual consumer phone brand, owned by Lenovo since 2014 — carries *Lenovo's* ledger (the Infinidat acquisition completed April 2026; SentinelOne's Singularity embedded in ThinkShield on new Lenovo PC shipments; the Ben-Gurion University cybersecurity center inside the CyberSpark ecosystem that co-locates Unit 8200; the Phoenix Technologies BIOS-unit acquisition closed April 26, 2026) but **none** of Motorola Solutions' military record. When a boycott guide or a news story says "Motorola," the distinction determines everything — one company makes prepaid phones; the other runs the occupation's communications backbone.

Quadrant IV — Corporate tie + tainted silicon (maximal entanglement)

Apple (BRS 582) · **Samsung** (BRS 639) · **Google Pixel** (Tensor; Israeli chip teams; Nimbus) · **Lenovo/Motorola Mobility** (Infinidat acquisition completed April 2026; SentinelOne's Singularity embedded in ThinkShield on new Lenovo PC shipments; the Ben-Gurion University Cybersecurity Innovation Center inside the CyberSpark ecosystem that co-locates Unit 8200; Lenovo's acquisition of Phoenix Technologies' BIOS unit closed April 26, 2026; Snapdragon or MediaTek silicon).

Defunct and absent

BlackBerry (hardware dead: TCL license ended, OnwardMobility collapsed February 2022 after the license was canceled, patents sold for a reported ~$600 million) · LG (exited phones 2021) · HTC (dormant) · Essential (assets to Nothing) · Vertu (niche luxury, unverified, immaterial volume) · Energizer/Philips licensees (ODM-built, immaterial). **Unverified minors, stated honestly:** Doro (Swedish, acquired by Norway's Xplora in autumn 2024 — no ties found), Lively/Jitterbug (US — GreatCall was co-founded by Martin Cooper, the inventor of the cell phone; sold to Best Buy for $800 million in 2018; rides Verizon), Swissvoice (Swiss), Emporia (Austrian), Snapfon and RAZ (US), Consumer Cellular (US) — none with Israeli corporate ties found.

The Honor dossier, in full

The only Tier E company in the investigation. **BRS 192; Military 0.00, Economic 0.00, Political 0.00; Digital 2.30 driven solely by the Honor Health app being listed as available in Israel.** The audit's complete negative ledger: no Israeli subsidiary, office, warehouse, importer of record, authorized distributor, R&D center, data center, investment, or settlement operation; Israel absent from Honor's country selector; Statcounter records **0% Israeli market share** (grey-market listings only); the only named foreign subsidiary is in Mexico. Entity hygiene the audit explicitly performed: Huawei's Israeli assets (Toga, HexaTier) are *not* attributed to Honor after the November 2020 divestiture, and an unrelated "Honor Construction and Investment Company Ltd." in Ramat Gan was identified and excluded. The tradeoff stated plainly: Honor's ownership is Chinese state-linked (upstream Shenzhen SASAC; China Mobile, China Telecom, BOE among investors), and Reporters Without Borders separately flags Chinese-device data-governance risk. A zero-Israeli-ties phone that is transparent to a *different* state's data-compulsion regime — the criterion is satisfied; the trust calculus inverts. And the chipset caveat: **every Magic-series flagship and premium Honor ships Snapdragon** — the clean-SoC choices are the X-series, the 200 Lite, and (tipped) the Magic 8 Mini with Dimensity 9500. The same model name can even carry different silicon by market (the Honor 600: Dimensity in China, Snapdragon globally).

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12. The "Clean Phone" Teardowns — Honest Results

- **PINE64 PinePhone:** Allwinner A64 SoC (Chinese), Realtek Wi-Fi, OmniVision camera (clean), Quectel EG25-G modem (Chinese — one of the only clean modems in existence), Goodix touch controller, hardware kill switches for modem/mic/camera. **No Israeli-authored component was found anywhere in the build** — the single cleanest phone in the investigation. Performance is entry-level at best; the ecosystem is hobbyist; and US carriers barely tolerate it (Part 16).
- **Purism Librem 5:** NXP i.MX8M SoC — NXP has an Israeli design center in Yokne'am — **Micron RAM** (tainted), Samsung rear camera (tainted), BroadMobi modem (clean), Kioxia storage (clean). The privacy community's flagship fails the criterion at three layers.
- **Fairphone 5:** Qualcomm QCM6490 — the "ethical phone" shares the Android flagship's core taint.
- **Punkt MC02** ($749, Swiss; Apostrophy OS built on GrapheneOS): Dimensity 900 — the minimalist-privacy flagship runs the investigation's cleanest-audited SoC. RAM/flash vendors undocumented.
- **Mudita Kompakt** ($439–519, Warsaw): MediaTek MT6761 on a de-Googled AOSP-based OS with hardware kill switches; FCC/PTCRB certified and tested working on AT&T, T-Mobile, and Verizon. A genuinely new clean option with proper US carrier certification — the strongest European entry.
- **Light Phone 3:** Qualcomm inside — tainted.

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13. Prepaid, Part One — The Shelf, the Flips, and the Corporate Lineages

Prepaid solves a *billing and identity* problem — no contract, no credit check, often no name at point of sale — but it changes neither the hardware criterion nor the network criterion. What it does offer is a shelf of cheap hardware from which genuinely cleaner silicon can be selected, and a set of corporate lineages worth mapping.

The shelf. Current US prepaid inventory (Tracfone, Straight Talk, carrier prepaid at Walmart and elsewhere) resolves into four brands: BLU, TCL, Motorola, and Nokia. BLU is a private American company (founded 2009, Doral/Miami, by Samuel Ohev-Zion; 70 million-plus devices sold; focus on the US, Latin America, and the Caribbean; "designed in Miami" branding over Chinese manufacturing) — with one caveat that belongs in the record: the **2018 FTC settlement** over China-based ADUPS firmware that collected personal data without disclosure, binding BLU to a 20-year third-party security-monitoring program. BLU is clean on the Israeli criterion and flagged on a *different* state's data practices — the same inversion pattern as Honor and Transsion. TCL and Motorola follow the ledger above; Nokia's Unisoc models are the clean flips.

The flip-phone chipset split, model by model — the cheapest prepaid tier hides its silicon on the spec sheet, so it is documented here: **Nokia 2760 Flip and 2780 Flip: Qualcomm Snapdragon 215 — tainted.** Nokia 2660 Flip: Unisoc T107 —

clean. New HMD 2760 Flip 4G: Unisoc T310 — clean. CAT S22 Flip (T-Mobile): Snapdragon 215 — tainted. Alcatel Go Flip: Qualcomm — tainted. The senior-phone channel follows the same split: Europe's Doro (Swedish), Swissvoice (Swiss), and Emporia (Austrian) carry no Israeli corporate ties, while every US rugged handset (Sonim's XP8/Xp10/XP Pro/XP5plus; Kyocera's DuraForce) is 100 percent Qualcomm.

Tracfone — the complete corporate lineage. Founded 1996 in Miami as Topp Telecom by David Topp and F.J. Pollak — the company that effectively invented the US prepaid phone model. February 1999: Telmex invested $57.5 million for a 55% controlling stake; in 2000 Telmex spun off its mobile unit as América Móvil (Carlos Slim's vehicle). Growth via acquisition: Straight Talk (Walmart partnership, 2009), Simple Mobile (2012), Page Plus (2013) — a family reaching roughly 20 million subscribers through 90,000 retail locations. September 14, 2020: Verizon announced acquisition for approximately $6.25 billion in cash and stock (headline value up to $6.9 billion); FCC approval November 22, 2021 carried demanding conditions (Lifeline participation for 7–20 years, customer migration to the Verizon network, device unlocking); the deal closed November 23, 2021 ($3,125 million in cash plus 57.6 million Verizon shares, with up to $650 million more). Every Tracfone-family prepaid SIM now terminates in Verizon — venture-level Israeli ties included.

The full MVNO ownership map. Mint Mobile (founded 2015 by David Glickman and Rizwan Kassim; Ryan Reynolds took 25% in 2019) was acquired by **T-Mobile US in May 2024 for up to $1.35 billion — parent company Ka'ena Corporation (Mint,**

Ultra Mobile, Plum) went with it — Mint is now wholly inside T-Mobile, which is ~51.4% owned by Deutsche Telekom. Tello is an independent US MVNO whose parent KeepCalling was founded by Romanian entrepreneur Florin Miron in 2002 — **the only independent, no-ties MVNO identified**. Boost rides AT&T/T-Mobile under EchoStar (no ties found at EchoStar's own level). Cricket belongs to AT&T; Metro to T-Mobile; Visible to Verizon; Consumer Cellular is a US company with no ties found at its own level. **Net finding: independent MVNO corporate layers surfaced zero Israeli ties of their own — every tie in the US prepaid stack enters through the underlying network owner or the billing substrate below.**

The refill-card layer. The physical prepaid refill cards sold at Walmart, Target, and grocery registers are distributed through Blackhawk Network and InComm — both American companies, with no Israeli ties found at the distribution layer. The card, like the SIM inside the phone's tray, is one of the few genuinely clean artifacts in the entire stack.

The kosher-phone channel. A US distribution layer most audits never examine: TAG Israel (a community filtering operation run from Jerusalem, installing content filters on smartphones and the Nokia 4G feature-phone line), TAG US (Lakewood, NJ), and GenTech (Brooklyn/Lakewood, 20-plus years of filtered-device sales "for local and Israeli use"). The hardware is ordinary; the Israeli connection is the *service layer* — installation, filter policy, and remote administration anchored in Israeli-based organizations. A distinct channel-layer tie, recorded separately from the handset ledger.

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14. SIM and eSIM — Genuinely Clean, With Two Caveats

- **Physical SIM cards:** the global supply is dominated by Thales (Gemalto), IDEMIA, G+D (Giesecke+Devrient), Kona I (Korea), and Chinese makers (Datang, Wuhan Tianyu, Watchdata, Eastcompeace) — French, German, Dutch, Korean, Chinese. **No Israeli SIM manufacturer exists or was found anywhere in the supply chain.** Your SIM card is not Israeli.
- **The secure chips inside cards and eSIMs** come from STMicroelectronics (clean), Infineon (German, clean — the same corporate family that gave the iPhone its only clean modem chapter), NXP (**tainted** — its Yokne'am design center), and Samsung (tainted via SIRC).
- **eUICC software** (the SIM operating system inside eSIMs) is written by Thales, G+D, IDEMIA, and **Kigen** — a UK company but an **Arm spinoff**, inheriting ARM's Ra'anana taint indirectly. **Practical takeaway: prefer a physical SIM.**
- **iSIM** — the SIM moving inside the main SoC — is the industry direction, which folds this layer back into the silicon layer (Qualcomm/MediaTek/Apple). Note it as a trend, not a current constraint.

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15. The Carrier Layer — All Three US Networks Are Tied; AT&T Is the Deepest

- **AT&T:** Israeli R&D since the 2007 Interwise acquisition; **~500 employees across two centers** (Tel Aviv's ToHA Tower and Airport City) as of the most recent reporting; scope includes **5G network software, cloud transformation, and "advanced products for managing first responder systems worldwide" — FirstNet, the US national public-safety network**. The center's own description: it "leads strategic initiatives for the entire company." An AT&T-based prepaid plan means calls traversing network software with Israeli authorship.
- **Verizon:** Verizon Ventures' Tel Aviv office (2017–March 2023) invested in **13 Israeli startups** (Cellwize — later acquired by Qualcomm; Vdoo — to JFrog; Iguazio — to McKinsey); the office closed, existing investments continue. Verizon's *network* itself shows no Israeli R&D center — lighter than AT&T by a full grade.
- **T-Mobile / Deutsche Telekom:** T-Venture has invested in Israeli venture funds since at least 2002 (per Deutsche Telekom's own SEC filings); DT's M&A records list an investment in Israeli software company Teridion; T-Mobile Ventures launched in 2020. **T-Mobile-based prepaid is the lightest-tied network option.**
- **MVNO mapping recap:** Tracfone family (Tracfone, Straight Talk, Total, Walmart Family Mobile) → Verizon only since the 2022 migration; Cricket → AT&T; Metro → T-Mobile; Boost → AT&T/T-Mobile; Mint, US Mobile (white SIM), Tello, Simple

Mobile → T-Mobile.

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16. The VoLTE Whitelist — Why the Cleanest Phone Barely Attaches to US Networks

Since the 3G shutdowns, US carriers authenticate voice service through VoLTE whitelists — carrier-maintained lists of approved device models — and this wall hits exactly the phones this investigation rates cleanest. AT&T moved to an explicit whitelist and excluded the PinePhone from March 2021; T-Mobile maintains a "compatible device" list that also excludes uncertified hardware, though workable service is reported on T-Mobile and its MVNOs, sometimes via the documented workaround of activating with an approved IMEI and moving the SIM. Purism now ships the Librem 5 with VoLTE enabled, but community threads document the same reality: hardware support plus **carrier certification** is the gate. The whitelist is a *usability* wall, not an Israeli tie — but it is the binding practical constraint on how far the clean-hardware answer can go in the American market.

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17. The Back Office — Amdocs, Comverse, and a Century of Israeli Carrier Software

- **Amdocs:** founded **1982 in Israel** as an offshoot of the Golden Pages directory business; headquarters now in Missouri; NASDAQ: DOX; record **$5 billion revenue in 2024**; billing and CRM for **~90% of the world's top carriers; 1.7 billion customer interactions daily**; roughly **25% of total revenue from AT&T alone**; customers include AT&T, Verizon, T-Mobile US, Comcast, Altice, Deutsche Telekom, and Vodafone; the Verizon relationship dates to a 2003 contract reported at $700 million to $1 billion over five years. **There is no US carrier — and no US prepaid brand — whose customer-facing back office is documented as Amdocs-free.** Switching costs are described by analysts as near-prohibitive; the layer is stable, not trending away. Every refill-card purchase, plan activation, and balance check runs on Israeli-founded software with Israeli R&D.
- **Comverse/Verint — the historical layer.** Founded 1982–84 by three Israelis (Kobi Alexander, Boaz Misholi, Yechiam Yemini) as Efrat Future Technologies, R&D in Israel, funded partly by Israeli government grants. Products: voicemail, SMS centers, and billing — the 1997 Boston Technology acquisition ($843 million in stock) made it the voice-messaging supplier to 12 of the world's top 20 carriers, with Deutsche Telekom among its earliest big customers; its Trilogue systems served prepaid markets in developing countries — functionally the first great prepaid-billing vendor. Its interception-recording product line became **Verint** (2002). A 2001 Fox News

investigation alleged Comverse/Amdocs wiretap access — **recorded only as a disputed, never-substantiated claim**; it was never proven and is flagged, not adopted. Comverse dissolved in 2012–13; its telecom business became Xura (2015) then **Mavenir** (2017) — so today's US open-RAN champion carries Comverse's Israeli engineering lineage — and part of Comverse went to Amdocs in 2015. The interception line lives on directly: Verint today sells lawful-intercept and analytics systems into carrier and government markets worldwide — the corporate descendant of Comverse's 1990s interception products, still serving the same function a generation later.

- **Cellebrite — the point of sale.** The machine the store employee uses to move your contacts, photos, and messages to a new prepaid phone is Israeli. Cellebrite (Petah Tikva, founded December 1999) began with the UME phone-to-phone transfer device sold to telecoms and phone stores; by 2005 the company said it was in **more than half of all Verizon and T-Mobile stores in the US, plus Best Buy and Walmart phone departments**; a Verizon-premium-retailer (Russell Cellular) was still announcing Cellebrite rollouts in 2016. The same company is, per The Intercept, "the FBI's go-to hackers" — its UFED forensics line extracts data from seized phones for law enforcement worldwide. The consumer transfer box and the police forensic box are one company's two product lines. **Mitigation is trivial: decline the in-store transfer and move data yourself.**

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18. The Backup-Cloud Layer — Samsung and Microsoft, Ending This Week

Samsung's 2019 Microsoft partnership made OneDrive the built-in default backup for Galaxy photos; Samsung Cloud's own sync shut down in 2021, herding users to Microsoft servers; and **September 30, 2026 — the week of this document's compilation — Samsung Gallery's OneDrive sync ended**, with Samsung moving to its own cloud (existing files remain on Microsoft servers; OneDrive and Microsoft 365 app preloads continue). For five years, every Samsung phone's camera roll defaulted to **Microsoft Azure — BRS 711 in the laptop investigation, the Unit 8200 cloud**. That default is now retired.

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19. The App and SDK Strata — Israeli Code Reaches the Phone Through Three Doors

Door one — the apps themselves. Meta's Tel Aviv office is a documented R&D center for "Facebook and Instagram's core advertising and [AI]" systems — and Meta's apps ship preinstalled or near-mandatory on virtually every prepaid Android. Spotify runs a Tel Aviv R&D office of roughly 300 staff working on recommendation and AI. Waze is Israeli-born and Google-owned. Via — the transit/ride platform used by cities worldwide — is Israeli-founded, with two wholly-owned Israeli subsidiaries and 250-plus Israel staff, per its own SEC filing. (Uber's own direct Israeli footprint was not confirmed; search results conflated it with Via, and the distinction is preserved rather than collapsed.)

Door two — the SDKs embedded inside other people's apps. This is the deepest software finding of the investigation. **AppsFlyer — Israeli-founded (2011; CEO Oren Kaniel; headquarters effectively "San Francisco, Israel"), ~$500 million ARR as of January 2026, profitable since 2024 — is the dominant mobile-attribution SDK: per its own industry data it leads the Android market with 48.51% of all installs** (third-party trackers put its share near 44%, ahead of Germany's Adjust at ~27% and the US Branch at ~22%). Its Series C was led by Israeli VCs Qumra and Pitango alongside Goldman Sachs and Deutsche Telekom Capital Partners. The average Android phone — including every clean Honor, Transsion, or BLU device — carries Israeli-authored tracking code inside its ordinary apps, reporting install behavior and in-app events. There is no

consumer setting that removes an SDK from an app someone else wrote; the only mitigations are app selection (F-Droid's free-software catalog largely lacks it) and network-level blocking of its endpoints (imperfect). The same layer includes **ironSource** (Israeli, merged into Unity in a deal reported around $4.4 billion) — the monetization stack inside a large share of free mobile games (reported-deal grade, flagged for filing-grade confirmation).

Door three — the games. Israeli companies author a remarkable share of the world's top-grossing mobile games: **Playtika** (Herzliya, 2010; NASDAQ: PLTK; Slotomania, Bingo Blitz, June's Journey, Best Fiends), **Moon Active** (Tel Aviv; Coin Master with $6 billion-plus lifetime revenue and a #9 global revenue rank in H1 2024; a 22,000-square-meter expansion signed June 2026), **Plarium** (RAID: Shadow Legends, roughly $1 billion grossed), plus CrazyLabs, TabTale, SciPlay, and Playstudios — a sector generating about $9 billion, with 74% of its mobile-game workforce in the social/casual genres that dominate app-store charts. Game content is a consumption choice, unlike the SDK layer — a user who never installs Coin Master is untouched by Moon Active — but the two together complete the picture: Israeli-authored code, analytics, and content reach a typical phone through three independent doors.

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20. Israel's Domestic Market — the "Domestic" Half of the Criterion

- **Operators and prepaid:** Cellcom (whose prepaid brand "TalkMan" became the generic Hebrew word for prepaid card), Partner, Pelephone (Israel's first operator, 1986), Hot Mobile, and Golan Telecom (merged into Cellcom in 2020 for roughly $216 million); MVNOs include Rami Levy, 012, Cellact, and Home Cellular. Tourist prepaid is passport-based; eSIM rollout is partial (Cellcom consumer eSIM from about 2022; Partner and Pelephone largely business-only).
- **5G (launched September 29, 2020):** licenses to Partner+Hot Mobile (joint) and Pelephone.
- **Network equipment vendors: Cellcom runs Nokia** for LTE (2014) and 5G edge slicing (a world-first live deployment in Netanya, 2022), plus a cloud-native 5G SA core — and conducted an Open RAN pilot with **Parallel Wireless**, a US-registered but Israeli-founded and Israeli-engineered vendor. **Pelephone's 5G was built with Ericsson** (an "independent 5G core"). The IDF's network is Motorola Solutions' exclusive domain.
- **Huawei's status in Israel: unresolved** — searches surfaced only UK/US ban material; no primary source confirming or denying an Israeli Huawei exclusion was found. Recorded as an open item rather than asserted from memory.

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21. The Minimum-Ties Phone — the Practical Result

Maximum mainstream cleanliness

Layer	Choice	Residual taint
Handset	**Honor MediaTek model** (X-series, 200 Lite; Magic 8 Mini when it ships)	MediaTek Tier-D venture portfolio (incl. OneView/RDC adjacency); ARM IP; EDA tooling
Memory/storage	SK Hynix / Kioxia (teardown-dependent)	None found
Camera	Sony or OmniVision sensor	None found (avoid Samsung ISOCELL; Corephotonics zoom IP is licensed industry-wide)
Fingerprint	Goodix (avoid Qualcomm 3D Sonic)	None found
OS	Stock MagicOS with Meta apps uninstalled; FOSS apps from F-Droid where possible	GrapheneOS unavailable (Pixel-only); AppsFlyer-class SDKs in third-party apps
SIM	**Physical SIM** on Tello or T-Mobile-based prepaid	Card itself clean; network = T-Mobile/DT venture-level ties
eSIM	Declined	Avoids NXP chip + Kigen/Arm paths
Activation	Online self-activation; no in-store transfer	Avoids Cellebrite
Backup	Self-hosted or local	Avoids cloud substrate

Irreducible floor, stated plainly: MediaTek's venture portfolio; ARM and Synopsys/Cadence tooling; T-Mobile/DT venture-level network ties; the **Amdocs billing substrate (unavoidable at any US carrier)**; AppsFlyer-class SDKs inside third-party apps; the ~1,400-veteran people layer; and the NSO/Intellexa/QuaDream offensive-cyber ecosystem (number-based, undodgeable by any purchase).

Maximum literal cleanliness

PinePhone + T-Mobile-based prepaid with the IMEI workaround — no Israeli-authored component found in the build, at hobbyist performance, against the VoLTE certification wall. The Mudita Kompakt is the certified, warranty-backed middle path: European company, MediaTek silicon, de-Googled OS, kill switches, works on all three US networks.

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22. Methodology and Evidence Grading

How the investigation was conducted. Six sequential research passes: breadth mapping; targeted gap-closure; primary-source deep reading (full articles and audit dossiers rather than snippets); the prepaid/carrier/SIM sweep; the complete brand sweep (~40 families); and the app/SDK strata. Sources include forensic corporate audits (OpenIntel's BDS-1000 dossiers for Honor, MediaTek, Apple, Samsung, GrapheneOS — cited with their methodology *and* their exculpatory findings), SEC filings (Mobileye, AMD, Via), company press releases and spec sheets (Lenovo/Infinidat, Honor's own device comparison tables), named-executive interviews (Srouji/Ynet), Israeli business press (Calcalist, Globes, CTech), quality international press (NYT, The Intercept, Light Reading, FDI Intelligence), FCC records (the Tracfone approval), court testimony (the Apple–Qualcomm litigation), and community/technical documentation (VoLTE whitelists, PinePhone wiki, device spec databases).

Evidence grades used throughout:

- **Audit-grade:** documented in a forensic dossier with explicit "no public evidence identified" statements.
- **Filing-grade:** corporate disclosure (SEC, subsidiary schedules).
- **On-record:** named executive or company statement.
- **Reported:** quality press, single or convergent sourcing.
- **Unresolved:** flagged where evidence is thin, dated, or contested.

Caveats the record requires:

- "No public evidence identified" records the result of specified public-source checks; it does not prove non-public activity impossible. It applies symmetrically — to MediaTek's lack of military contracts and to Apple's lack of them alike.
- The BDS-1000 framework is one human-vetted scoring system, reproduced here as a summary of *documented activity*, not as an endorsement of its politics; the underlying facts stand independently of the scores.
- Where the user's criterion is "zero ties in any way," even venture-portfolio adjacency (MediaTek's OneView/RDC round) and distributor channels count; where it is "nothing Israeli inside the machine," MediaTek silicon qualifies as clean. Both readings are maintained side by side throughout.
- Personal details about private individuals (a founder's background, an executive's faith) are treated as irrelevant to the corporate criterion and excluded; companies are what they do.

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23. What Remains Unresolved — the Complete List

1. **Apple Secure Enclave provenance** — no sources found in repeated attempts.
2. **Apple U1/U2 ultra-wideband provenance** — no sources found.
3. **Punkt MC02 RAM/flash vendors** — no teardown exists; industry norms suggest Samsung or Micron (inference, flagged).
4. **Ericsson's Israeli R&D presence** — two dedicated search rounds produced nothing either way; its network *operations* in Israel are documented.
5. **Israel's Huawei 5G stance** — searches surfaced UK/US material only.
6. **MediaTek's post-2024 venture portfolio** — 2025 filings do not confirm continuation or exit.
7. **Xiaomi's own direct Israel footprint** — now actively unresolved: the documented R&D center belongs to Xiaoyi, not Xiaomi; only retail sales presence is confirmed.
8. **Uber's direct Israeli operations** — conflated with Via in search results; recorded separately.
9. **Any US MVNO running non-Amdocs billing** — none found; per-carrier confirmation is impractical.
10. **Honor's Magic 8 Mini chipset** — tipped (Dimensity 9500), not yet shipping-confirmed at compilation time.

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24. Key Figures at a Glance

Entity	Score / grade	Governing fact
Honor	**BRS 192 — Tier E**	Cleanest entity audited; only "tie" is an app's Israel availability listing; state-linked Chinese ownership
MediaTek	BRS 248 — Tier D	No office/subsidiary/contracts; venture portfolio incl. OneView/RDC adjacency; distributor channels
Apple	BRS 582 — Tier C	Israeli CPU cores; C1X/N1; storage controllers; Q.ai ~$1.5–2B (Jan 2026); Benevity controversy; near-zero military/digital
Samsung	BRS 639 — Tier B	SIRC flash/UFS + Herzliya cameras + Transchip; Exynos contributions
Qualcomm	(silicon-level taint)	Haifa/Hod Hasharon since 1993; Autotalks 2023; $1B Apple exclusivity era
Microsoft (laptop ref)	BRS 711 — Tier B	Azure/Unit 8200 record; $133M IMOD agreement
Intel (laptop ref)	BRS 712 — Tier B	Kiryat Gat; Mobileye 96.9% voting control
Dell (laptop ref)	BRS 691 — Tier B	Israeli M&A >$1B; $150M+ IMOD tender
Huawei	No score; corporate tie confirmed	Toga Networks ~400–500 staff; HexaTier; Kirin silicon clean
GrapheneOS	Audit-clean (software)	Pixel-only hardware binds it to Tensor
AppsFlyer	Not scored; market leader	48.51% of Android installs; $500M ARR; Israeli-founded
NSO Group	Entity-listed (US, Nov 2021)	Pegasus as state weapon; DECA licensing; $8M/25 numbers

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25. Glossary

- **SoC (System on Chip):** one chip fusing processor, graphics, camera processing, security, and modem interface — the reason a phone cannot mix-and-match parts the way a laptop can.
- **Baseband / modem:** the phone's second, secret computer — its own closed operating system, the most privileged attack surface in mobile hardware.
- **ISP:** image signal processor — the silicon that turns raw sensor data into photos.
- **UFS / eMMC:** the storage interface standards inside phones; Samsung's Israeli center architects them.
- **SDK:** software development kit — code an app developer embeds; the AppsFlyer attribution SDK sits inside roughly half of Android apps.
- **Attribution:** the analytics discipline of determining which ad caused an install.
- **SEP:** standard-essential patent — a patent a standard (like 5G) cannot be practiced without.
- **MVNO:** mobile virtual network operator — a carrier that owns no towers and rides a major network (Mint, Tello, Cricket).
- **VoLTE whitelist:** a carrier's approved-device list for voice-over-LTE; uncertified phones cannot make calls on US networks.
- **eSIM / eUICC / iSIM:** software SIM (eSIM) and its operating system (eUICC); iSIM moves the SIM into the main processor.

- **CALEA:** the US law requiring carriers to build lawful-intercept capability into their networks.
- **BSS/OSS:** business and operations support systems — billing, CRM, provisioning: Amdocs' domain.
- **Open RAN:** a movement to open cellular network hardware interfaces; vendors include Mavenir (Comverse lineage) and Parallel Wireless (Israeli-founded).
- **Tier E/D/C/B:** OpenIntel's complicity bands — E (Minimal/None) through B (Severe); BRS is the 1,000-point score.
- **Unit 8200:** the IDF's signals-intelligence unit and the industry's talent pipeline.
- **IMOD:** Israel Ministry of Defense. **DECA:** its export-control arm, which licenses Pegasus exports.
- **Entity List:** the US Commerce Department blacklist (NSO added November 2021; Toga added August 2020).

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26. The Conclusion, in One Paragraph

A cell phone with zero ties to Israel cannot be bought — not because of one component but because Israeli engineering sits inside every flagship silicon architecture, every Western modem, the tools that design all chips, the billing software that runs nearly every carrier, the dominant advertising SDK inside half of all apps, and an export-licensed offensive-cyber industry built to attack the devices themselves. What a buyer *can* do is now mapped to the last layer: an Honor phone with a MediaTek chip — the only Tier E company and the cleanest silicon audited — on a T-Mobile-based prepaid plan with a physical SIM, activated online, with Meta apps uninstalled and F-Droid apps preferred; or, for maximal literal cleanliness, a PinePhone with all the compromises that entails, or the certified European middle path of the Mudita Kompakt. Every choice above that floor leaves a documented residue — a venture portfolio, a distributor channel, a billing substrate, an SDK in someone else's app, a people layer no purchase can reach — and this document's purpose is that none of those residues is hidden from the buyer who wants to know. Every layer was audited, every brand was searched, nothing was adopted from anyone's theory, and everything unresolved is marked unresolved.

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Compiled 2026-09-20. Six research passes; primary sources opened in full; all monetary figures as reported by cited sources, dated figures labeled; "no public evidence identified" statements preserved verbatim from the underlying audits.

Part Five: The Zero-Ties App: A Complete Audit of Israeli Entanglement in the Cell Phone App Ecosystem

> *A fully-sourced investigation into the apps on a cell phone — messaging, payments, productivity, AI, VPNs, creative tools, games, and the invisible advertising infrastructure beneath them — and which can pass a zero-ties-to-Israel criterion, as of September 2026.*

1. The Question

The device investigations established that no phone and no laptop can be bought with zero ties to Israel. But a phone is also a container of apps — and apps are different from hardware in one crucial way: **they are individually choosable.** You cannot choose your Snapdragon chip away, but you can delete an app and install a different one in ninety seconds.

So the question sharpens: **among the apps a phone user actually touches — messaging, social, payments, maps, music, work, security, AI — which are entangled with Israel, at what depth, and what are the clean substitutes?** And beneath the apps themselves lies a second question most users never think to ask: what invisible code is embedded *inside* the apps other people wrote?

How to read this document

- **Part 2** gives the short answer.
- **Part 3** explains the three axes of an app audit — the framework everything below uses.

- **Parts 4–11** walk the app categories one by one: messaging and social, payments and finance, productivity, AI, VPN and security, navigation and travel, creative tools and education, and streaming.
- **Part 12** is the invisible infrastructure layer — the advertising and analytics code embedded inside apps, where three of the finding pillars live.
- **Part 13** is the consolidated Israeli-app catalog.
- **Part 14** is the clean column — the substitution table.
- **Parts 15–18** cover method, open questions, key figures, and a glossary.
- **Part 19** is the conclusion.

The framing note from both previous reports applies unchanged: this document argues no political position. The user's criterion is a supply-chain standard — zero ties to Israel — and the market is measured against it. Where involvement is found it is documented with sources; where it is not found, that is documented too; where evidence is thin, the ambiguity is preserved.

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A thirty-second orientation for readers new to how apps are built

When you tap an app, you are running code written by that app's company — but rarely *only* by that company. Modern apps are assembled like a meal from ingredients: the developer writes the core, then bolts in **SDKs** (software development kits) — pre-built code for analytics, advertising, crash reporting, payments, maps. A single weather app can contain a dozen companies' code, most of them invisible to you. This matters because it means an app audit has three separate questions: **who made the app** (the company axis), **what is bolted inside it** (the infrastructure axis), and **what it shows you** (the content axis). A clean company can ship an app containing an entangled SDK; an entangled company can stream clean content. All three axes are audited separately below.

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2. The Short Answer

Most of the app categories a phone user touches daily are led — at company level, infrastructure level, or both — by Israeli-authored software; the clean substitutes exist but are consistently second-tier in market share. The deepest findings, in plain terms:

1. **Some of the apps themselves are Israeli companies.** Viber (~a billion users, built from Tel Aviv), eToro (an Israeli-headquartered brokerage, NASDAQ-listed), the Lightricks editing suite (Facetune, Photoleap, Videoleap — built in Jerusalem), the Waze/Via navigation pair, and a

remarkable share of the world's top-grossing mobile games (Playtika, Moon Active's Coin Master, Plarium's RAID).

2. **The "privacy" category is owned by the wrong people.** ExpressVPN, CyberGhost, Private Internet Access, and ZenMate — four of the most recommended VPN apps on Earth — are all owned by **Kape Technologies**, a British-Israeli company wholly owned by Israeli billionaire Teddy Sagi, founded by a Unit 8200 veteran, whose review-site arm (vpnMentor, Wizcase) recommends its own products. The tools people download to escape tracking are, at ownership level, tracked to the same place as everything else.
3. **The advertising infrastructure underneath everything is a three-pillar Israeli layer. AppsFlyer** (Israeli-founded; its attribution SDK sits inside roughly half of all Android apps), **ironSource** (Israeli, merged into Unity — the monetization stack inside free games), and **Taboola/Outbrain** (both Israeli-founded; their merger, completed in 2024, left one company running recommendation modules across 20,000 publisher properties reaching a claimed 2.6 billion people — including the news apps on your phone). No consumer choice removes an SDK from an app someone else wrote.
4. **The payment rails are deeply tied.** PayPal and Venmo's fraud-detection and security systems are built in Israel (three acquisitions: Fraud Sciences, CyActive, Cymbio); Coinbase's wallet cryptography is Israeli-authored (the Unbound Security acquisition); the card networks behind Apple Pay and Google Pay (Visa, Mastercard, Citi) all run Israeli R&D centers.

5. **The clean column is real but narrower:** Signal and Telegram (messaging), the Match Group dating portfolio (Tinder, Hinge — a fully clean acquisition ledger), Proton and Mullvad (VPN), Duolingo and Canva and Notion (education/design/notes), X, Robinhood, Binance, Zelle (finance), and — notably — **OpenAI, which has deliberately declined to open an Israeli office** (Globes, July 2026: it manages Israel operations "from abroad"), though its reported talks to acquire Israeli-founded Decart could change that.

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3. The Three Axes of an App Audit

Axis one — the company. Is the app maker itself Israeli-founded, Israeli-headquartered, or operating an Israeli R&D center that writes the app's code? This is the strongest tie: the code is the product.

Axis two — the infrastructure inside. Acquisitions and SDKs that place Israeli-authored components in the app's data path. The AppsFlyer attribution SDK (48.51 percent of Android installs per its own 2024 industry data) is the canonical case: the app belongs to someone else, but Israeli code inside it tracks the install.

Axis three — the content. What the app streams or hosts. Content is a consumption fact, not a code fact: Netflix's Israeli connection is content-level (Fauda, Tehran, Shtisel, backing for the Sam Spiegel Series Lab) with no Israeli engineering office found — the exact opposite of Spotify, whose Tel Aviv office of roughly 300 staff builds recommendation systems. Same "tie"

word, entirely different mechanisms, and the distinction is maintained throughout.

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4. Messaging and Social

- **Viber — Israeli app.** Founded in Tel Aviv in 2010 by Talmon Marco (former IDF Central Command chief information officer) and Igor Magazinnik; core functionality developed from the Tel Aviv office; Rakuten acquired it for **$900 million in 2014**; half its employees are in Israel; 950 million to 1.2 billion registered users. A billion-person messaging network that is Israeli-authored at the core.
- **Meta (Facebook, Instagram, WhatsApp, Messenger) — deep R&D tie.** The Tel Aviv R&D center works on core advertising and AI systems for Facebook and Instagram (per Meta's own descriptions, recently in the news for layoffs of roughly 5 percent). These apps ship preinstalled or near-mandatory on virtually every Android phone, so the ads-ranking and AI inference monetizing them includes Israeli-authored engineering regardless of device.
- **Snapchat — corporate tie.** Herzliya R&D center built on the **Cimagine** (augmented reality) and **Voca.ai** acquisitions (about $70 million, November 2020); sites in Tel Aviv and Kfar Yehoshua.
- **Reddit — acquisition tie.** Acquired **Spell**, an Israeli natural-language-processing startup, in 2024 — its first acquisition — followed by AI ad-tech purchases.

- **X — no Israeli ties found.** Zero results across dedicated searches; recorded no-evidence-found.
- **Tencent / WeChat — light, audited.** OpenIntel's audit: no office or subsidiary; but **GateHub, a Tel Aviv firm, is Tencent Cloud's first strategic partner in Israel**, marketing Tencent Cloud, WeChat Pay, and facial-recognition products; Tencent's venture record in Israel includes Phytech ($11 million round, 2017), Future Crops (2022; bankrupt 2023), and ONE ZERO Digital Bank ($120 million round, December 2021). The audit's own verdict: "no public evidence identified" of any Israeli state or military contract.
- **TikTok (ByteDance) — no ties found, at consumer-guide grade** (no audit dossier exists for ByteDance).
- **Match Group (Tinder, Hinge, OkCupid, PlentyOfFish) — clean ledger, the dating category.** The full acquisition record contains nothing Israeli: Match.com itself (1999, ~$50 million, via IAC), OkCupid (2011, ~$50 million), PlentyOfFish (2015, $575 million), Hinge (majority 2017, remainder 2019), and Hyperconnect (2021, $1.73 billion — Korean, maker of Azar and Hakuna). Tinder was incubated inside IAC's Hatch Labs, not acquired. No Israeli R&D center, acquisition, or office surfaced across dedicated searches — the cleanest major app category identified. (Bumble, the remaining major dating app, is unresolved — see Part 16.)
- **Signal, Telegram, LINE — clean.** Signal is US-based; Telegram is Dubai-based; LINE is Japanese-Korean. No Israeli operations found in any pass.

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5. Payments, Commerce, and Finance

- **PayPal and Venmo — the deepest fintech tie.** PayPal acquired **Fraud Sciences for $169 million in January 2008** — founded by Shvat Shaked and Saar Wilf, both intelligence veterans; the deal seeded PayPal's Israeli operation — then **CyActive of Beersheba for about $60 million in March 2015** (founders from military intelligence; opened a second development center), and **Cymbio in May 2026**, with PayPal Israel now described as a "strategic technology hub" for agentic commerce, including Microsoft Copilot and Perplexity integrations. The fraud-detection, cybersecurity, and AI risk systems that score **every PayPal and Venmo transaction** are Israeli-authored, built by an Israel R&D operation of roughly 200-plus engineers.
- **eBay — five Israeli acquisitions.** Shopping.com ($620 million, 2005); The Gifts Project (~$30 million, 2011 — became eBay's Israeli development center); SalesPredict (~$40 million, 2016); plus Fraud Sciences via PayPal and CyActive via PayPal.
- **eToro — the app itself is an Israeli company.** Headquartered in Bnei Brak (its principal executive offices are at 30 Sheshet Hayamim Street per its own SEC filings); incorporated in the British Virgin Islands; NASDAQ: ETOR since May 14, 2025 (an IPO raising roughly $620 million at a ~$4.2–4.3 billion valuation); founded 2007 by Yoni and Ronen Assia and David Ring; about 1,520 employees; 3.81 million funded accounts across 75 countries, with about $18.5 billion in assets under administration and roughly 1,520 employees.

A user's brokerage, order routing, and crypto custody run on an Israeli-headquartered platform.

- **Coinbase — Israeli-authored wallet cryptography.** Acquired **Unbound Security** in November 2021 (Petah Tikva; founded 2015 by cryptographers Yehuda Lindell, Nigel Smart, and Guy Peer; reported at up to $150 million) and announced a "tech center of excellence in Israel." The multi-party-computation technology securing Coinbase's keys is Israeli-authored.
- **The card rails:** Visa, Mastercard, and Citi all operate Israeli R&D centers — so the card networks underneath Apple Pay and Google Pay carry an engineering-level tie regardless of phone choice.
- **Robinhood, Binance — no Israeli corporate ties surfaced.** Recorded alongside a symmetry note to prevent confusion: Binance is *sued by* October 7 victims' families over alleged laundering for Hamas — entanglement with Israel's adversaries, a different axis entirely.
- **Zelle (the bank consortium) — no ties found**; cash remains the only fully clean payment rail.

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6. Productivity and Work

- **Slack / Salesforce — a major and current tie.** Salesforce entered Israel in 2011 (Navajo Systems) and now runs roughly **900 employees across three R&D centers** (Tel Aviv, Nazareth). Its Israeli acquisition ledger: BlueTail, Kernesen, Implisit, Datorama, Bonobo, ClickSoftware, **Zoomin and Own ($1.9 billion, 2024)**, and **Doti (~$100 million, November 2025)** — whose team "will join Salesforce's AI R&D hub in Israel," with Doti's founders leading the Israeli branch of Slack. Slack's agentic search and AI features are being written in Israel.
- **Microsoft 365 and Google Workspace** — the parent-company dossiers from the laptop and phone reports apply in full (Microsoft BRS 711, the Unit 8200 cloud; Google Nimbus and Israeli chip teams).
- **Zoom Video — no Israeli R&D surfaced**, with one critical naming warning: **do not confuse it with ZoomInfo**, the business-data company whose global R&D is centered in Ra'anana with 400-plus Israel employees. Search results conflate them constantly; the distinction is preserved here.
- **Yahoo apps:** Yahoo's Israeli R&D (ClarityRay, Dapper, RayV) persists under Apollo ownership.
- **Notion, Basecamp — no Israeli ties found** (no-evidence grade). FOSS alternatives (CryptPad, Nextcloud) are structurally cleaner still.
- **The continuing B2B pattern — Zendesk→Unleash.** December 2025: Zendesk acquired **Unleash**, an Israeli AI enterprise-search startup, for its agent products. Not a

consumer phone app, but recorded here because it exemplifies the mechanism by which ties enter the app economy quietly: one startup acquisition at a time, the acquired team's code becoming the parent product's core (the same mechanism as Salesforce/Own and Salesforce/Doti above, and Reddit/Spell in Part 4).

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7. The AI Apps

- **ChatGPT (OpenAI) — no Israeli office, deliberately.** Globes (July 2026): "OpenAI to manage Israel operations from abroad." But OpenAI is reported to be in talks to acquire **Decart** — Israeli-founded, Tel Aviv-based, about 89 of its 106 staff in Israel, previously valued near $4 billion — at a reported **$6 billion** valuation. If completed, an engineering-level tie; at compilation time, talks only. Also of record: **Irregular**, a Tel Aviv AI-security testbed company ($450 million valuation; Sequoia and Redpoint-backed), was the named host in the August 2026 "rogue AI" security-testing episode involving OpenAI, Anthropic, and Meta — an adversarial-testing relationship, not ownership, and flagged as such.
- **Claude (Anthropic) — no Israeli development center** (a representative sits in Ireland). Two recorded ties: its **Palantir partnership** (Palantir operates an Israeli R&D center and holds Israeli military contracts), and an **Intercept report (October 2025)** that **Rafael — the Israeli arms prime — obtained Claude via Amazon Web Services and**

specifically sought to test it, despite Anthropic's acceptable-use policy barring weapons development. How that squares remains unresolved.

- **Gemini (Google) and Copilot (Microsoft)** — the parent dossiers apply.

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8. The VPN and Security Category — the Defining Finding

Kape Technologies is the story of the category. The company is British-Israeli, formerly **Crossrider** — an ad-injection platform that Malwarebytes classified as malware — founded in 2011 by Shmueli Achdut and **Koby Menachemi, a Unit 8200 veteran**. Teddy Sagi bought it in 2012 for $37 million; it listed on London's AIM at a $250 million valuation; it rebranded Kape in 2018; Sagi took it fully private in June 2023 (it is now wholly owned by his Unikmind vehicle), followed by roughly 180 layoffs. Its portfolio:

- **ExpressVPN — acquired for $936 million in 2021** · **CyberGhost — €9.1 million, 2017** · **Private Internet Access — $127.5 million, 2019** · **ZenMate — €4.8 million, 2018** (merged into CyberGhost, 2023) · **Intego antivirus**.
- **Webselenese — the review-site arm:** owner of **vpnMentor and Wizcase**, two of the largest VPN-review sites on the internet, which rank its owner's products for the users trying to research them.
- Development center in Tel Aviv's Azrieli Sarona Tower; CEO Ido Erlichman is a Duvdevan veteran. Sagi's own record: a

1996 fraud conviction (roughly nine months served), a ~$3 million Friends of the IDF scholarship donation (2019), a 1-million-shekel soldier-transport donation (2023), Panama Papers appearances, ~$6.4 billion net worth.

The honest counterweight, stated because the record requires it: VPN-industry reporting — including competitors with an interest, flagged as such — notes that **no evidence of misuse has surfaced**; ExpressVPN runs RAM-only servers and has commissioned audits. The documented facts are the ownership chain and the review-site circularity; wrongdoing is not claimed, only the chain. Buyers who find the chain disqualifying under the criterion have a clean, well-audited set: **Proton (Swiss), Mullvad (Swedish), IVPN (Gibraltar), and Nord Security's NordVPN/Surfshark (Lithuanian — which also owns Saily, the clean travel-eSIM from the phone investigation). HolaVPN**, the other famous Israeli VPN, is Israeli-founded with its own bandwidth-reselling history — avoid under this criterion.

***

9. Navigation, Travel, and Food

- **Waze — Israeli, Google-owned** (covered in the phone report).
- **Via — Israeli-founded, filing-grade:** two wholly-owned Israeli subsidiaries and 250-plus Israel staff, per its own SEC filing.
- **Uber — resolved this investigation:** an OpenIntel dossier identifies an R&D/engineering footprint in Israel, and in June

2026 Uber partnered with Israeli **Autobrains** (Tel Aviv; 300-plus patents) and Nvidia on its Munich robotaxi program. Corporate tie confirmed; no Israeli-authored app core asserted.

- **Booking.com — Tel Aviv AI Innovation Center** (documented since 2017).
- **Google Maps and Apple Maps** — parent dossiers.
- **Lyft — no Israeli ties found.** DoorDash and Instacart — none surfaced (unverified grade).
- **The clean navigation option: Organic Maps** and other OpenStreetMap-based free-software apps.

***

10. Creative Tools and Education

- **Lightricks — the editing category's crown is Israeli, fully confirmed.** Jerusalem-based, founded 2013 by Hebrew University postdoctoral researchers in computer graphics — Zeev Farbman, Nir Pochter, Yaron Inger, Amit Goldstein; founder profiles record Unit 8200 service. **Facetune** was Apple's #1 best-selling paid app of 2013, topping charts in 120-plus countries; the suite (Photoleap, Videoleap, Motionleap, the LTX Studio AI-filmmaking platform, Popular Pays) has passed 500–730 million downloads with multiple millions of paying subscribers; $335 million raised, including a **$130 million Series D at a $1.8 billion valuation (2021)** led by Insight Partners; Getty Images and Shutterstock training partnerships (December 2024); the 2024–2026 generative-AI pivot came with repeated layoffs (70 in 2024, 85 earlier, 75 more in June 2026) and a split of the profitable consumer apps from the costly LTX AI-video division. The alternatives are Adobe (US), Canva (Australian), and CapCut (ByteDance) — both clean at no-evidence grade.
- **Duolingo — clean.** Its first acquisition was Gunner, a US design agency (2022); no Israeli ties found. The adjacent fact: **Loora**, an Israeli AI English-learning app founded by two former IDF commandos (Roy Mor and Yonatan Levin), serves 15 million users and competes directly with Duolingo — Israeli-founded entrants define the category's frontier even where incumbents are clean.

***

11. Streaming and Content

- **Netflix — content-level tie only.** Fauda, Tehran, Shtisel, and backing for the Sam Spiegel Series Lab (2022); no Israeli engineering office found. The cleanest major streaming service on the engineering axis.
- **Spotify — engineering-level tie:** the Tel Aviv R&D office (roughly 300 staff) builds recommendation and AI systems.
- **YouTube and Twitch** — parent dossiers (Google; Amazon, whose Annapurna Labs chip division is Israeli-founded and which co-holds Project Nimbus).
- **Music alternative at no-evidence grade:** Bandcamp and local files; Apple Music inherits its parent's dossier.

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12. The Invisible Infrastructure Layer — Where Three Pillars Live

Beneath every category above sits the layer the user never chooses: SDKs embedded in other people's apps, and the ad systems that monetize the free tier of the mobile world. Three pillars, all Israeli-founded, dominate it.

- **AppsFlyer — attribution.** Founded 2011; CEO Oren Kaniel; headquartered in effect "San Francisco, Israel"; **about $500 million ARR as of January 2026 and profitable since 2024**; the dominant mobile-attribution SDK — per its own industry data leading Android with **48.51 percent of all installs** (third-party trackers put its share near 44 percent, ahead of Germany's Adjust at ~27 percent and the US Branch

at ~22 percent). Its Series C ($56 million) was led by Israeli VCs **Qumra and Pitango** alongside Goldman Sachs and Deutsche Telekom Capital Partners. The SDK inside roughly half of Android apps reports installs, ad sources, and in-app events to an Israeli-founded analytics company. No consumer setting removes an SDK from an app someone else wrote; the mitigations are app selection (F-Droid's free-software catalog largely lacks it) and network-level blocking of its endpoints (imperfect).

- **ironSource — monetization.** Israeli; merged into **Unity** in a deal reported around $4.4 billion; the mediation and monetization stack inside a large share of free mobile games (reported-deal grade, flagged for filing-grade confirmation). Together with AppsFlyer it means the free-app economy's two money pipes are Israeli-authored.
- **Taboola/Outbrain — content recommendation.** The sponsored-content modules at the bottom of articles across the mobile web and news apps — the "around the web" boxes — are an Israeli duopoly that became a monopoly: **Taboola** (founded Tel Aviv 2007 by Adam Singolda; NASDAQ: TBLA; investors including Pitango, Baidu, Fidelity, Comcast Ventures, DailyMail, and Groupe Arnault; $632 million revenue in H1 2021 alone, roughly a fifth of it from Microsoft) and **Outbrain** (founded 2006 by Israelis Yaron Galai and Ori Lahav; backed by Carmel, Gemini Israel, GlenRock Israel, Rhodium, Index, and Lightspeed). The two announced an $850 million merger in 2019; after years of negotiation it **closed in mid-2024**, leaving one Israeli-founded company

running recommendation modules across **20,000 publisher properties with a claimed audience of 2.6 billion people** — customers include CNN, the BBC, The Washington Post, USA Today, The Guardian, Sky News, and MSN. Every ad-funded news app monetizes through it.

Why this layer matters most: the company axis can be dodged by app choice; the content axis by taste; but the infrastructure axis attaches to apps the user merely *uses* — a bank app, a airline app, a local newspaper app — and it follows them regardless of every other choice they make.

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13. The Consolidated Israeli-App Catalog

Across all passes, the apps a phone user can hold that are **Israeli-authored at the core**:

Category	**Israeli app / company**
Messaging	**Viber** (Rakuten-owned; ~1 billion users; Tel Aviv-built)
Navigation	**Waze** (Google-owned)
Transit	**Via** (SEC: two wholly-owned Israeli subsidiaries, 250+ staff)
Brokerage	**eToro** (HQ Bnei Brak; NASDAQ: ETOR)
Photo/video editing	**Lightricks suite** (Facetune, Photoleap, Videoleap, LTX Studio; Jerusalem; $1.8B valuation)
Mobile games	**Playtika** (NASDAQ: PLTK), **Moon Active** (Coin Master, $6B+ lifetime), **Plarium** (RAID, ~$1B), plus Reworks ($400M, via Playtika) and Beach Bum (via Voodoo)
Education	**Loora** (AI English learning; 15M users)
Attribution SDK	**AppsFlyer** (~half of Android installs)
Ad monetization	**ironSource** (merged into Unity)

Category	Israeli app / company
Content recommendation	**Taboola + Outbrain** (merged 2024; 2.6B-reach)
VPN (ownership)	Crossrider→**Kape** (ExpressVPN, CyberGhost, PIA, ZenMate, vpnMentor)
App-adjacent SaaS	**Fiverr, Monday.com, Wix** (and Payoneer) — Israeli companies used through their phone apps

The catalog's breadth is itself the finding: **from the phone's photo filters to the ads under its news to the VPN protecting it, each category's leading independent traces to Israel.** The counter-catalog is real but consistently second-tier in market share, category by category. (Fiverr, Wix, MyHeritage, and Moovit — the latter two Intel-owned — are included at well-established-public-record grade; they were not re-verified in this investigation's searches and are labeled accordingly.)

The colonization pattern, stated once because it generalizes: where a platform is clean, Israeli industry builds on top of it. Roblox Corporation surfaced no Israeli corporate ties — yet Israel has a booming Roblox creator economy (a live championship selling out the Menorah Arena six times), and Israeli studios keep being acquired into global gaming portfolios (Beach Bum to Voodoo; Reworks to Playtika for $400 million). The tie migrates to whatever layer Israeli startups colonize.

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14. The Clean Column — the Substitution Table

Category	Tied	Clean substitute (grade)
Messaging	WhatsApp/Messenger/Viber (Meta R&D / Israeli)	**Signal, Telegram**
Social	Instagram, Snapchat (Cimagine/Voca), Reddit (Spell)	**X (no-evidence)**
Dating	—	**Tinder, Hinge — Match Group ledger fully clean**
Payments	PayPal/Venmo, eToro (Israeli), Coinbase (Unbound), card rails	**Zelle, Robinhood, Binance (no-evidence), cash**
Brokerage/crypto	eToro	Robinhood, Binance (no-evidence)
Productivity	Slack (Doti→Slack AI Israel), Microsoft 365, Google Workspace	**Notion, Basecamp, FOSS (no-evidence)**
AI	Gemini, Copilot; ChatGPT (Decart pending)	**Claude (no office; Palantir/Rafael caveat)**, local models
VPN	**ExpressVPN, CyberGhost, PIA, ZenMate, Hola (Kape)**	**Proton (CH), Mullvad (SE), IVPN (GI), Nord/Surfshark (LT)**
Creative	**Facetune/Photoleap/Videoleap (Lightricks)**	**Canva (AU), CapCut, Adobe**
Music	Spotify (Tel Aviv R&D)	Bandcamp, local files
Navigation	Waze, Google/Apple Maps, Uber, Via, Booking	**Organic Maps (FOSS), Lyft (no-evidence)**
Education	Loora	**Duolingo (no-evidence)**
Email	Gmail/Outlook/Yahoo	**Proton Mail, Tuta, Fastmail (AU)**
Cloud	Drive/OneDrive/iCloud	**MEGA (NZ), pCloud (CH), self-hosted**
Streaming	YouTube/Twitch; Spotify (eng.)	**Netflix (content-level only)**
In-app SDKs	**AppsFlyer, ironSource/Unity, Taboola/Outbrain**	F-Droid FOSS catalog; DNS blocking (imperfect)

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15. Methodology and Evidence Grading

How the investigation was conducted. Three research passes over the app layer: breadth mapping of every major category; gap-closure on the open items (Lightricks, Match Group, Uber, Canva/Duolingo/Notion); and the infrastructure sweep (Taboola/Outbrain, Roblox, the Israeli catalog). Sources include forensic corporate audits (OpenIntel dossiers for Tencent and Uber), SEC filings (eToro, Via), company press releases and funding announcements, Israeli business press (Globes, Calcalist), quality international press (NYT, The Intercept, Digiday), and market-share datasets (AppsFlyer's own industry data, third-party trackers — flagged as self-reported where applicable).

Evidence grades: audit-grade (OpenIntel, with their explicit "no public evidence identified" statements preserved); filing-grade (SEC); on-record (company statements); reported (press); no-evidence-found (nothing surfaced, not proof of absence); unresolved (flagged). Where a review site or market-share dataset has an interest in its own finding, the interest is flagged.

Caveats the record requires:

- "No ties found" means no Israeli office, acquisition, R&D center, or investment surfaced in the searches conducted; it is a statement about the evidence, not a guarantee about the world.

- The Kape counterweight is stated in full: ownership documented, misuse not claimed, audits exist.
- The Anthropic/Rafael finding is a report about a customer relationship, not about Anthropic's corporate structure; it is recorded with that framing.
- Self-reported market shares (AppsFlyer's 48.51 percent) are labeled as such and corroborated where third-party trackers exist.

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16. What Remains Unresolved

1. **Bumble** and the **password-manager set** (1Password, Bitwarden, Dashlane, LastPass) — two query rounds timed out; unresolved, not asserted either way.
2. **Uber's numeric BRS score** — the dossier confirms an Israeli footprint; the score was not extracted.
3. **Pinterest's** acquisition of The Yes (2022) — unconfirmed in results.
4. **ByteDance/TikTok** at audit grade — no dossier exists.
5. **Decart/OpenAI** — talks at compilation time; completion unconfirmed.
6. **ironSource/Unity** merger terms at filing grade.
7. **Roblox** — no corporate ties found, unverified grade.
8. **Health/fitness, smart-home, and kids' app categories** — not yet individually swept (Fitbit, Calm, MyFitnessPal, Roblox-adjacent titles).

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17. Key Figures at a Glance

Entity	Figure	Context
Viber	$900M (2014)	Rakuten acquisition; ~1B users; Tel Aviv-built
eToro	~$4.2–4.3B valuation	NASDAQ IPO May 2025; HQ Bnei Brak
Lightricks	$1.8B valuation (2021)	Facetune et al.; 500–730M downloads
PayPal/Israel	$169M + ~$60M + Cymbio	Fraud Sciences 2008; CyActive 2015; Cymbio 2026
Coinbase/Unbound	~$150M (2021)	MPC wallet cryptography
Kape/ExpressVPN	$936M (2021)	Plus CyberGhost, PIA, ZenMate, Intego, vpnMentor
AppsFlyer	~$500M ARR; 48.51% of Android installs	Market leader; Qumra/Pitango on cap table
Taboola/Outbrain	$850M merger (closed 2024)	20,000 properties; 2.6B claimed reach
Salesforce/Israel	$1.9B (Own) + ~$100M (Doti)	~900 staff; Slack AI in Israel
Moon Active/Coin Master	$6B+ lifetime	#9 global game revenue H1 2024
Match Group	$1.73B Hyperconnect (2021)	Clean ledger — no Israeli acquisitions

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18. Glossary

- **SDK (software development kit):** pre-built code an app developer bolts into their app — analytics, ads, payments, maps. The AppsFlyer SDK is inside roughly half of Android apps.
- **Attribution:** the analytics discipline of determining which ad caused an app install — AppsFlyer's business.

- **Mediation / monetization:** the auction system that decides which ad network fills an app's ad slots — ironSource's business.
- **Chumbox:** the "around the web" sponsored-content module at the bottom of articles — Taboola/Outbrain's business.
- **MPC (multi-party computation):** cryptography that splits a secret key into pieces so no single party holds it — Unbound's technology, now inside Coinbase's wallet security.
- **RAM-only servers:** VPN servers that keep no data on disk — ExpressVPN's architectural defense (cited in the counterweight).
- **R&D center:** an engineering office that writes production code, distinct from a sales office.
- **BRS / Tier:** OpenIntel's 1,000-point complicity score and bands (E minimal through B severe).
- **AUP:** acceptable-use policy — Anthropic's rules against weapons-development use.
- **FOSS:** free and open-source software — the F-Droid catalog, Organic Maps, Signal's stack.
- **Entity List:** the US Commerce blacklist (NSO added November 2021).

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19. The Conclusion, in One Paragraph

An app is the one layer of the phone stack where the user holds real power — ninety seconds to delete and replace — and this audit maps exactly where that power reaches. The apps themselves divide into three camps: Israeli-authored category leaders (Viber, eToro, the Lightricks suite, Waze, Via, and a remarkable share of top-grossing games), deeply tied Western services (Meta, PayPal, Slack, Spotify, Coinbase, Booking, Uber), and a genuine clean column (Signal, Telegram, the Match dating portfolio, Proton, Mullvad, Duolingo, Canva, Notion, X, Robinhood, Binance, Zelle, and — by deliberate choice — OpenAI, which manages Israel "from abroad"). Below the apps sits the layer no choice reaches: AppsFlyer inside half of Android installs, ironSource inside free games, and Taboola/Outbrain underneath the news — three invisible pillars, all Israeli-founded. And the cautionary tale of the category is Kape: four of the world's most recommended VPNs, plus the review sites that recommend them, owned by one Israeli billionaire's holding company — the place where a user trying to escape the layer most visibly landed inside it. Every claim above carries a source; every ambiguity is marked; and everything unresolved is listed in Part 16, because an honest map of what cannot be known is worth more than a confident one.

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Compiled 2026-09-20. Three research passes; audit-grade, filing-grade, and on-record sources separated throughout; self-reported figures labeled; "no evidence found" statements preserved for what they are.

Part Six: The Zero-Ties Router: A Complete Audit of Israeli Entanglement in Home Routers and Mesh Wi-Fi

A fully-sourced investigation into the device that sees everything your home does online — home routers and mesh Wi-Fi systems — and the closest configurations to a zero-ties-to-Israel standard, with the protections that matter, as of September 2026.

1. Why the Router Is Different

Every investigation in this series has circled the same truth from a different angle: the most important device is the one you never think about. The laptop, the phone, and the apps have all been audited. But every one of them, in every home, passes through a single box — **the router** — and that box sees everything: every website request (as DNS lookups), every device that joins the network, every hour of use, every smart-TV binge and camera feed. If a phone is a diary, the router is the librarian who reads every checkout slip.

A router is also three devices in one: a **modem or gateway** (talks to the ISP), a **router** (directs traffic), and a **Wi-Fi radio** (speaks to your devices). Each layer can be entangled independently. And in most American homes, the ISP chose at least one of those layers for you.

How to read this document

- **Part 2** gives the short answer.
- **Part 3** orients new readers (router/mesh/cloud basics).

- **Parts 4–8** are the audit: the corporate ledger, the silicon layer, the software layer, the mesh cloud-dependency spectrum, and the attack-surface record.
- **Part 9** is the regulatory shock: the FCC's 2026 router ban and what it does to your choices.
- **Part 10** is the protections chapter — the practical hardening stack.
- **Part 11** is the minimum-entanglement build.
- **Parts 12–15** cover method, open questions, key figures, glossary.
- **Part 16** is the conclusion.

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2. The Short Answer

No mainstream home router is fully clean — but the router is the one device category where a near-clean, locally-controlled, non-cloud configuration genuinely exists, and where the 2026 US regulatory environment has accidentally narrowed the market to exactly one clean mainstream choice. The findings:

1. **The three companies making essentially all router Wi-Fi chips are Broadcom (entangled at the silicon level, Israeli R&D), Qualcomm (entangled, two Israeli R&D centers since 1993), and MediaTek (the clean audited path — BRS 248, Tier D, no Israeli office or contract).** Realtek is the secondary clean option. Every router you can buy runs on one of these three; the choice of chip is the single biggest determinant of the router's

entanglement.

2. **The cloud is the real divide.** Mesh systems are not created equal in who they report to: **Amazon's eero has no local interface at all** — every setting flows through Amazon's cloud on an app-only design with no opt-out for its data collection — while **Netgear's Orbi is the only major mesh system that runs entirely from a local web interface with no account required**. In this category, "no cloud account" is worth more than any brand promise.
3. **The FCC banned foreign-produced consumer routers in 2026**, and as of May 13, 2026 the only exemptions granted are **Netgear and Eero**. The practical choice set for a new US router has collapsed to one clean US brand (Netgear) versus Amazon's eero.
4. **The attack-surface record proves the stakes**: VPNFilter (2018, ~500,000 routers in 54 countries, attributed to Russia's GRU-linked Fancy Bear) and Slingshot targeted virtually every brand in this document — Netgear, Linksys, MikroTik, ASUS, TP-Link, Ubiquiti, ZTE, D-Link, Huawei. The router is the most-attacked device in the home, and the security research industry that defends it is, in significant part, Israeli (Check Point Research publishes major router threat reports).
5. **The protection pattern is consistent and short**: choose a local-first system (Orbi, or an OpenWrt-flashed device, or a Turris Omnia), put it behind a bridged ISP modem, harden the settings, and take your DNS off your ISP entirely (Unbound locally, or Quad9 — the Swiss non-profit with no logging).

Every one of those steps is a half-hour of work.

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3. A Thirty-Second Orientation

A **modem** talks to your internet provider over cable, fiber, or phone line. A **router** directs traffic between your devices and the modem. A **mesh system** is several routers working as one, blanketing a larger home in Wi-Fi. A **cloud-dependent router** stores your settings — and often your Wi-Fi password, device list, and usage metadata — on the manufacturer's servers, which means the manufacturer (and, legally, the government where those servers sit) can be compelled to turn them over. A **local-first router** keeps everything on the box in your house. And **firmware** is the router's own operating software — the layer where implants live, and the layer you update (or fail to).

4. The Corporate Ledger

- **Netgear (San Jose, founded 1996).** The only US-headquartered major router brand; FCC-exempt under the 2026 rules. **No Israeli ties surfaced in any pass** — its recent R&D expansion is in Chennai, India (via VAAG Systems, 2025). The **Orbi** line runs fully from a local web interface with no Netgear account required — documented across the user community as "the only major home mesh system not tied to the cloud." Netgear Armor (optional security subscription) is Bitdefender-powered — Bitdefender is Romanian, clean at corporate level. Netgear is the cleanest mainstream choice in the category.

- **eero (Amazon).** The category's market leader by simplicity — and its entanglement is deep. eero hardware runs on **Amazon's Annapurna Labs chips — the Israeli-founded, Tel Aviv-designed silicon arm Amazon acquired in 2015** — and the service sits on Amazon's cloud, which co-holds the Israeli government's Project Nimbus contract. The privacy record: **no web interface at all** (app-only), **no opt-out** from data collection; eero collects network speeds, email, device IP addresses, IP-based geolocation, connected-device lists, and stores network settings in the cloud — where, as router-security analysts note, the US government can compel their production. Privacy reviewers' verdict: "a tougher brand to recommend for privacy-focused consumers."
- **Google Nest WiFi.** Inherits the full Google dossier (Israeli R&D, Nimbus, Israeli chip teams) — and is cloud-managed by default.
- **Linksys (Belkin/Foxconn).** Foxconn-owned since 2018; the laptop-audit Foxconn finding (assembly clean, Apple-partnered supply chain) applies; Linksys keeps a web interface (Velop) — middle of the cloud spectrum.
- **ASUS.** Clean corporate ledger from the laptop audit (no Israeli R&D, acquisitions, or subsidiaries found). ZenWiFi mesh runs local web UI. ASUSWRT firmware is mature; MerlinWRT (third-party) exists. A strong clean alternative.
- **TP-Link (Deco).** A security record, not just a corporate one: **Check Point Research (Israeli firm) discovered a firmware implant in TP-Link routers in May 2023**; Microsoft's CovertNetwork-1658 botnet takedown (October

2024) centered on TP-Link devices; the Texas Attorney General opened a probe in October 2025; the company moved its headquarters to Irvine, California the same month. Market-share claims vary wildly (12% per Lansweeper telemetry; ~65% per disputed industry figures). Under the FCC's foreign-router rules, TP-Link's consumer future in the US is the question mark of the category.

- **Ubiquiti (NYSE-listed, US).** No Israeli ties surfaced. Professional-grade gear with a local controller option; steeper learning curve.
- **MikroTik (Latvia).** No Israeli ties found; enthusiast favorite; runs RouterOS with full local control; OpenWrt-flashable models exist. Note: on both the VPNFilter and Slingshot target lists (as was nearly everyone).
- **Firewalla (Santa Clara, 2016, unfunded).** No Israeli ties surfaced; security-first consumer firewall/router.
- **Turris Omnia (CZ.NIC, Czech non-profit).** Open hardware, open software (Turris OS, auto-updating, ~$334). The maximal-cleanliness enthusiast choice.
- **AVM Fritz!Box (Germany).** Clean corporate; the European standard; not a US-market product.
- **D-Link, Tenda, ZTE (Chinese/others).** No Israeli ties surfaced (unverified grade), but FCC-rule exposure.

5. The Silicon Layer

Router Wi-Fi chips are a triopoly: **Broadcom, Qualcomm, and MediaTek control essentially 100%** of the market. From the earlier investigations: Broadcom carries Israeli R&D; Qualcomm has run Israeli R&D centers since 1993 with ~700 Israel staff; **MediaTek is the audited clean path (BRS 248, Tier D — venture-portfolio residue only, no office, no contracts)**; Realtek (Taiwan) is the clean secondary. Practical consequence: a MediaTek Filogic-based router (many ASUS, Netgear, and TP-Link models) or Realtek-based unit minimizes silicon-level entanglement; Broadcom-based units (most high-end) carry the deeper tie. ARM and the EDA duopoly floor applies everywhere, as established in the laptop report.

6. The Software and Firmware Layer

- **OpenWrt** (open-source router OS, community-governed) and its cousins (DD-WRT, Tomato; Turris OS) are the cleanest software path: inspectable, no vendor telemetry, WireGuard-capable, ad-blocking (AdGuard Home/Pi-hole) installable.
- **Stock firmware** varies: Netgear and ASUS keep local UIs; eero has none; ISP-issued firmware (Technicolor, Compal) is often locked and update-delayed — a standing security risk documented across the ISP-hardware literature.
- **The Wi-Fi protocol itself** has been broken twice in the modern era: **KRACK (2017)** — Mathy Vanhoef's break of WPA2's handshake, patched everywhere; and **Dragonblood (2019)** — Vanhoef and **Eyal Ronen's** five-flaw demolition of

WPA3 (downgrade attacks on transition mode, timing and cache side-channels, denial-of-service, and an EAP-pwd invalid-curve authentication bypass), attackable for under $125 of cloud computing and initially downplayed by the Wi-Fi Alliance. Recorded, with the note that Ronen is Israeli — a defensive-research fact and a people-layer footnote, not a corporate tie.

- **DNS** — the router's most intimate data (every domain you visit) — defaults to your ISP. The protection stack: run **Unbound** locally (your router asks the internet's root servers directly; nothing leaves), or **Pi-hole/AdGuard Home** filtering plus **Quad9** (Swiss non-profit foundation; no logging; blocks malicious domains) as upstream; NextDNS (US, configurable) is the runner-up. Every step away from ISP DNS is a step away from a subpoena-able log of your household's life; encrypt the lookups themselves with DoH or DoT so the ISP cannot read them in transit either.

7. The Attack-Surface Record

- **VPNFilter (May 2018):** Cisco Talos and Symantec documented **~500,000 routers across 54 countries** infected by malware attributed to Russia's Fancy Bear (APT28 — the DNC hackers). Targets included Linksys, MikroTik, Netgear, QNAP, and TP-Link models. Three stages: a persistent stage-1 loader that survives reboots, a stage-2 payload for traffic theft and device management, and a **kill switch that could brick half a million devices at once**; a packet sniffer specifically watched **Modbus/SCADA protocols** — Ukraine infrastructure was the evident interest. The FBI seized the command domain (ToKnowAll.com) and publicly advised: reboot, disable remote management, update firmware. CISA later (2022) tracked its successor, **Cyclops Blink** (Sandworm).
- **Slingshot (2018, Kaspersky-discovered):** a second router-implant campaign hitting ASUS, D-Link, Huawei, Linksys, MikroTik, Netgear, TP-Link, Ubiquiti, UPVEL, and ZTE.
- The meta-lesson: **every clean brand in this document was on a nation-state target list** — because routers are the internet's most useful beachhead. Hardening (below) is not paranoia; it is the documented defense against a real, repeated class of attack.

8. The Cloud-Dependency Spectrum — the Category's Real Divide

From most to least cloud-bound: **eero** (no local interface exists; app and cloud account mandatory; no data opt-out) → **Deco/TP-Link** (cloud account mandatory; minimal web UI) → **Velop/Linksys** (web interface available) → **Orbi/Netgear** (full local web UI; no account needed — the only major mesh system operable entirely offline) → **OpenWrt/Turris/MikroTik** (fully local, inspectable). The practical rule this document repeats across all three of its companion investigations applies here with special force: **in networking, the clean answer is a shorter wire — the device that works with no account at all.**

9. The Regulatory Shock — the FCC's 2026 Ban

On March 23, 2026, the FCC's rules excluding foreign-produced consumer routers from the US market took effect; **as of May 13, 2026, only Netgear and Eero hold exemptions**. Whatever its security rationale (Chinese-manufactured routers were the stated concern), the practical effect on this document's criterion is stark: the new-router choice set has collapsed to **one clean US brand (Netgear) versus Amazon's eero**. Existing routers are unaffected; the used/OpenWrt ecosystem is unaffected; but the mainstream retail shelf is now a two-horse race with exactly one non-Amazon horse.

10. The Protections Chapter

In priority order, for any household:

1. **Take the ISP's box out of the driver's seat.** Set the ISP gateway to **bridge mode** (or passthrough/DMZ as fallback) and put your own router behind it. Fiber ONTs usually can't be removed (some GPON SFP workarounds exist); cable modems must come from the provider's approved DOCSIS 3.1 list; DSL needs PPPoE credentials — but the Wi-Fi your family uses can be entirely yours.
2. **Choose local-first hardware.** Orbi (mainstream), ASUS ZenWiFi, MikroTik, Turris Omnia, or any OpenWrt-supported unit. Avoid cloud-mandatory systems if privacy is the criterion.
3. **Harden the settings:** unique admin password; WPA3 (or WPA2-AES minimum); **WPS off** (a brute-forceable design from 2011); **UPnP off** (lets devices punch holes in your firewall); remote management off; firmware auto-update on; guest network for IoT/smart-home devices.
4. **Take DNS off the ISP:** Unbound locally (maximal), or Quad9 upstream (Swiss, no logging, threat-blocking).
5. **Filter at the network:** AdGuard Home or Pi-hole for whole-house ad/malware blocking.
6. **Verify your exposure:** run Gibson's ShieldsUP port scan; check for unexpected open ports.
7. **If you VPN, do it at the router:** WireGuard on OpenWrt/Turris covers every device that joins — with the VPN choices from the apps investigation (Proton, Mullvad, IVPN; avoid the Kape-owned brands).

11. The Minimum-Entanglement Build

Layer	Choice	Residual
Gateway	ISP modem in bridge mode	ISP firmware layer (unavoidable; minimal exposure)
Router/mesh	**Netgear Orbi** (local-first) or Turris Omnia / MikroTik + OpenWrt	Broadcom or MediaTek silicon (prefer MediaTek models; ARM/EDA floor)
DNS	Unbound local, or Pi-hole → Quad9	—
Wi-Fi security	WPA3, WPS/UPnP/remote-mgmt off	Protocol-layer history (KRACK/Dragonblood) patched
IoT	Guest VLAN/network	—
VPN (optional)	WireGuard at router; Proton/Mullvad	—

Irreducible floor: the ISP's modem (you can bridge it, not remove it); the silicon triopoly (choose MediaTek); ARM/EDA; and the fact that the global router-security research ecosystem — including Check Point Research, which published the TP-Link implant finding — is substantially Israeli. The floor here is lower than in any other device category in this series, because the router is the one box you can fully own.

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12. Methodology and Evidence Grading

Five research passes: corporate sweeps (brand by brand), silicon and chipset mapping, software/firmware/cloud-dependency analysis, the attack-surface record (VPNFilter, Slingshot, protocol breaks), and the protection stack. Grades: audit-grade (OpenIntel MediaTek dossier carried from the laptop investigation), S/A-grade (FCC records, Cisco Talos, Vanhoef's papers), community-verified (Orbi local-first, B-grade), no-evidence-found (Ubiquiti, Firewalla), and flagged B-grade (TP-Link market-share disputes). Self-reported claims (eero's data practices as described in its policy) are labeled as such.

13. What Remains Unresolved

1. Netgear and ASUS clean status: no-evidence-found, not audit-grade (no OpenIntel dossier exists for either).
2. Ubiquiti/Firewalla: no-evidence-found.
3. The FCC exemption process for further brands: ongoing as of compilation.
4. Whether TP-Link's Irvine restructuring achieves exemption: open.
5. D-Link/Tenda/ZTE corporate: unverified.
6. The identity of Check Point Research's sources for the TP-Link implant: protected, as standard.

14. Key Figures at a Glance

Figure	Value	Context
Router Wi-Fi chip triopoly	Broadcom / Qualcomm / MediaTek ≈100%	MediaTek = audited clean (BRS 248)
VPNFilter	~500k routers, 54 countries	2018; Fancy Bear; kill switch
FCC foreign-router ban	March 23, 2026	Exemptions: Netgear, Eero only
eero data posture	No opt-out; no local UI	Amazon cloud; Annapurna silicon
Orbi distinction	Full local web UI	Only major cloud-free mesh
Dragonblood attack cost	<$125	WPA3's five flaws, 2019
Quad9	Swiss foundation	No logging; threat-blocking DNS

15. Glossary

- **Modem/gateway:** talks to your ISP; often ISP-owned.
- **Router/mesh:** directs traffic; mesh = multiple units, one network.
- **Bridge mode:** ISP box forwards everything to your router; your router runs the network.
- **WPS/UPnP:** convenience features (button pairing; automatic port opening) that are security liabilities.
- **DNS:** the directory lookups that reveal every site you visit.
- **Unbound/Pi-hole/Quad9:** local resolver / network filter / clean Swiss DNS.
- **OpenWrt:** open-source router operating system; the maximal-control path.
- **KRACK/Dragonblood:** the 2017 and 2019 breaks of WPA2 and WPA3.
- **VPNFilter/Slingshot/Cyclops Blink:** nation-state router malware campaigns.

- **ASIL-D:** automotive safety certification (QNX's domain; relevant to the cars companion document).
- **BRS/Tier:** OpenIntel complicity scoring from the laptop investigation.

16. The Conclusion, in One Paragraph

The router is the one device in the home that everything else trusts, and therefore the one where ownership matters most — and it is also, by luck of its open-software heritage, the one device where a private citizen can still achieve near-complete control. The audit finds a category split not by brand virtue but by architecture: cloud-mandatory systems (eero, Deco) that report your household's metadata to a manufacturer's servers by design, versus local-first systems (Orbi, OpenWrt, Turris) that keep it in the house; silicon that is entangled at the top two suppliers (Broadcom, Qualcomm) and audited-clean at the third (MediaTek); and an attack history that proves the box is contested territory among nation-states. The 2026 FCC ban has narrowed the retail shelf to Netgear and Amazon; the citizen's answer is to take the shorter wire — a bridged modem, a local-first router, a private DNS, and thirty minutes of hardening — and thereby turn the most surveilled device in the house into the most defensible one. Every claim above is sourced and graded; every gap is named; the pattern that held across laptops, phones, apps, and the Lebanon file holds here too: the clean answer is never a better company. It is a shorter wire.

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Compiled 2026-09-21. Five research passes; grades preserved; self-reported data practices labeled; no-evidence-found statements kept for what they are.

Part Seven: The Zero-Ties Car: A Complete Audit of Israeli Entanglement in the Connected Automobile

A fully-sourced investigation into the modern car — the phone's big sibling — its Israeli-built safety, security, and data layers, the documented hacking record from the university lab to Vault 7, and the realistic protections available to a driver, as of September 2026.

1. Why the Car Is the Phone's Big Sibling

A modern car is a data center on wheels: dozens of computers networked together, a cellular modem, GPS, Bluetooth, Wi-Fi, cameras, microphones, radar, and — in connected models — a continuous uplink to the manufacturer. The average new car runs more lines of code than a fighter jet. And it sits in your driveway, unlocked by radio, updated over the air, and increasingly monetized as a data source. When WikiLeaks published the CIA's Vault 7 files in 2017, one passage chilled the automotive world: the agency's capabilities against vehicle systems would permit "**nearly undetectable assassinations**." That sentence is this document's starting point — not as a claim about any specific death, but as the standing proof that the car's attack surface was understood, at nation-state level, years before most drivers knew their car had an attack surface.

How to read this document

- **Part 2** — the short answer. **Part 3** — orientation for non-specialists.

- **Parts 4–7** — the audit: silicon, the security backbone, the data layer, and the OEM ledger.
- **Part 8** — the documented hacking record. **Part 9** — EV charging. **Part 10** — privacy.
- **Part 11** — protections. **Part 12** — the minimum-entanglement configuration.
- **Parts 13–17** — method, open questions, key figures, glossary, conclusion.

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2. The Short Answer

No clean new car exists — the automotive safety, security, and data layers are structurally Israeli-built across every major brand. Mobileye (Intel-owned, Israeli) supplies the driver-assistance eyes in a majority of the world's vehicles. Qualcomm's automotive silicon — designed across Israeli R&D centers running since 1993 — runs the dashboards and modems of much of the industry. The automotive-cybersecurity industry that guards it all is an Israeli cluster: Argus (acquired by Continental for $430 million, founded by three Unit 8200 veterans), Upstream, Karamba (General Motors' partner), C2A, GuardKnox, and Cymotive — Volkswagen's joint venture with former Israeli intelligence leadership. The connected-car data marketplace was built in Herzliya (Otonomo). And every major OEM either has documented military/settlement entanglement profiles (GM, Volkswagen, Toyota, Mercedes-Benz — per Who Profits) or Israeli venture operations (Hyundai's CRADEL Tel Aviv). The realistic answer for a driver is therefore **not elimination but minimization**: the simplest, least-connected vehicle that meets your needs, hardened keys, no data-sharing subscriptions, and awareness of the relay-theft epidemic that has made keyless cars the easiest property crime in the industrialized world.

3. A Thirty-Second Orientation

A car's computers are divided into domains: the **powertrain** (engine and brakes), the **infotainment head unit** (screen, radio, navigation), the **telematics control unit or TCU** (the cellular modem that phones home), and increasingly **ADAS** (driver assistance — cameras, radar, lane-keeping). These domains talk over internal networks (CAN bus). The security question is always the same: who can cross from the outside world (Bluetooth, cellular, key fob) into the safety-critical inside? Everything in Parts 4–10 is an answer to that question.

4. The Silicon Layer

- **Mobileye (Intel-owned, Jerusalem-founded; ~96.9% Intel voting control):** the ADAS leader — eyeQ chips and SuperVision in vehicles from dozens of brands; SuperVision builds on QNX. If your car has lane-keeping or automatic emergency braking, the odds are excellent the eyes are Israeli.
- **Qualcomm (Snapdragon Digital Chassis; Israeli R&D since 1993, ~700 Israel staff):** infotainment and TCU silicon across the industry. The automotive ledger includes **Autotalks** (V2X, ~$350–400 million, 2023), **Cellwize** ($350 million), **Wilocity** (~$300 million, 2014), **CSR's Israeli imaging unit** ($45 million). At CES 2026, Qualcomm and Google announced pre-integrated Android Automotive on Snapdragon — tightening the pairing of the two most entangled vendors.

- **Nvidia** (Mellanox-derived automotive networking) and **NXP** (its Yokne'am, Israel design center) supply supporting silicon.
- **The clean alternative at the OS layer: QNX (BlackBerry, Canadian)** hosts 235 million vehicles' safety systems at a 38.2% share with automotive's highest safety certification — Canadian-authored, clean at corporate level. Android Automotive (Google) is the entangled, fastest-growing rival (~18% by 2027 projected). Automotive Grade Linux (open) and Wind River (Aptiv-owned) sit between. Elektrobit — Continental's software arm — inherits the Argus tie below.

5. The Security Backbone — an Israeli Cluster

The industry that *guards* the connected car is, to a first approximation, Israeli:

- **Argus Cyber Security** (founded 2013 by three **Unit 8200** veterans plus Zohar Zisapel; renamed **PlaxidityX** in 2024): acquired by **Continental for ~$430 million in 2017**; Argus technology sits inside Continental's automotive software, which ships to most major OEMs; Argus also maintains a partnership with Check Point. Continental's entire connected-car security stack is Israeli-authored.
- **Upstream Security** (Herzliya, 2017): cloud automotive monitoring, a category leader.
- **Karamba Security:** ECU hardening; **partnered with General Motors** for vehicle-SDV security.
- **Cymotive Technologies:** co-founded by **Volkswagen's CARIAD** with former Israeli intelligence leadership; being acquired by India's KPIT for up to **$120 million** (May 2026)

— VW's software-defined-vehicle security is Israeli-built.

- **C2A Security, GuardKnox:** Israeli automotive-cyber startups. PlaxidityX (the ex-Argus team) also found the EVerest charger vulnerability (Part 9).

The plain-language summary: **the locks on the car's doors are largely made in Israel.** Under this investigation's criterion that is the decisive layer — and it cannot be optioned out of a new car.

6. The Data Layer

- **Otonomo (Herzliya, 2015):** the pioneering connected-car data marketplace — ingesting **4 billion data points daily from more than 40 million vehicles**, with OEM customers including Daimler, BMW, and Mitsubishi; SPAC-listed at $1.4 billion in 2021, collapsed to roughly $70 million by 2023, and acquired by Urgent.ly in October 2023 — after weathering a documented scandal in which the startup Smartcar showed Otonomo's technical materials had copied its own. For years, the broker between your car's telemetry and its commercial buyers was an Israeli-founded platform. The category it founded now runs through larger brokers and OEM in-house programs — the data still flows; the HQ changed.
- **NissanConnect runs on Microsoft Azure** — inheriting the laptop investigation's Azure/Unit 8200 record — with **Valens Semiconductor** (Israeli; also a MediaTek portfolio company) chipsets validated into its architecture.
- **Stellantis** expanded its **Palantir** partnership (five years, March 2026) — Palantir holds Israeli R&D and Israeli military

contracts (per the apps investigation).

- The **telecommunications pipes:** Harman (Samsung) builds Ready Connect 5G TCUs on **Qualcomm** modems; SiriusXM's connected-vehicle services layer serves Stellantis-era fleets (and surfaced in the hacking record, Part 8).

7. The OEM Ledger

OEM	Documented position
General Motors	OnStar in-house (12M subscribers); Who Profits profile; Karamba security partner; Cruise employs Chris Valasek as director of cybersecurity
Ford	Who Profits profile; a Tel Aviv research center is reported but unverified in this investigation — flagged
Volkswagen	Cymotive JV (Israeli-built SDV security); Who Profits profile
Toyota	Electreon (Israeli wireless-charging) partnership, March 2025; Who Profits profile
Mercedes-Benz / BMW	Mobileye partnerships; Mercedes holds a Who Profits profile
Hyundai	**CRADEL Tel Aviv** (2018) — venture ties incl. Autotalks, Allegro.ai, Percepto, OPSYS, D-ID; on-record praise of Israel's automotive sector
Stellantis	Palantir expansion (2026); Qualcomm connected-tech rollout; no Who Profits profile (per the Maserati audit's negative finding)
Nissan	Azure telematics; Valens chipsets; Alliance Tel Aviv innovation lab; Mobileye-powered ProPilot

The mirror-image finding: Israel's own military now excludes Chinese connected vehicles — the IDF's November 2025 tender for 10,000 leased vehicles excluded Chinese brands (Chery was dropped from contention) on the same supply-chain logic this series applies outward. The principle is universal; the

application is selective.

8. The Documented Hacking Record

- **2011 — the academic foundation:** University of Washington and UC San Diego researchers (Koscher, Czeskis, Roesner, Patel, Kohno) published the first comprehensive experimental analysis of a modern automobile, demonstrating that internally compromised components could commandeer brakes and engine — the paper that created automotive cybersecurity as a field.
- **2015 — the highway demonstration:** Charlie Miller and Chris Valasek remotely hijacked a **Jeep Cherokee** through its Uconnect head unit: an unauthenticated service on the Sprint-connected infotainment computer reached the CAN bus, letting them cut the transmission on a live highway (with a Wired reporter at the wheel), later steering and braking at low speed. **Fiat Chrysler recalled 1.4 million vehicles** — the first cybersecurity-driven vehicle recall — while maintaining "no defect" in the traditional sense. The industry response was to hire the hackers: Valasek became Cruise's director of cybersecurity; Miller went to GM. A ten-year retrospective (VehicleSec, 2025) marked how far the field has come — UNECE regulations R155/R156 (July 2024) now make cybersecurity and OTA-management systems legally mandatory for new vehicle types, a direct descendant of this work.
- **2017 — Vault 7:** the CIA's files described vehicle-systems access enabling "nearly undetectable assassinations" — the

capability statement that frames every later car-hacking headline.

- **2023 — the telematics-broker flaw:** a vulnerability in SiriusXM's connected-vehicle services allowed remote unlock and engine start on multiple brands (Honda, Nissan, Acura, Infiniti reported) via nothing but the VIN — B-grade sourcing, flagged; and Nissan separately surfaced an unauthenticated-API exposure of vehicle data (B-grade). The lesson of both: the *broker layer* between you and the car is itself attack surface.
- **The everyday epidemic — relay theft:** the most common car "hack" needs no malware at all: two criminals with a signal-amplifying relay box extend your key fob's sleeping radio to the parked car, open it, and drive off in under a minute. ADAC testing found only **about 15% of vehicles** resisted the technique; in the Netherlands, **59% of 2025's stolen cars** were taken keylessly. Defenses: Faraday pouches for keys at home, OBD-port locks, software immobilizers (e.g., xProtect), kill switches, trackers, and — simplest — a physical steering lock.

9. The EV Charging Layer

The charger is the car's strangest appendage — a 240-volt, internet-connected computer on your wall. Its protocol, **OCPP**, has shipped with plaintext communications and weak authentication; six OCPP zero-days surfaced in April 2024; the UK suspended a Spanish-made charger from sale in February 2024 over security; 116,000 user records were breached in November 2024. **Pwn2Own Automotive** made it a contest: the 2025 edition produced 49 flaws (Tesla's Wall Connector hacked five ways; the Ubiquiti EV Station shipped with its cryptographic key extractable from firmware; a Phoenix Contact chain earned $91,750), and the 2026 Tokyo edition produced 76 zero-days including full control of Alpitronic's HYC50. In the wild, **CVE-2026-4156** grants root over ChargePoint's Home Flex via its OCPP interface, and **CVE-2024-37310** (EVerest, found by PlaxidityX — the Israeli ex-Argus firm) shows the open charger stack needs its own guardians. Researchers have demonstrated grid-destabilization scenarios — mass synchronized demand spikes — turning home chargers into infrastructure weapons. Israeli firms (C2A's EVSec, PlaxidityX's AutoTester) are, again, prominent in the defense of this layer.

10. The Privacy Verdict

Mozilla's *Privacy Not Included* reviewed 25 car brands in September 2023 and failed **all of them** — calling cars "the worst product category we have ever reviewed," worse than mental-health apps: every brand collects more data than needed, 84% share or sell it, Tesla scored worst, and the data flows through sensors, paired phones, navigation, and third parties (SiriusXM, Google Maps). The car is the most data-hungry consumer device in this series — and, unlike the phone, it cannot be carried into a Faraday pouch.

11. The Protections Chapter

What a driver can actually do, in order of impact:

1. **Buy less connected car.** Trim levels matter: decline connected-services subscriptions, navigation packages with modems, and in-car commerce. The simplest drivetrain that meets your needs is the strongest privacy decision available.
2. **Harden the keys.** Faraday pouch at home; store keys away from doors; consider an OBD lock and software immobilizer in relay-theft hotspots; a $30 steering lock defeats the entire relay industry.
3. **Audit the account.** Factory app accounts: minimize permissions, decline marketing consent, request data deletion where offered (84% of brands share data; deletion policies vary).
4. **Keep software current** — now a legal requirement for makers (UNECE R155/R156) and a real defense for you.

5. **Don't jailbreak or plug untrusted devices into the OBD port**; the port is the master key.
6. **For the threat-model-extreme:** the used, pre-connectivity car (roughly pre-2014 for minimal attack surface) is the maximal-clean answer — with the honest tradeoff of losing modern safety systems.

12. The Minimum-Entanglement Configuration

There is no zero-ties new car: Mobileye's eyes, Qualcomm's dashboards, and the Israeli-built security backbone come with the territory. The realistic floor: the least-connected trim of a chosen model; keys in a Faraday box; no telematics subscriptions; OBD port locked; steering lock if local theft rates warrant; safety systems left intact (they are the QNX-hosted, Canadian-authored layer and should never be disabled). For the strictest criterion, the pre-connected used car remains the only true exit — accepting older crash-safety economics as the price.

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13. Methodology and Evidence Grading

Four research passes: corporate and silicon; the security and data clusters; the hacking record and relay epidemic; EV charging and privacy. Grades: audit-grade (OpenIntel references where they exist), S/A-grade (UNECE texts, Pwn2Own results, UW/UCSD and Miller/Valasek work), B-grade flagged (SiriusXM VIN flaw, Nissan API, Ford Tel Aviv center), no-evidence-found (QNX/BlackBerry corporate), and on-record quotes preserved verbatim.

14. What Remains Unresolved

1. Ford's Tel Aviv research center: reported, unverified here.
2. The SiriusXM connected-vehicle flaw's full brand list: B-grade.
3. OEM in-house data platforms' current suppliers (post-Otonomo): partially opaque by design.
4. Chinese OEMs (BYD et al.): outside US-market scope, unverified.
5. Whether Cymotive's sale to KPIT completes: pending at compilation.

15. Key Figures at a Glance

Figure	Value	Context
Mobileye voting control	96.9% Intel	ADAS in a majority of new cars
Argus/Continental	~$430M (2017)	Three Unit 8200 founders
Cymotive/KPIT	Up to $120M (2026)	VW's SDV security, Israeli-built
Jeep recall	1.4M vehicles	First cybersecurity-driven recall, 2015
Relay-theft protection rate	~15% (ADAC)	59% of 2025 Dutch thefts keyless
Pwn2Own Automotive 2026	76 zero-days	Tokyo; incl. Alpitronic full control
QNX share	38.2%; 235M vehicles	Clean Canadian safety OS
Otonomo	4B data points/day, 40M vehicles	Collapsed $1.4B → ~$70M → Urgent.ly

16. Glossary

- **TCU:** telematics control unit — the car's cellular modem.
- **ADAS:** advanced driver assistance — cameras/radar/lane-keeping.

- **CAN bus:** the car's internal network connecting its computers.
- **OBD port:** the diagnostic port under the dash — the master key.
- **QNX/AAOS/AGL:** the car's operating systems (Canadian/Google/open).
- **UNECE R155/R156:** 2024 mandates requiring cybersecurity and OTA-management systems.
- **Relay theft:** keyless entry defeated by radio extension — no malware, under a minute.
- **V2X:** vehicle-to-everything communication (Autotalks' domain).
- **OCPP:** the charger network protocol with the checkered security record.
- **Who Profits:** the Israeli-occupation corporate database referenced for OEM profiles.

17. The Conclusion, in One Paragraph

The car inherits everything this series established about phones and doubles it: a single fused architecture, a privileged second computer (the TCU, the modem of the road), an attack surface documented from the university lab to the CIA's own files, and — uniquely — an entanglement so structural that no option sheet escapes it: the eyes (Mobileye), the dashboards (Qualcomm), the locks (Argus/Continental, Karamba/GM, Cymotive/VW), and for years the data broker (Otonomo) are all Israeli-built, and every major OEM's ledger carries a documented tie. The protections that remain are real but modest: the least-connected trim, hardened keys against the relay epidemic, locked OBD ports, kept-current software, and — for the strictest criterion — the pre-connected used car as the only true exit. The same pattern that closed the Lebanon file closes this one: when the holder cannot verify the authorship of the layers beneath them, the rational response is minimization — shorter wires, fewer radios, older simplicity — and the car, more than any device yet audited, punishes the alternative. Everything above is sourced, graded, and gap-marked; the open questions are listed in Part 14, and nothing unresolved is presented as resolved.

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Compiled 2026-09-21. Four research passes; B-grade items flagged; on-record quotes verbatim; the Ford Tel Aviv center and SiriusXM flaw explicitly bounded.

Part Eight: The Zero-Ties Implant: A Complete Audit of Israeli Entanglement in Connected Medical Devices

A fully-sourced investigation into the devices we cannot switch off — pacemakers, defibrillators, insulin pumps, continuous glucose monitors — their Israeli corporate apparatus, the hacking record from Barnaby Jack to the FDA's recall files, and the honest limits of what a patient can do, as of September 2026.

1. Why Medical Devices Are the Hardest Target in This Series

Every device audited so far — laptop, phone, router, car — shares one mercy: you can put it down. A pacemaker, an insulin pump, an implanted defibrillator cannot be switched off, Faraday-pouched, or left in a drawer. It is inside you, its radio is always near, and the person who might care most about its vulnerabilities is also the person least able to do anything about them: the patient, who chose it in a clinic, under time pressure, from a shortlist written by someone else.

This document therefore has two duties the others did not: to map the entanglement completely, and to be brutally honest about the protection ceiling — because in this category, more than any other, the honest answer to "how do I protect myself" is partly "you cannot, fully — but here is everything that does work."

How to read this document

- **Part 2** — the short answer. **Part 3** — orientation for patients and families.
- **Parts 4–7** — the audit: cardiac devices, diabetes devices, the hack record, and the regulatory bedrock.
- **Part 8** — the patient-defense ecosystem (the people doing this work for free).
- **Part 9** — protections: what actually works and what does not.
- **Part 10** — minimum-entanglement choices by device type.
- **Parts 11–15** — method, open questions, key figures, glossary, conclusion.

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2. The Short Answer

The cardiac-device corporate ledger is dominated by Medtronic — whose Israeli apparatus is the largest of any medical company in this series — while the diabetes-device stack is, remarkably, the cleanest major hardware category yet found. Medtronic operates roughly **1,200 employees across eight Israeli sites** (Jerusalem, Yokneam, Herzliya) and built its surgical-robotics, respiratory-monitoring, transcatheter-valve, and capsule-endoscopy businesses on Israeli acquisitions — Mazor Robotics ($1.6 billion, 2018, the largest medtech acquisition in Israeli history), Oridion ($346 million), Ventor ($325 million, plus a court-ordered NIS 674 million tax ruling finding its IP had effectively transferred to Medtronic without full compensation), Given Imaging's PillCam (via Covidien, 2014), CathWorks, and a $20 million stake in Israeli insulin-pump startup Triple Jump — active and growing through 2025–2026 under Medtronic Israel CEO Arik Corcos. Boston Scientific holds an Israeli tie (SoniVie, up to $540 million, March 2025). **Biotronik — the Berlin company whose home-monitoring data flows to Germany — is the cleanest corporate in the cardiac stack.** Abbott's St. Jude business owns the largest device-cybersecurity recall on record. The diabetes stack — Dexcom, Tandem, Insulet, Abbott's Libre, Beta Bionics — surfaced no Israeli corporate ties at all. And the open-source Tidepool Loop pathway offers something no other category in this series has: a community-built, FDA-cleared algorithm the patient can actually read.

3. A Thirty-Second Orientation for Patients and Families

An implanted cardiac device — pacemaker, ICD (implantable cardioverter-defibrillator), or ILR (implantable loop recorder, a long-term heart monitor) — contains a radio. At home, a **transmitter or monitor** (a bedside box or a smartphone app) reads it regularly and forwards the data to the manufacturer, who forwards it to your clinic. None of these systems can change your device's programming remotely — by design, implant programming happens only in the clinic, with a wand on the skin. That design boundary is the single most important safety fact in this document: every documented attack could *read* data or, in the worst theoretical case, exhaust a battery or deliver an inappropriate shock — appalling possibilities — but no one has ever changed a device's pacing prescription from across the internet. Hold that fact; it bounds everything else.

4. The Cardiac Ledger

- **Medtronic — the entangled giant.** Beyond the Israeli apparatus in Part 2, the cardiac-specific facts: the **Reveal LINQ** loop recorder pairs with a proprietary cellular bedside monitor (MyCareLink) that uploads nightly; its pacemakers and ICDs pair with the CareLink ecosystem. Medtronic's devices have also been the community's benchmark target: the DEF CON BioHacking Village has tested Medtronic devices three years running, with Medtronic reporting "zero vulnerabilities discovered" — a claim the Village's coordinated-disclosure culture neither confirms nor disputes publicly.
- **Abbott (St. Jude Medical) — the recall record.** In August 2016 the short-seller Muddy Waters published a report (commissioned from the security firm MedSec) alleging the Merlin@home RF communicator could be attacked; in January 2017 the FDA issued a safety communication and ICS-CERT an advisory, and Abbott pushed an automatic patch to nearly all units; in April 2017 the FDA sent Abbott a warning letter over its cybersecurity risk-assessment process; and in **August 2017 came the largest device-cybersecurity recalls in history: a firmware update for ~465,000 US implanted RF-enabled pacemakers** (requiring an in-person clinic visit, ~3 minutes) **and a recall of ~495,000 Merlin@Home transmitters.** The stated attack consequences: rapid battery depletion and inappropriate pacing or shocks. The system's defense history since: no public exploitation.

- **Boston Scientific — the SoniVie tie.** Acquired Israeli Renal-denervation firm SoniVie (Rehovot) for up to $540 million, March 2025 ($360 million plus milestones); its LATITUDE home-monitoring ecosystem otherwise clean.
- **Biotronik — the clean standard.** Berlin-based; its **Home Monitoring** system (CardioMessenger GSM unit, data to a center in **Germany**) pioneered remote cardiac monitoring (FDA clearance 2001) and remains the most privacy-preserving major platform. No Israeli corporate ties surfaced in any pass.
- **The home-monitoring layer, mapped:** four manufacturer ecosystems — Biotronik (GSM to Germany), Medtronic CareLink, Boston Scientific LATITUDE, Abbott Merlin.net — all manufacturer-tied, none permitting remote programming, with Abbott's holding the incident record. Device choice at implant time is, functionally, a choice among these four data paths.

5. The Diabetes Ledger — the Clean Corner

- **Continuous glucose monitors: Dexcom** (San Diego — no Israeli ties surfaced; note its own FDA warning letter of March 2025 and a Class I recall of one Android app build — an app-layer defect, fixed; acquired Nutrisense in 2026), **Abbott FreeStyle Libre** (US; corporate history in Part 4), **Medtronic's CGM** (inherits the Part 4 apparatus).
- **Insulin pumps: Tandem** (San Diego; SEC-grade corporate, no ties found), **Insulet's Omnipod** (Massachusetts; tubeless; no ties found), **Medtronic MiniMed** (Ireland/US operating

history; Medtronic corporate apparatus), **Ypsomed** (Swiss; partners with Cambridge's CamDiab on the MyLife Loop — the Swiss-British clean path), **Beta Bionics' iLet** (US, rising fast).

- **The open-source pathway:** the DIY closed-loop movement (#WeAreNotWaiting, 2013) built **Loop** (an iPhone app connecting pump and CGM over Bluetooth LE), which the non-profit **Tidepool** (backed by the Helmsley Charitable Trust and JDRF, $6 million) carried through the FDA — cleared **January 23, 2023**, the first open-source algorithm and the first interoperable automated-insulin-dosing app ever authorized; partners include Insulet's Omnipod and (future) Medtronic Bluetooth pumps, with Dexcom G6 as CGM. The caveat, consistent with this whole series: the open algorithm runs on an iPhone — the cleanest software on the most audited-tainted host. It is the only category where the patient can read the code that keeps them alive.

6. The Hack Record — from Demonstration to Regulation

- **Barnaby Jack, the founding dossier.** In 2011, the New Zealand hacker demonstrated at McAfee's FOCUS conference a wireless takeover of an insulin pump: **no serial number needed** (improving on Jay Radcliffe's 2011 Black Hat work, which had required one), a range of ~300 feet with a high-gain antenna, commanding maximum 25-unit doses repeatedly until the 300-unit reservoir ran dry — multiple lethal doses, delivered by radio. In 2012, at BreakPoint in Melbourne, he went after a pacemaker: querying the device returned its model and serial number, and he delivered an **830-volt shock** to a test unit — and showed the attack could propagate, device to same-vendor device. He was scheduled to present "Implantable Medical Devices: Hacking Humans" at Black Hat 2013; **he died on July 25, 2013**, a week before the talk (a coroner ruled accidental drug overdose; he was 35). The FDA's device-safety leadership publicly credited his demonstrations with forcing the issue; the FDA's first cybersecurity guidance followed in 2013. DHS analyses separately noted implant vulnerability to strong electromagnetic pulses.
- **The St. Jude sequence** (Part 4) — the market's stress test.
- **The infrastructure flaws:** URGENT/11 (VxWorks TCP/IP) and Ripple20 (Treck TCP/IP) — bugs in the third-party network stacks buried inside medical devices — drove recalls and advisories across the industry, motivating the SBOM regime below.

- **The DEF CON record:** the BioHacking Village has tested hundreds of devices across a decade; at DEF CON 34 (August 2026) researchers found **68 vulnerabilities** across medical devices, all under coordinated disclosure — "zero headlines," which is the point.

7. The Regulatory Bedrock

The **PATCH Act's Section 524B** (enacted in the Consolidated Appropriations Act, 2023; effective March 29, 2023) made medical devices **the first product category in the world with a mandatory Software Bill of Materials** — manufacturers must submit an SBOM (SPDX or CycloneDX formats, per NTIA minimum elements), maintain a vulnerability-monitoring and patching plan with defined cadence, offer coordinated vulnerability disclosure, and (per the June 2025 guidance) notify customers of exploitable vulnerabilities within 30 days; FDA guidance has iterated (September 2023 → June 2025 → February 2026, aligning with the quality-management overhaul QMSR/ISO 13485). The EU's Cyber Resilience Act (2024) converges on the same architecture. From patch-or-perish to paper-or-RTA (refuse-to-accept enforcement began October 2023): the industry that Barnaby Jack embarrassed in 2011 now has the strictest software-supply-chain market entry on earth.

8. The Patient-Defense Ecosystem

The people protecting patients without being asked: the **BioHacking Village** (DEF CON's medical arm, run by Nina Alli; 5,000-plus hackers trained; its #WeHeartHackers coordinated-disclosure program partnered with the FDA itself in 2019); **I Am The Cavalry** (the grassroots human-safety initiative whose volunteer researchers forced manufacturers to the table); the **Hippocratic Oath for Connected Medical Devices** (3,274 signatures from researchers pledging responsible disclosure); and the **2016 DMCA exemption** that legalized good-faith medical-device security research — won, in part, by that community. Medtronic, Philips, Abbott, BD, and Roche now send engineers to the Village. The inside of the industry has quietly adopted its adversaries' methods; patients are the beneficiaries.

9. Protections — What Works and What Does Not

What does not work, said plainly: no RF-shielding pouch, bag, or garment is a recognized mitigation for an implanted device's radio (searched across rounds; no standard exists, and strong magnets near some devices are themselves contraindicated). You cannot firewall an implant. Anyone selling implant "EMF protection" is selling comfort.

What does work:

1. **The pre-implant conversation** — the only true choice point. Ask the cardiologist or endocrinologist: what radio does this device have, what is its range, what is the home-monitor path, and which manufacturer ecosystem does it join? Where the clinical options are equivalent, the ledger is a legitimate

tiebreaker: **Biotronik for cardiac remote-monitoring; the Tandem/Insulet/Omnipod/Dexcom stack or Ypsomed-CamDiab for diabetes; Loop/Tidepool for algorithm transparency** (accepting the iPhone caveat).

2. **Attend every recall and firmware update.** The 2017 Abbott recall was resolved in ~3-minute clinic visits; the population-level defense is procedural, and it works — treat recall letters like prescriptions.
3. **Register the device** with the manufacturer so advisories reach you.
4. **Use only the manufacturer's transmitter/app** for home monitoring; the phone inherits its own ledger (per the phone investigation), so the app choice is a second, quieter decision.
5. **Clinic checks on schedule** — interrogation sessions confirm battery and settings regardless of the network.
6. **For the threat-model-extreme** (a handful of public figures, abuse survivors, journalists): discuss with the care team whether a device with wireless functionality disabled or minimized is clinically appropriate — some devices support clinician-controlled RF limits, and the question costs one appointment.

10. Minimum-Entanglement Choices by Device Type

Need	Cleanest mainstream option	Residual
Pacemaker/ICD + remote monitoring	**Biotronik** (Berlin; data to Germany)	Clinic decision governs; silicon supply chain universal
Heart rhythm monitoring (ILR)	Abbott Confirm Rx (runs through *your* phone — choose the phone per the phone report) or Biotronik	Phone-ledger inheritance
CGM	**Dexcom** or Abbott Libre	App-layer; Dexcom's own 2025 FDA letter noted
Insulin pump (tubed)	**Tandem**	—
Insulin delivery (tubeless)	**Insulet Omnipod**	—
Automated insulin dosing	**Tidepool Loop** (open source, FDA-cleared) on the cleanest available phone	iPhone caveat
Emergency (AED, public)	**Philips HeartStart** (offline, no connectivity) vs WiFi-enabled rivals	—

Irreducible floor: the silicon and supply chain inside any implant (the Lebanon file's lesson applies in its darkest form here — authorship of the thing inside you); the clinic's formulary; the regulatory regime that now at least forces disclosure; and the fact that in this category, unlike every other in this series, the final choice is made with a physician, not a shopping cart.

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11. Methodology and Evidence Grading

Four research passes: corporate (manufacturer by manufacturer), the security-research record, the regulatory layer, and protections. Grades: S-grade (peer-reviewed clinical reviews of the monitoring ecosystems, FDA texts), A-grade (Calcalist/CTech for Medtronic's Israeli operations; AP; Reuters), on-record (FDA officials' statements on Jack; corporate DEF CON participants), B-grade flagged (Ypsomed inferred via MyLife Loop; Dexcom recall details). Searched-and-not-found items are stated as such (RF shielding).

12. What Remains Unresolved

1. **RF-shielding for implants:** no recognized mitigation exists; the honest floor is procedural.
2. Ypsomed corporate: Swiss, inferred via partnership; unverified independently.
3. Cochlear implants and deep-brain stimulators: category noted; manufacturer-level audit incomplete (Medtronic DBS inherits the corporate ledger).
4. The 2022-protesters medical-sanctions thread: outside scope, noted once, not pursued.
5. Whether Tidepool adds Android hosting: open at compilation.

13. Key Figures at a Glance

Figure	Value	Context
Medtronic Israel	~1,200 staff, 8 sites	Mazor $1.6B; Ventor $325M + NIS 674M ruling; PillCam; CathWorks; Triple Jump
SoniVie/Boston Scientific	Up to $540M (Mar 2025)	Renal denervation, Rehovot
Abbott recalls	465,000 implants + 495,000 transmitters	Aug 2017; largest device-cyber recalls
Barnaby Jack	Insulin pump 2011; pacemaker 830V 2012; died July 2013	Forced FDA's first cyber guidance
Tidepool Loop	FDA-cleared Jan 23, 2023	First open-source AID algorithm
Section 524B	Effective Mar 29, 2023	World's first mandatory SBOM
DEF CON 34	68 vulnerabilities, coordinated	Aug 2026; "zero headlines"
Biotronik	Berlin; GSM data to Germany	Cleanest cardiac corporate

14. Glossary

- **Pacemaker/ICD/ILR:** rhythm devices; ILR = long-term monitor.
- **Home monitor/transmitter:** bedside box or app forwarding device data to the clinic.
- **Interrogation:** the clinic wand-session that reads (and only in-clinic can program) the device.
- **CGM:** continuous glucose monitor; **AID:** automated insulin dosing (the "artificial pancreas").
- **SBOM:** software bill of materials — the ingredient list the FDA now requires.
- **Coordinated disclosure:** the researcher-to-manufacturer reporting pipeline (the BioHacking Village's model).

- **DMCA exemption:** the 2016 legal carve-out permitting good-faith implant research.
- **RTA:** FDA refuse-to-accept — the enforcement teeth behind 524B.

15. The Conclusion, in One Paragraph

A medical device is the only object in this series that the owner cannot put down, cannot shield, and mostly did not choose alone — and that asymmetry defines everything the audit found. The corporate map splits cleanly: one giant built substantially in Israel (Medtronic's eight sites, its robotics and valves and capsules and insulin stakes), one giant holding the industry's cybersecurity scar tissue (Abbott's 2017 recalls), one acquisition-marked newcomer (Boston Scientific–SoniVie), and one quiet German standard for the privacy-conscious (Biotronik); while the diabetes stack — Dexcom, Tandem, Insulet, Beta Bionics — emerged as the cleanest hardware category this series has examined, crowned by the only FDA-cleared open-source therapy algorithm on earth. The hack record runs from a New Zealander's 300-foot insulin-pump takeover through an 830-volt pacemaker shock to a recall of nearly half a million implanted hearts — and its product, finally, is the strictest software-disclosure regime in any industry, defended nightly by volunteers at DEF CON. The protections, stated without decoration: choose at the implant conversation when the choice exists; keep every firmware appointment; register the device; ask the one hard question about radios. And the shadow over all of it, from the Lebanon file: the day supply-chain authorship becomes lethality, the devices that cannot be put down are the ones that matter most — which is why this document, like the implant itself, keeps working after you stop thinking about it. Everything above is sourced, graded, and gap-marked; Part 12 lists what could not be resolved, and nothing there is presented

otherwise.

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Compiled 2026-09-21. Four research passes; clinical reviews and FDA texts at S-grade; corporate reporting at A-grade; no recognized shielding mitigation exists and none is claimed.

Colophon

Compiled October 2026 from the nine constituent documents, each researched across multiple passes, written on instruction, and delivered only after adversarial audit. Reading order in this edition: the purpose, the attack, the letter, then the evidence. Every constituent document carries the series' mark beneath its title; in this edition the mark stands once, above, for all of them. Nothing has been altered from the audited originals except heading hierarchy, order, and the foreword's inventory-table sequence.